

SPATIAL CATASTROPHE RISK INTELLIGENCE

Spatial Catastrophe Risk Intelligence

Crowd Intelligence, Artificial Intelligence and Actuarial Modelling
for Continuous Catastrophe Underwriting

Narrative and Technical White Paper

Nevil Maloba

A Kenya-first architecture for continuous catastrophe underwriting

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Part 1

The Kenya Multi-Hazard Intelligence Thesis

Spatial Catastrophe Risk Intelligence

At sunrise on the lower Nzoia plains, a farming household may have maize in the field, animals on higher ground, produce expected at a market and schoolchildren who cross the same road used by traders and milk collectors. Upstream precipitation is not yet a household loss, and a rising river is not yet an insurance claim. The catastrophe takes financial form when water leaves the channel, enters a cultivated field, closes the primary road, contaminates stored food and forces the family to spend scarce cash on transport, shelter or replacement supplies. An insurer encounters the same water later and through disconnected files: a motor claim arrives with one photograph, a property policyholder reports damaged contents, and a crop portfolio reveals a concentration of affected farms.

The river does not experience those consequences as separate events. The household, county engineer, lender and insurer do. That separation is the commercial problem addressed by **Spatial Catastrophe Risk Intelligence (SCRI)**: an insurer-facing product that turns observations of one evolving physical event into an uncertainty-bearing view of exposed policies, likely damage, claims workload and portfolio loss.

1. Six catastrophes, six livelihood pathways

Kenya's flood risk unfolds through six Water Resources Authority basin areas, each shaped by its topography, settlement, infrastructure and economic activity [1]. In the Lake Victoria North basin, the Nzoia links highland rainfall to low-lying agricultural exposure. In the Tana basin, a river originating around Mount Kenya and the Aberdares passes through steep upper catchments, major infrastructure and flatter middle and lower reaches before the delta [2]. The Athi system carries metropolitan, industrial and downstream exposure; Ewaso Ng'iro North connects water scarcity, pastoral mobility and episodic flooding. SCRI preserves those directed relationships through probability distributions over depth, extent, velocity and duration on river-and-drainage networks.

Drought emerges gradually for a pastoral household in Wajir, Mandera or another arid and semi-arid county. Rainfall may arrive late, fall unevenly or stop before pasture and water sources recover. Loss accumulates through longer walks to water, declining livestock body condition, lower milk production, altered migration, unfavourable terms of trade, reduced food consumption and the sale of productive animals. Kenya's National Drought Early Warning System recognises this progression through biophysical, production, access and utilisation indicators [3]. SCRI translates that governed drought state into the exposures and contractual terms of an insurer's livestock, crop, health, credit-linked or business portfolio.

Wildfire around Mount Kenya presents a different chain. The forest supports water, grazing, firewood, cultivation and tourism-linked livelihoods; research with surrounding communities also shows that fire knowledge, deliberate land use and preparedness are inseparable from the physical hazard [4]. A camera model can detect smoke or flame earlier. That detection creates economic value when a responsible responder receives the alert, reaches the fire and changes its spread. The product therefore models **ignition -> detection -> dispatch -> intervention -> conditional severity**, connecting signal quality to response and avoided-loss evidence.

The 2019-2020 desert locust invasion makes the mobile biological peril equally concrete. Swarms crossed borders and counties while crops and pasture remained fixed. The loss depended on swarm density, lifecycle, wind, vegetation, crop stage and the timing and reach of control operations. The response combined movement surveillance with food-security protection, control, livelihood support and restoration of productive assets [5]. SCRI helps an agricultural insurer identify insured crops

and grazing areas within forecast swarm corridors, then combines surveillance with field evidence and crop-stage vulnerability to estimate loss.

Rainfall-induced landslide is intensely local. A household on an escarpment may lose a dwelling; a road blockage may isolate farms, schools and health facilities well beyond the slide footprint. Kenyan evidence describes recurrent shallow failures in Murang'a and destructive debris flows in West Pokot under combinations of intense rainfall, slope, land cover and drainage conditions [6]. A national susceptibility map supports regional screening, while reports of cracks, blocked culverts or slope movement raise the priority of local inspection and conditional failure analysis.

Extreme heat and severe storm often reveal themselves through distributed effects rather than one continuous visible scar. Heat loss can appear as reduced working hours, health expenditure, lower productivity, refrigeration demand and service disruption. Kenyan research with community health workers shows how exposure, long outdoor journeys, low income and limited access to cooling combine to affect work and livelihoods [7]. A severe storm creates another pattern: roof and crop damage from wind or hail, road interruption from intense rain, power failure, injury and business interruption. The two hazards may share meteorological observations while retaining distinct vulnerability functions and insurance terms.

Kenya's recent climate reports describe the coexistence of heatwaves, floods, intense rainfall and strong winds, together with disruption to agriculture, infrastructure, health and displacement [8]. SCRI therefore combines one shared financial interface with six physical adapters. The product communicates uncertainty and loss consistently while preserving the science of river routing, drought persistence, fire spread, swarm movement, slope failure and heat stress.

2. Three Kenyan events seen through the product

The architecture becomes credible when it reconstructs what was knowable during a real event as well as the finished disaster. A reconstruction is therefore written on two timelines. **Event time** records when rain fell, a river rose or a swarm moved. **Knowledge time** records when a usable observation reached the insurer. Later official totals enrich the historical record, while each model replay remains frozen to the information genuinely available at its cut-off.

The March-to-May 2024 floods: a national emergency made of local pathways. Kenya's 2024 State of the Climate report records severe flooding during the long-rains season. By 19 May, 41 counties and 412,763 people had been affected; 55,676 households, representing 278,380 people, had been displaced; and the official situation record reported 295 deaths, 188 injuries and 78 people missing. Sixty-four health facilities in 12 counties were affected [9]. Those totals establish the human scale. SCRI adds the event footprint, exposure intersection and insured-loss view required by an insurer.

For a lower-Nzoia portfolio, a point-in-time reconstruction would begin before the first claim. The prior state would contain seasonal forecasts, recent basin rainfall, river-level history, drainage and terrain, and the policy snapshot in force before the event. As upstream rainfall and gauge evidence arrived, the state model would update inundation probability along the directed river network. Radar water detections and independently corroborated reports would then sharpen the footprint. Only after intersection with buildings, crops, roads and policy terms would the insurer see a range of possible claims. A road closure could delay market access and claims inspection without being covered physical damage; an inundated field could be a livelihood loss even where no crop policy existed. The product

must show both facts without adding the uninsured loss to the insurer's booked amount.

The operational story is also temporal. Before overtopping, the accumulation desk sees insured value by possible depth band and the claims team prepares accessible reporting routes. During inundation, uncertainty may narrow in some cells and widen in others when gauges fail or roads become inaccessible. During recession, water extent can shrink while mould, contamination, crop loss and business interruption continue. The version history explains every revised loss estimate through a new footprint, improved depth, corrected exposure, reported claims or policy interpretation.

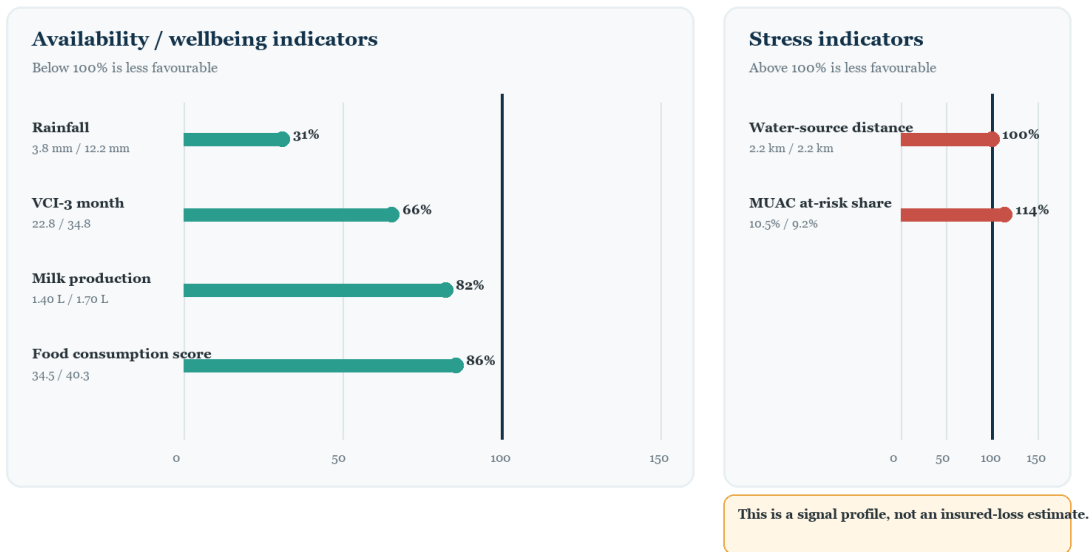
The 2019-2020 locust invasion: movement, surveillance and control. The World Bank described the invasion as Kenya's worst in 70 years. Swarms spread to 28 counties; about 175,000 hectares of crop and pasture were affected; nearly 164,000 households were within the response population; and roughly three million vulnerable people were placed at food-security risk [5]. The financial pathway extended from damaged crops to pasture, livestock feeding and mobility, public surveillance and control costs, household income, food access and restoration of productive assets.

An SCRI reconstruction clusters reports, aircraft observations and imagery into possible biological entities, with wind, vegetation and lifecycle information shaping movement scenarios. Each scenario is intersected with crop stage, grazing dependence and insured units. Control activity is recorded as an intervention with time, location and coverage. Weather, swarm behaviour, reporting coverage and control selection enter the counterfactual analysis of whether intervention changed the subsequent loss distribution. The insurer receives a corridor of possible exposure and claims, while the responsible agencies maintain the public-response record.

Isiolo in January 2026: drought as a sequence rather than a date. The county drought bulletin placed Isiolo in Alert and deteriorating conditions. It reported rainfall of 3.8 mm against a normal reference of 12.2 mm, a three-month Vegetation Condition Index of 22.8 against 34.8, milk production of 1.40 litres against a reference above 1.70, a food-consumption score of 34.5 against a reference above 40.3, and an at-risk mid-upper-arm-circumference share of 10.5 per cent against a reference below 9.2 per cent [10]. The indicators move at different speeds and use different units. Together they reveal a livelihood pathway from insufficient rainfall to weaker vegetation and water access, reduced livestock production and declining household consumption.

Isiolo drought signal, January 2026

Current indicator as a percentage of the bulletin reference; direction is shown explicitly



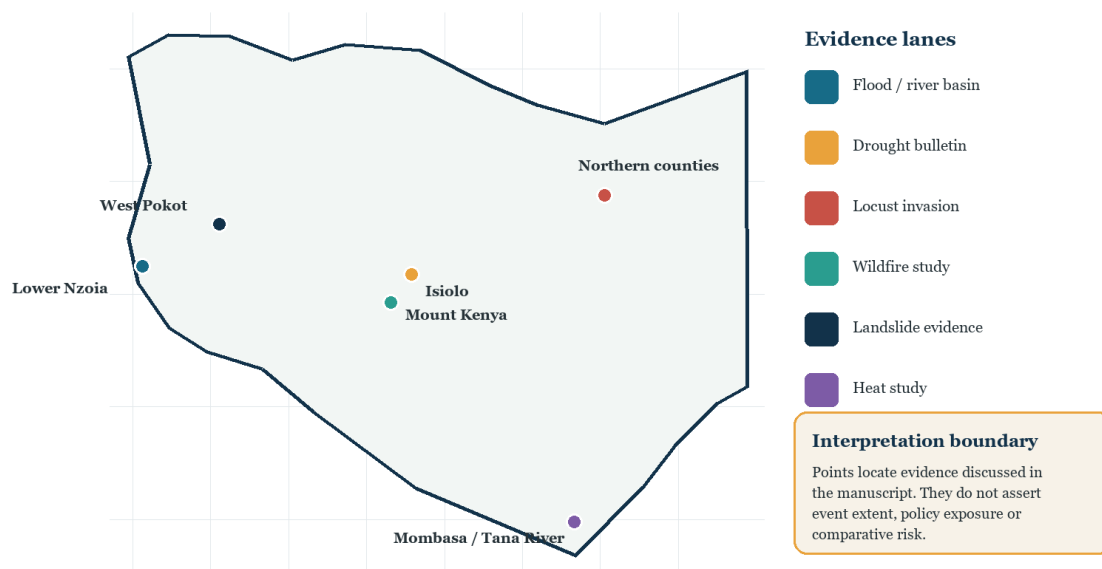
Source: NDMA Isiolo County Drought Early Warning Bulletin, January 2026.

Figure 1.1: Published Isiolo indicators as a drought-signal profile. Values are normalised to their bulletin reference for visual comparison. The chart feeds drought-state assessment; exposure, vulnerability and contract terms produce the insured-loss view.

For an insurer, this profile updates a latent drought state and directs attention to exposed livestock or livelihood products. Field evidence establishes mortality and crop damage; policy terms establish coverage; and any parametric contract calculates payout from its pre-agreed index. Household experience described by milk, nutrition and water access then supports basis-risk evaluation.

Kenyan evidence geography used in the sample

Reported or studied locations; orientation only, not a hazard boundary or pricing map



Source: manuscript evidence register. Coordinates are approximate orientation points.

Figure 1.2: Evidence geography used in this sample edition. The points are approximate orientation locations for reported or studied evidence. They are neither hazard boundaries nor a comparative risk map.

These reconstructions expose the same product discipline in three different physical systems. Flood intelligence routes water through a basin; locust intelligence tracks a mobile biological population; drought intelligence accumulates persistent and sometimes contradictory indicators. Each can use crowd and AI observations. The shared product path connects physical state to exposed livelihood, damage mechanism, contract and owned decision.

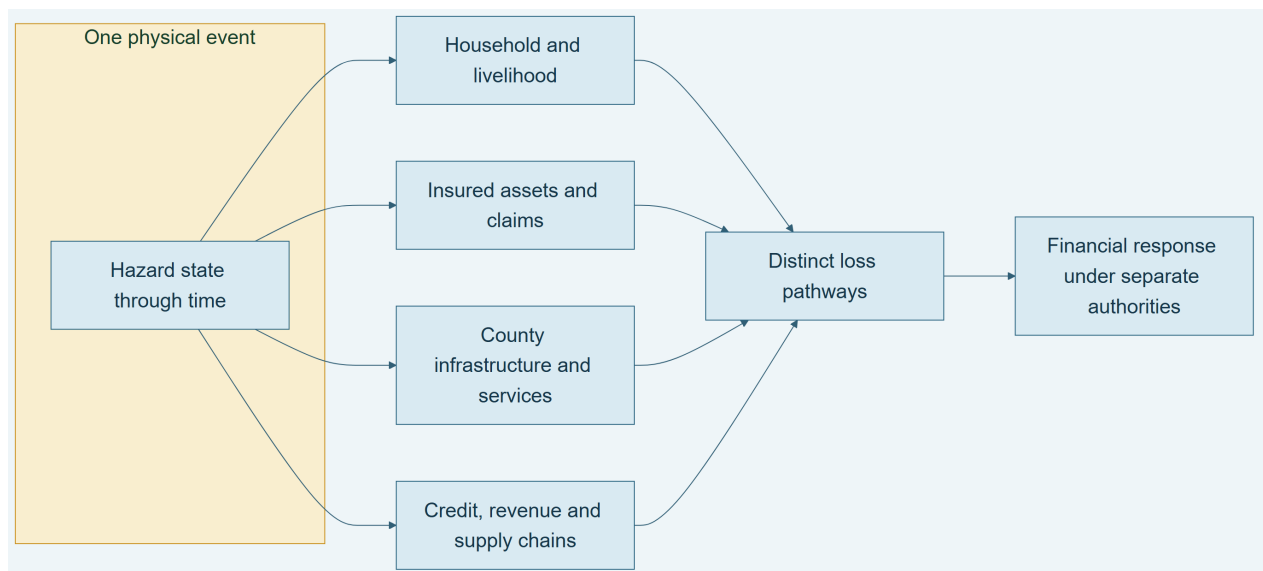


Figure 1.3: One catastrophe, several loss pathways

3. The product and the information gap

SCRI performs four connected jobs. It observes the event through authoritative sensors, earth observation, cameras, field reports and institutional records. It estimates the unobserved state while retaining uncertainty and provenance. It intersects that state with a versioned exposure portfolio and vulnerability functions. It then produces institution-specific intelligence for accumulation monitoring, claims triage, event-loss nowcasting, reinsurance communication and future underwriting review.

SCRI preserves the meaning of every transformation. A gauge reading informs flood probability; the hazard state drives a damage distribution; exposure and policy terms produce insured-loss samples; authorised teams interpret those samples for claims, reserving, capital and underwriting. Each stage carries its own data, model, owner and audit trail, allowing users to move from a portfolio number back to the evidence that shaped it.

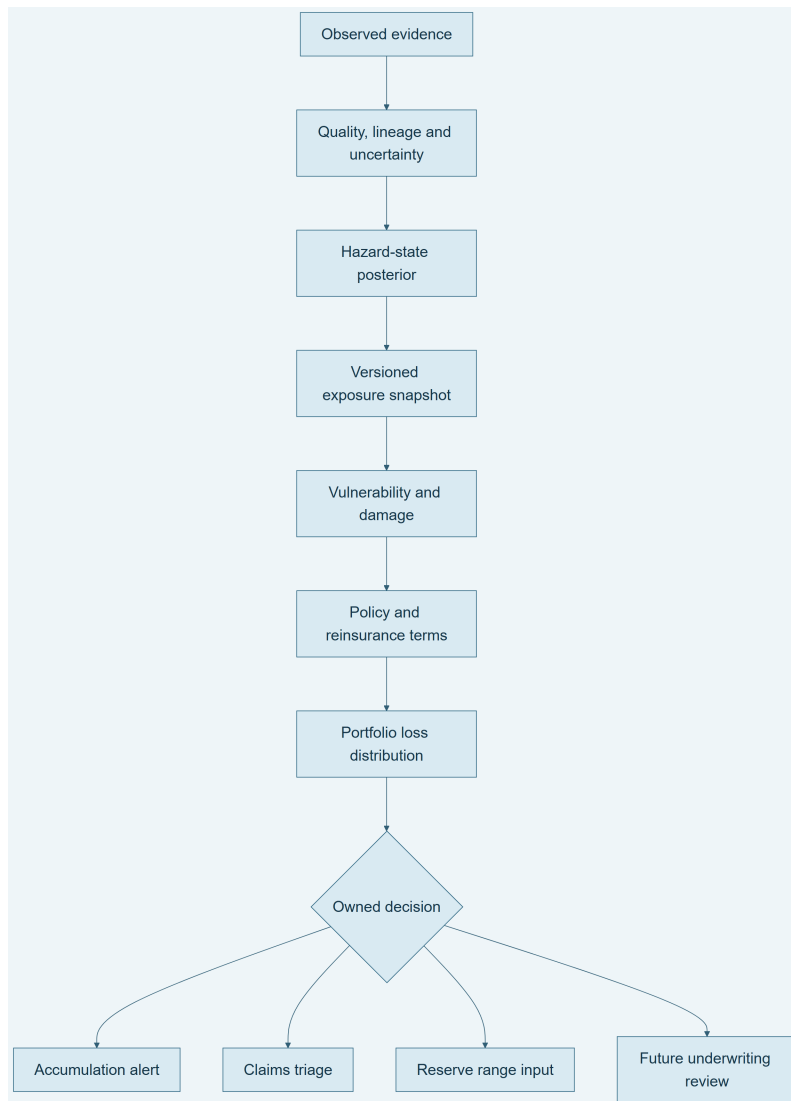


Figure 1.4: The SCRI product chain

The immediate product is a continuously refreshed catastrophe view. During an event, the insurer can identify policies that may be affected, estimate a range of gross and insured loss, prepare claims capacity, notify reinsurers and test liquidity. Validated experience then enters the insurer's established prospective process for pricing and risk appetite. The Insurance Regulatory Authority's insurance-risk guideline already treats product design, pricing, underwriting, claims management, reserving and reinsurance as connected but distinct responsibilities [11]. SCRI carries one evidence trail into those different workflows while preserving the decision rights of each team.

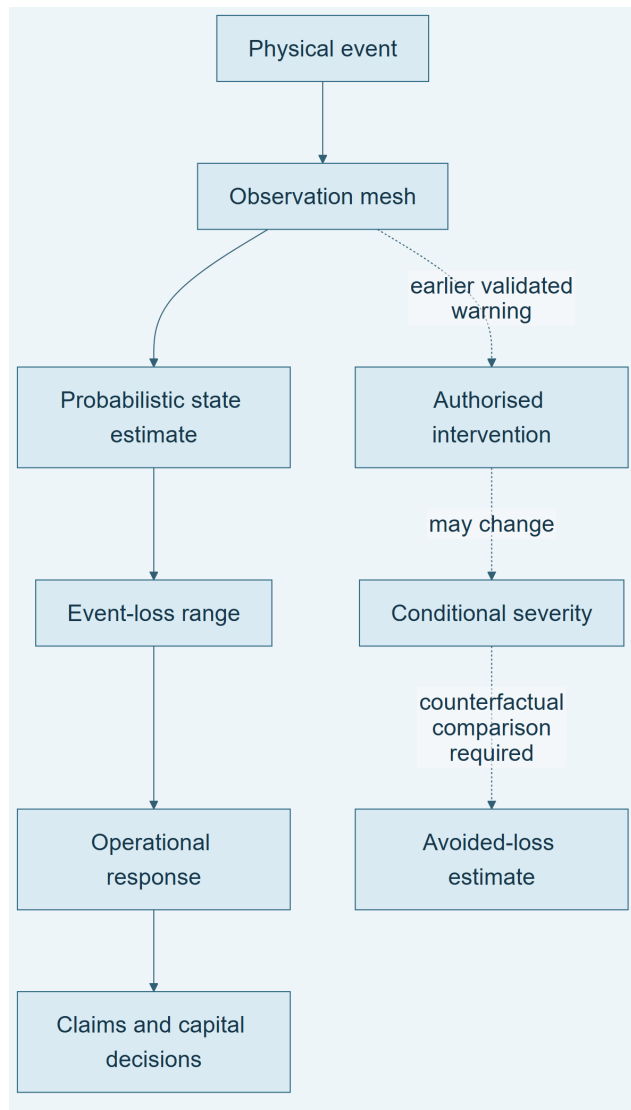


Figure 1.5: From better estimation to effective loss-reducing action

4. Actuarial modelling as the commercial bridge

The product's centre is **Actuarial modelling**. Machine perception structures observations; hazard science estimates the physical event; actuarial modelling determines what that event means for an insurance portfolio. The calculation begins at asset level:

$$L_{e,i}^{GU} = V_i D_{h,c(i)}(I_{e,h}(g_i), M_i). \quad (1.1)$$

where V_i is the replacement value or other relevant exposure measure, $I_{e,h}(g_i)$ is the uncertain hazard intensity at the asset location, M_i contains vulnerability modifiers, and $D_{h,c(i)}$ is a damage-ratio distribution for hazard h and exposure class $c(i)$. Policy deductibles, limits, exclusions and coverage conditions are then applied to obtain insured loss. Aggregating those distributions across correlated assets produces an event-loss range rather than a single exact number. Appendix C derives the conditional moments, bounds and portfolio aggregation implied by Equation (1.1).

Real-time observations principally update the current event. They may change the estimated footprint, intensity or confidence, and therefore the event-loss distribution. Long-term annual catalogues, vulnerability calibration and tariff indications change through controlled model-development cycles using sufficient experience, exposure normalisation and independent validation.

The second claim - the value of action - asks when earlier intelligence changes the event outcome. Let a denote an intervention and τ the elapsed time between detection and effective response. Lower latency creates economic value when a responsible team can act and the action changes conditional severity. For some hazards:

$$\Delta EL(a) = \mathbb{E}[L \mid \text{available response}, a_0] - \mathbb{E}[L \mid \text{available response}, a]. \quad (1.2)$$

may be positive. Estimating $\Delta EL(a)$ requires a credible counterfactual, an intervention record and uncertainty about response effectiveness. It may be material for suppressible fires, movable livestock, pest control or asset protection. It may be small when flood defences are already overwhelmed, communications fail or no responder has authority and resources to act. Appendix C expresses Equation (1.2) in potential-outcomes notation, derives its identification conditions and distinguishes the value of information from the value of intervention.

This calibrated distinction strengthens the commercial proposition. SCRI offers a traceable way to reduce uncertainty earlier, direct scarce operational attention and learn which interventions change claims and losses. Observation gaps, trigger error and response constraints become measurable product questions rather than hidden assumptions. That is the thesis the remaining parts now make implementable.

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Part 2

Crowd AI and Observation Architecture

Spatial Catastrophe Risk Intelligence

1. The distributed sensor network

During a flood, the first useful observation can arrive through many channels: a gauge rise upstream, a radar image acquired hours later, a matatu driver reporting an impassable bridge, or a farmer photographing a waterline against a familiar wall. Each source contributes a different fragment. SCRI combines those fragments into a progressively clearer event view.

Spatial Catastrophe Risk Intelligence (SCRI) begins with an observation contract that gives artificial intelligence a defined job. Kenya already has authoritative hydrometeorological infrastructure, including manual and telemetric surface-water monitoring across basin areas [1]. Sentinel-1 adds day-and-night, all-weather radar imagery, with emergency products that delineate inundation at the sensor's operational resolution [2]. Community reports add effects that remote sensing may miss: a road is unusable, livestock cannot reach water, a health facility has lost power, or a crop was at a vulnerable stage when inundation occurred.

The commercial advantage comes from joining those perspectives while preserving their meaning. A river gauge is a calibrated measurement at a known location. A satellite-derived mask is a model output with acquisition and processing uncertainty. A photograph contributes frame-specific evidence. A text report is a claim by a contributor, expressed in language and affected by access, incentives and memory. The platform stores each source according to what it is.

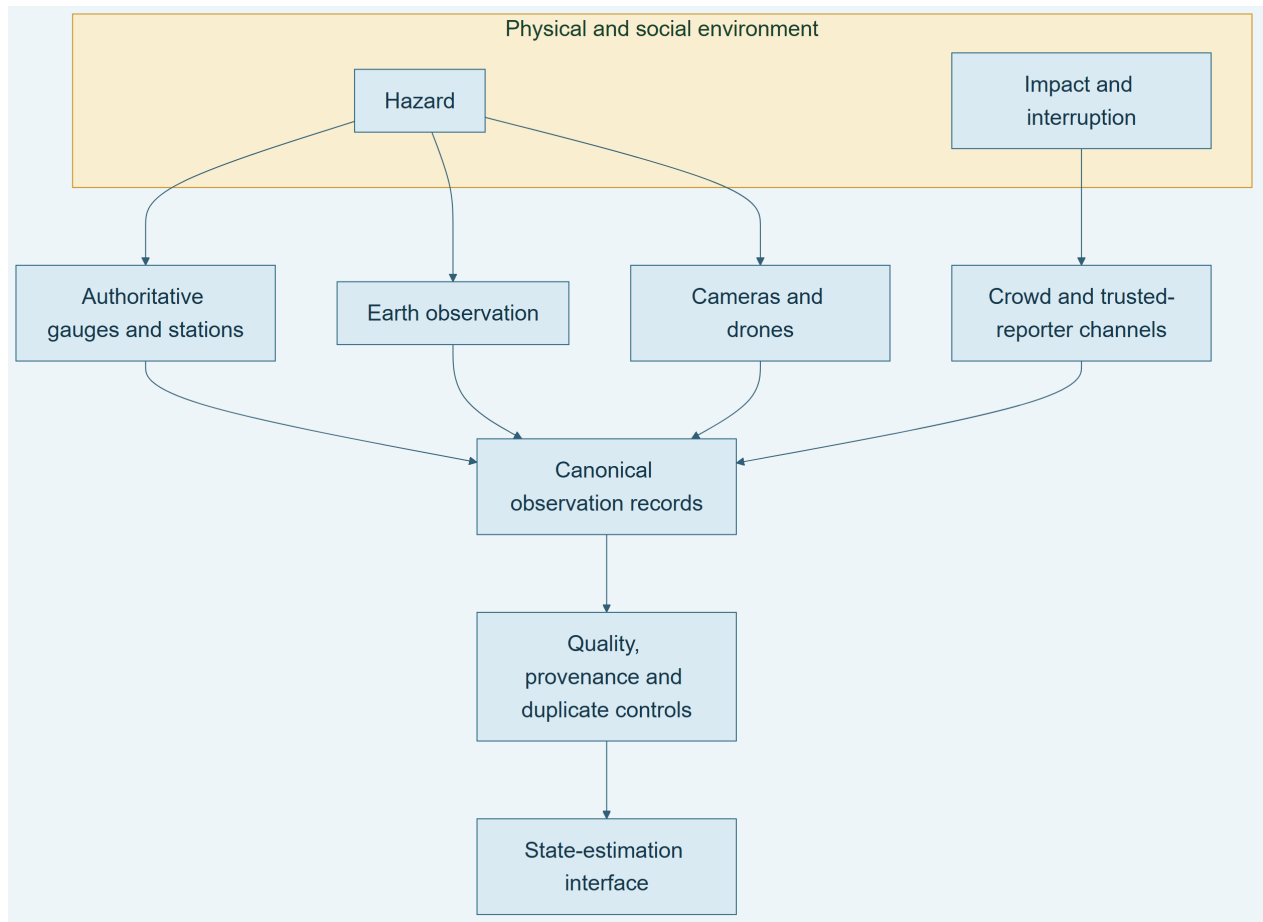


Figure 2.1: Multimodal evidence enters through one governed interface

The minimum record is deliberately explicit:

```
observation_id
hazard_type
source_type
source_id_or_pseudonym
event_time
knowledge_time
ingestion_time
geometry
coordinate_reference_system
observed_variable
value
unit
model_or_report_confidence
source_reliability
provenance
duplication_cluster
consent_and_permitted_use
quality_flags
correction_or_predecessor
schema_version
```

Three times are required because catastrophe reconstruction is a point-in-time problem. **Event time** records when the condition was observed. **Knowledge time** records when the source could have made it available. **Ingestion time** records when SCRI received it. A 10:00 replay therefore uses only evidence available by 10:00, while a flood photograph uploaded at 15:00 enters later versions. This distinction protects model validation from hindsight and allows an insurer to ask the operational question that matters: *what could we reasonably have known then?*

Geometry is equally disciplined. A phone coordinate is a point with an accuracy radius; a road report may be a line segment; a satellite mask is a raster or polygon at a stated resolution; a drought indicator may be a zonal statistic over a livelihood area. Every value carries a unit. Every model-derived feature carries the model version and lineage to the raw input. Provenance follows the general principle expressed in the W3C PROV data model: a derived entity should remain traceable to the activities and sources that produced it [3].

2. Machine perception with defined jobs

YOLO has a useful and clearly defined role. Object detection produces classes, confidence scores and bounding boxes. An instance-segmentation model produces object masks. Semantic segmentation assigns a class to pixels across a scene. Current Ultralytics documentation exposes those as distinct tasks and outputs [4]. SCRI creates flood or burn polygons from suitable segmentation and geospatial processing, while bounding boxes remain object observations.

For wildfire, detection may identify smoke, flame, vehicles or people; segmentation may estimate a visible flame or burn boundary; thermal data may provide another signal. For flood, radar- or optical-image segmentation can estimate water extent, while depth usually requires terrain, gauges,

hydraulics or a separately validated inference model. For locusts, an object detector supports close-range survey imagery, while regional swarm density comes from multiple surveys, movement evidence and ecological state. For landslide, models may segment a scar after failure while precursor risk remains a state-estimation problem. For drought and heat, time-series anomalies are often more important than vision.

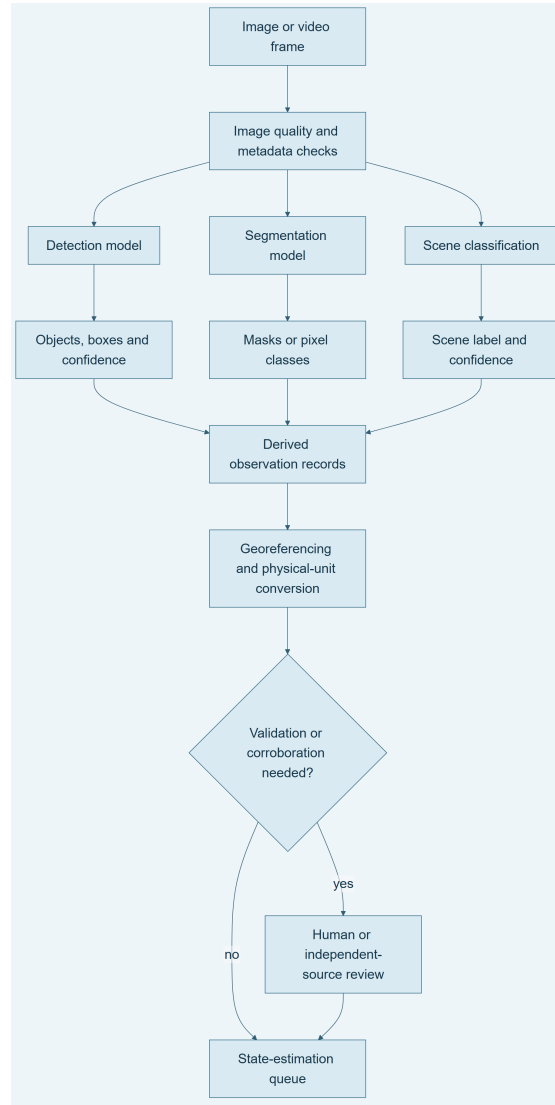


Figure 2.2: Detection, segmentation and state estimation remain separate

Model confidence and hazard probability answer different questions. A detector confidence of 0.86 describes support for a class prediction under the model’s training and calibration regime. The fusion layer combines that output with image time, location and independent observations to estimate hazard probability; the loss layer then evaluates financial consequences.

Performance thresholds are set during hazard- and use-case-specific validation. A target such as “mAP greater than 0.75 under occlusion” becomes meaningful when it is tied to a dataset, class distribution, camera geometry, decision threshold and cost of errors. Each model card states the intended use, excluded conditions, training geography, evaluation split and metrics. Detection and segmentation models are evaluated on event and geography holdouts, with precision-recall curves,

class-specific recall, calibration, intersection-over-union or Dice scores where appropriate, and operational latency. Error slices include darkness, glare, rain, smoke, low bandwidth, small objects and unfamiliar landscapes.

SCRI uses a hybrid edge-and-cloud design. Edge inference is valuable when latency or connectivity matters and the task can run on bounded hardware - for example, screening a camera stream for smoke and transmitting a short alert package. Cloud inference supports computationally intensive earth-observation processing, model comparison and retraining. The hybrid design maintains useful service through local queues, later synchronisation and explicit degraded modes when either side is unavailable.

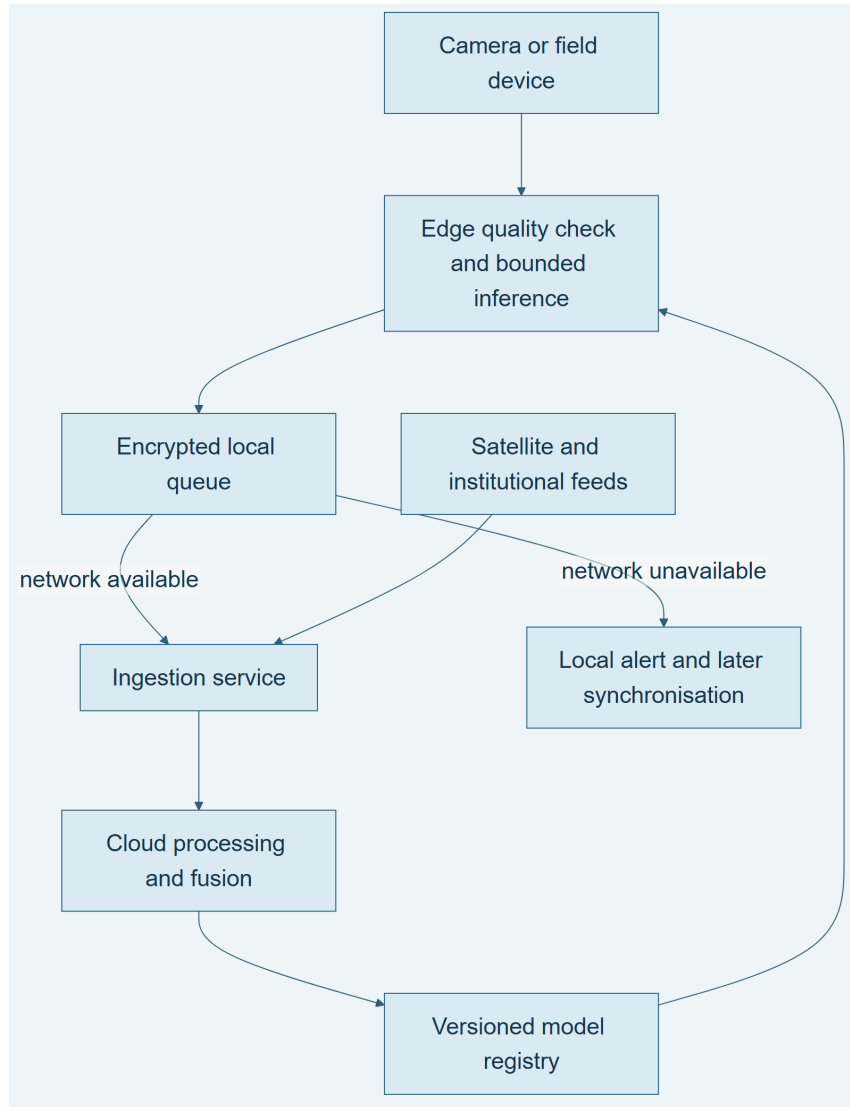


Figure 2.3: Hybrid inference with an offline path

3. From a model catalogue to a perception workbench

The production design uses **Ultralytics as a machine-perception workbench, not YOLO as a synonym for artificial intelligence**. The distinction matters because catastrophe imagery cre-

ates several different tasks. A detector locates bounded objects. A segmenter delineates a surface. A tracker associates evidence through time. A classifier describes the whole scene. An oriented box represents a rotated but bounded object. A pose model estimates keypoints. A depth model proposes scene geometry. An open-vocabulary model helps analysts discover a class that has not yet entered the production taxonomy. Ultralytics documents a common train, validate, predict, track, export and benchmark workflow across several of these tasks and model families [6]-[8].

SCRI places those implementations behind a stable interface. A current compact YOLO-family model may be the best edge detector for livestock near a threatened water point. An RT-DETR challenger may provide a better accuracy-latency trade-off for another object set; RT-DETR was designed as an end-to-end real-time detector with a transformer encoder-decoder architecture [9]. A SAM-family model may accelerate interactive masks. A Swin or SegFormer implementation may outperform both on a high-resolution semantic-segmentation problem. The institutional product should be able to promote the validated challenger without changing the loss engine, claims workflow or evidence schema.

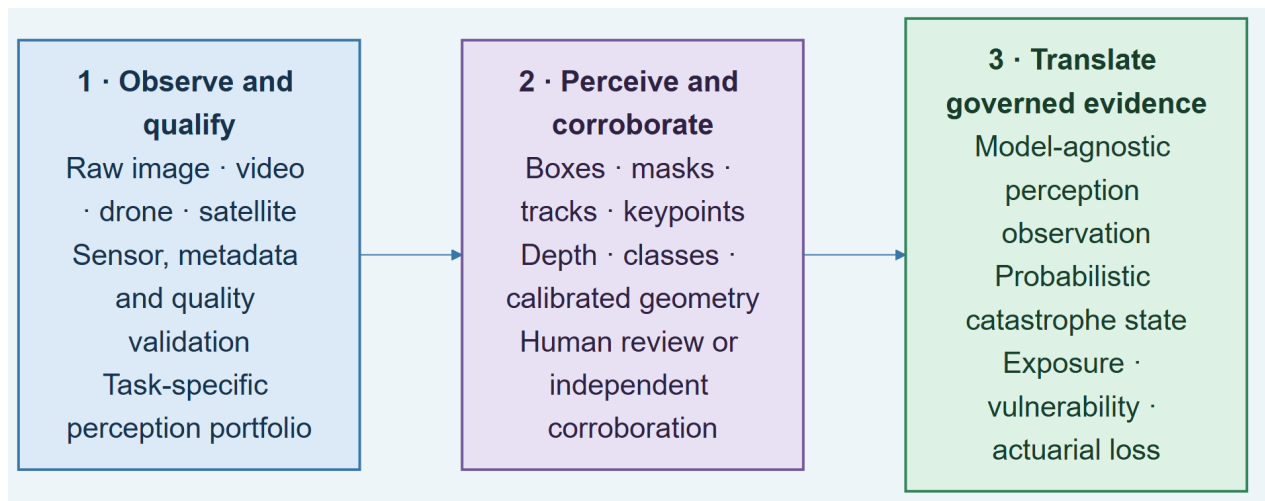


Figure 2.4: The perception workbench ends in governed evidence, not an autonomous financial decision

3.1. A model-agnostic perception-result contract

The contract prevents one vendor, architecture or model version from becoming the platform’s ontology. It extends the canonical observation record with fields that describe the visual inference itself:

```

perception_result_id
parent_observation_id
task_type
model_family and model_version
weights_version and training_dataset_version
input_asset_hash and frame_or_tile_reference
sensor_type and acquisition_geometry
class_taxonomy_version
prediction_geometry
coordinate_reference_system
score_type and calibrated_probability
  
```

uncertainty_components
 ground_sample_distance or image_scale
 spatial_support and boundary_tolerance
 correlation_group
 review_status and reviewer_provenance
 permitted_use
 runtime, hardware and export_format

prediction_geometry can contain a box, oriented box, polygon, raster probability surface, track, keypoint set or depth raster. score_type distinguishes a raw logit, confidence heuristic, calibrated class probability, mask probability or conformity score. correlation_group links frames from one video, overlapping drone tiles, one satellite acquisition or an ensemble created from the same source. The state engine can therefore recognise that fifty frame-level detections may describe one physical observation sequence.

The contract has a positive operational effect: a county analyst, catastrophe modeller and claims team can discuss the same evidence without discussing the neural-network internals. The analyst sees the image and review status; the state engine receives a likelihood-bearing observation; the actuary sees the resulting change in exposure and loss. Model detail remains available for audit and validation.

4. The perception portfolio by task

4.1. Detection: bounded and countable objects

Detection is appropriate where the object has a defensible class and bounded location: people, vehicles, livestock, locusts in survey imagery, flames, road obstructions, fallen poles and damaged components. The production metric follows the decision. A rescue workflow may prioritise recall and rapid human review. A claims-survey workflow may prioritise class precision and geometry. A locust counting workflow must report the smallest detectable object at the image's ground sample distance and how occlusion changes undercount.

Aggregate metrics remain useful for engineering comparison, but the model card reports class-level precision, recall, average precision, calibration and error cost. A model with attractive mean average precision can still be unsuitable if it repeatedly misses people in glare, dark livestock against burned ground, smoke against bright cloud or small vehicles in wide drone scenes. The validation set therefore includes hazard, county, sensor, season, lighting and acquisition-height slices.

4.2. Instance and semantic segmentation: from pixels to physical surfaces

Instance segmentation separates individual flooded buildings, damaged roofs, crop plots or discrete debris regions. Semantic segmentation is better for continuous water, burn, vegetation-stress, smoke or debris surfaces. The output becomes a physical quantity only through an explicit geospatial transformation:

1. Validate or estimate camera geometry and image quality.
2. Georeference, co-register or orthorectify the scene.
3. Preserve per-pixel probability and boundary uncertainty.
4. Convert pixels into area, length or intersection at the native support.
5. Reconcile sensor resolution with the exposure geometry.

6. Pass a probability surface or sampled boundaries to state inference.

If pixel j represents ground area a_j and has calibrated water probability p_j , an expected visible inundated area is:

$$\mathbb{E}[A^{vis}] = \sum_{j=1}^J a_j p_j. \quad (2.1)$$

This expectation does not claim that pixels are independent. Operational sampling should preserve spatially coherent mask uncertainty, sensor artefacts and common scene error. A hard polygon remains useful for display and routing, while the probabilistic surface remains available for state inference and loss sampling.

4.3. Multi-object and mask tracking: evidence through time

Tracking associates observations across frames. A vehicle moving through floodwater is a temporal object-association problem. A livestock group moving toward a water point may be counted without double counting. A road camera can reveal when water first crosses a carriageway. SAM 2 is a promptable image-and-video segmentation model with a streaming-memory architecture, making it a useful candidate for analyst-assisted mask propagation and video segmentation [10].

The physical interpretation remains peril-specific. A smoke plume deforms and advects; temporal segmentation, optical flow or an atmospheric dispersion representation is usually more faithful than a rigid-object tracker. Close-range tracking of locusts can improve counting in one survey clip, but regional swarm movement requires wind, ecological state and sequential field locations. Camera motion from a drone must be separated from object motion through stabilisation, pose estimation or georegistration.

SCRI converts a sequence into one correlated evidence object with time-indexed manifestations. This preserves onset and movement information while avoiding the fiction that 300 adjacent frames are 300 independent confirmations.

4.4. Oriented bounding boxes and non-rectangular infrastructure

Oriented bounding boxes are useful for bounded objects whose angle matters: roofs, vehicles, rail wagons, fallen poles, bridge elements or some aerial infrastructure. They can reduce background area relative to axis-aligned boxes and represent the orientation of damage. Roads, rivers, floodplains, fields and landslide scars often need polylines, polygons or segmentation because an oriented rectangle would discard their physical form. The task is chosen from the geometry of the evidence, not from the availability of a model endpoint.

4.5. Monocular depth as provisional geometry

Monocular depth contributes relative scene geometry and, under controlled conditions, provisional metric evidence. A visually plausible depth map is not automatically water depth. Flood depth normally requires a chain that may combine camera calibration, known reference objects, waterline detection, terrain elevation, gauge observations and hydraulic constraints.

SCRI therefore distinguishes:

- **relative depth**, which orders surfaces by apparent distance;
- **camera-referenced metric depth**, which uses calibration and scale evidence;
- **water depth above local ground**, which also needs water-surface and terrain geometry; and
- **hydraulically reconciled depth**, which conditions the visual estimate on gauge, terrain and flow information.

For camera c and location g , a fused depth observation can be represented as:

$$Y_{c,g}^{depth} = d_g + b_c + \epsilon_{c,g}^{scale} + \epsilon_{c,g}^{scene}, \quad (2.2)$$

where d_g is the physical depth of interest, b_c is camera-specific bias and the error terms represent scale and scene ambiguity. The likelihood passed to Part 3 carries these uncertainties. It does not turn a depth-model output into a precise hydraulic measurement.

4.6. Pose estimation for bounded rescue assistance

Pose estimation can support a narrow rescue-assistance workflow: a person may appear fallen, stranded or signalling. The system can prioritise a frame for a trained human, establish the camera and time, and route the evidence to the responsible response process. Necessity, image quality, occlusion, consent, retention and permitted purpose are explicit. The output remains a review cue, and operational teams confirm the situation through available channels.

This bounded design concentrates the capability where it creates clear human value. It also avoids repurposing sensitive human imagery for underwriting, pricing or unrelated surveillance.

4.7. Open-vocabulary detection and promptable segmentation

Open-vocabulary and promptable models are discovery and annotation accelerators. YOLOE and YOLO-World are examples of promptable or open-vocabulary capabilities exposed in the current ULtralytics ecosystem [11]. An analyst can search for an emerging damage concept, use a visual or text prompt to propose objects, or ask a SAM-family model to delineate a new surface.

The output initially enters a **candidate-review queue**. The queue records the prompt, model, image, proposed class and reviewer decision. Recurrent concepts then move through taxonomy definition, annotation guidance, Kenyan training examples, closed-set validation and production approval. This makes open vocabulary a disciplined route for discovering Kenya-specific catastrophe features rather than a shortcut around validation.

4.8. Vision Transformers remain in the architecture

Vision Transformers remain because context is often the deciding evidence in catastrophe imagery. The original ViT demonstrated image classification through sequences of image patches [12]. Swin introduced hierarchical shifted-window attention suitable for multiscale dense-prediction tasks [13]. SegFormer combines a hierarchical transformer encoder with a lightweight decoder for semantic segmentation [14]. These designs are useful challengers for contextual scene classification, high-resolution earth observation, pre/post-event change detection and semantic surfaces.

Their value is tested rather than assumed. Global context may help distinguish smoke from cloud, haze or dust; multiscale context may improve the delineation of water between buildings; change

models may detect roof loss or new landslide scars. Compact convolutional or hybrid models may still dominate on low-power devices. The product strategy is a portfolio:

Operational need	Baseline or candidate	Promotion evidence
Low-latency bounded objects at the edge	Compact YOLO-family detector	Class recall, calibration, watts, latency and degraded-mode performance
End-to-end real-time detection challenger	RT-DETR family	Event/geography holdout gain at an acceptable compute cost
Scene classification	CNN and ViT challengers	County, sensor and season transfer; smoke/cloud/dust discrimination
Semantic surfaces	U-Net-style baseline; SegFormer or Swin challenger	Boundary accuracy, calibration, spatial transfer and runtime
Promptable annotation	SAM-family model	Analyst time saved and corrected-mask quality
Video masks	SAM 2 or task-specific temporal segmenter	Drift, occlusion recovery and frame-sequence calibration
Open-vocabulary discovery	YOLOE, YOLO-World or comparable model	Reviewer precision and successful taxonomy conversion

5. Geospatial and remote-sensing imagery

Catastrophe imagery extends beyond ordinary RGB photographs. Sentinel-1 synthetic-aperture radar brings all-weather, day-and-night observation, while backscatter depends on acquisition geometry, surface roughness and the interaction between water and the built environment. Speckle, layover, shadow and double-bounce effects require remote-sensing treatment. A general RGB detector cannot simply be pointed at a radar raster and treated as validated.

Multispectral imagery carries physically meaningful bands beyond visible colour. Pre/post-event change requires co-registration, radiometric or sensor harmonisation and a statement of temporal difference. Drone imagery may need camera calibration, mosaicking and orthorectification. Ground sample distance sets the smallest defensible visible object. Cloud, shadow, smoke, glare, turbidity and vegetation generate different error modes by peril.

Specialised geospatial models therefore sit behind the same perception-result contract. A Sentinel-1 flood classifier, a multispectral vegetation-stress model and an Ultralytics detector can coexist because each publishes geometry, scale, provenance, uncertainty and model lineage in the same language.

6. Building the Kenyan catastrophe-vision dataset

Model architecture is only one part of performance. The durable product asset is a governed Kenyan dataset that reflects seasons, landscapes, building forms, roads, crops, livestock, cameras, languages and operational conditions.

The programme begins with a **hazard-task taxonomy**. Flood labels include visible water, passability, waterline, affected building, exposed vehicle and image usability. Fire labels distinguish smoke,

flame, burned surface, active perimeter evidence, firefighting assets and visibility. Locust labels specify lifecycle, survey scale and counting limitations. Landslide labels distinguish scar, debris, blocked road and precursor evidence where scientifically defensible. Drought and heat focus on visible crop, livestock and asset conditions while earth-observation and time-series models remain the primary state engines.

Annotation guidance defines positive, negative and ambiguous examples. Annotators can mark an uncertain boundary, occluded object, image-quality failure or class disagreement. A second reviewer checks high-consequence classes and a hazard specialist adjudicates difficult cases. Agreement statistics are reported by class and event; disagreement becomes annotation uncertainty rather than being silently removed.

Dataset splitting follows catastrophe reality:

- **event holdout** prevents frames from one flood or fire appearing in both training and test sets;
- **geography holdout** measures transfer to a county, basin or landscape not used for training;
- **sensor holdout** tests new cameras, drones or satellites;
- **season holdout** tests wet/dry, crop-stage and atmospheric changes; and
- **extreme-condition holdout** tests darkness, rain, smoke, glare, occlusion and low bandwidth.

Random image splitting would overstate performance because neighbouring frames, drone tiles and copied photographs share information. The dataset registry therefore records an origin cluster and event identifier before splitting.

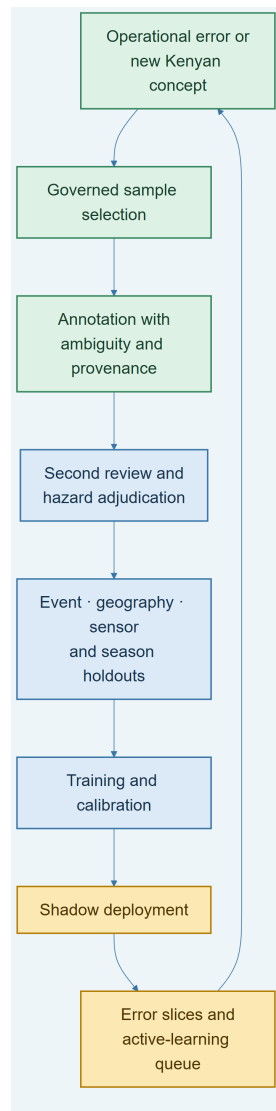


Figure 2.5: Kenyan dataset development is a continuous, event-aware learning cycle

Active learning prioritises examples that can improve an approved use case: low-confidence but high-exposure scenes, model disagreement, novel domains, repeated human correction and errors near decision thresholds. Sampling is balanced so connected urban areas do not consume the annotation budget while remote pastoral, agricultural or forest landscapes remain underrepresented. Analysts can also request deliberate negative examples, because a system trained mostly on disasters may generate false alarms during ordinary conditions.

7. Perception uncertainty as a product output

SCRI separates six uncertainties:

1. **Aleatoric uncertainty** from inherent ambiguity such as smoke, darkness, occlusion or a mixed water boundary.
2. **Epistemic uncertainty** from limited Kenyan examples or unfamiliar construction and land cover.

3. **Domain-shift uncertainty** from a new county, sensor, season, altitude or viewing geometry.
4. **Annotation uncertainty** from legitimate disagreement about a mask or damage class.
5. **Geometric uncertainty** from coordinate error, camera pose, orthorectification and raster resolution.
6. **Operational uncertainty** from compression, dropped frames, stale imagery and partial uploads.

Calibration is measured on the deployment-like holdouts, not only the training distribution. Reliability diagrams, Brier or log score, expected calibration error and class-specific threshold analysis accompany precision and overlap metrics. Ensembles can expose model disagreement. Conformal methods can be evaluated for prediction sets or mask uncertainty; recent work proposes conformal prediction sets for instance segmentation, while its coverage assumptions still need testing under event and geographic shift [15].

Uncertainty changes action constructively. A clear, calibrated road-obstruction detection may enter a rapid verification queue. A broad flood boundary produces an exposure range. A new sensor with strong domain-shift indicators routes more scenes to human review. A high-consequence rescue cue stays visible with its uncertainty and review status.

8. Deployment, export, licensing and maintenance

The hardware decision is empirical. Each candidate is benchmarked on representative phones, edge accelerators, drone computers, county servers and cloud instances using the intended image size, batch size and precision. The register records latency percentiles, throughput, memory, energy where measurable, export success, accuracy change after quantisation, start-up time and behaviour under thermal or network constraint. Ultralytics supports several export and benchmarking pathways, but each exported artefact remains a separately versioned model requiring equivalence checks [8].

The edge package contains the model, taxonomy, thresholds, quality checks, permitted use, expiry or review date and rollback version. A signed update is staged, tested and progressively deployed. If the model or camera degrades, the device can return to capture-and-forward mode or a transparent rule baseline.

The dependency register records source code, model weights, training data, licence, attribution, redistribution rights, export format, security updates and vendor-exit plan. This is particularly important where a research model, commercial workbench and proprietary Kenyan data are combined. The architectural contract allows SCRI to replace a dependency while retaining the observation and actuarial interfaces.

9. Six peril-specific perception products

The shared workbench becomes useful when each peril gives it a concrete observation job. SCRI therefore configures a peril adapter that specifies the scene, target variable, acceptable evidence, spatial support, temporal meaning, quality test and downstream decision. The adapter also states which physical model carries the main state-estimation responsibility.

9.1. Flood: from visible water to depth, access and accumulation

In the Nzoia basin, a satellite-derived water surface can contribute broad footprint evidence while a bridge camera records local onset and a field photograph anchors a waterline. In Nairobi, a short intense storm can overwhelm drainage between satellite acquisitions, so road cameras, community evidence and terrain-aware rainfall-runoff estimates may carry more immediate value. The flood adapter supports a probabilistic visible-water surface, building and road intersections, time-indexed passability, calibrated waterline evidence, provisional depth and post-event damage cues.

The perception output identifies what is visible and how certain it is. Terrain, drainage, gauges and hydrological or hydraulic models reconcile the surface into depth, duration and propagation. For an insurer, the first interface is exposure accumulation by credible depth band. For a county team, it is the road, facility or settlement whose access state changed. For claims, it is an inspection and contact queue linked to the original evidence.

9.2. Drought: visible condition inside a slow environmental state

Drought is primarily a time-series and livelihood-system problem. Rainfall, soil moisture, vegetation, water availability, markets and household or livestock indicators establish the evolving state. Machine perception adds focused evidence where visual assessment is meaningful: crop phenology and condition, surface-water availability, livestock body condition under a validated protocol, rangeland cover, borehole surroundings and exposed infrastructure.

A photograph of one animal cannot represent a county drought phase. The adapter records sampling design, species, age class, view quality, assessor protocol, location, season and representativeness. Computer vision can support consistent field measurement and triage, while the state engine aggregates it with NDMA and environmental evidence. The product can then show whether deterioration came from vegetation, water, livestock, prices or access and where anticipatory action may still protect livelihoods.

9.3. Wildfire and rangeland fire: detection, perimeter and operational access

Fire perception serves a sequence. A compact edge model screens a fixed camera for smoke or flame. A confirmed cue creates a short evidence package containing frames before and after detection, camera identity, bearing, weather and model version. Thermal or satellite evidence and independent reports support confirmation. Segmentation and time-series imagery then help estimate active perimeter, burned area and change; vehicles, water points, access routes and structures form the exposure context.

Smoke, dust, haze and cloud create a contextual classification problem suited to transformer challenges as well as compact detectors. Night, wind, camera shake and backlighting receive their own evaluation slices. A plume is interpreted through temporal masks and wind rather than as a rigid object. The response interface presents bearing, age, corroboration and routes. The actuarial interface receives footprint and intensity samples together with suppression evidence.

9.4. Locust and catastrophic crop pests: survey counting inside a mobile biological process

Close-range imagery can detect or count insects under a defined survey geometry. Drone or field imagery can also document crop condition and treatment evidence. The regional problem is a mobile

biological population with lifecycle, wind-assisted movement, vegetation response and control intervention. The locust adapter therefore preserves survey area, camera height, image scale, occlusion, lifecycle class, count uncertainty and sampling protocol.

Tracking prevents double counting within a short clip. Open-vocabulary review can help discover a visually similar pest or damage manifestation, after which the class enters expert taxonomy and validation. Swarm density and direction emerge from multiple surveys and ecological inference. Agricultural exposure adds crop type, stage, area, expected yield, prices, control cost and livelihood dependence. The field interface supplies a spatially prioritised survey queue while the actuarial layer models production and income loss.

9.5. Landslide: local change, blockage and slope context

Pre- and post-event aerial or satellite imagery can reveal a scar, debris field, blocked road or altered channel. Segmentation represents irregular scars and debris more faithfully than a rectangle. Oriented boxes remain useful for rotated vehicles, roofs, poles or bounded infrastructure components. Co-registration uncertainty is essential because apparent change can come from parallax, season, shadow or sensor geometry.

The perception product complements rainfall, saturation, terrain, geology and susceptibility. It can rapidly map access disruption and identify candidate affected structures after failure. Precursor monitoring uses appropriate deformation, ground-sensor or geotechnical evidence. The infrastructure interface follows the affected road or service network; the state model retains the intensely local spatial support.

9.6. Severe storm and extreme heat: damage evidence and visible vulnerability

After severe wind or hail, drone and street imagery can detect damaged roofs, fallen trees, poles, debris and blocked access. Oriented boxes and segmentation support aerial inspection, while pre-event imagery helps separate old condition from new damage. Scene classifiers can assess image usability and distinguish weather-related cues from ordinary deterioration.

Extreme heat is less visually observable. Weather, humidity, building physics, power demand, health and labour indicators carry the state. Vision can document shade, roof material, ventilation features, queues at water points, livestock condition or visible infrastructure stress where the purpose is defined. SCRI treats these as exposure or vulnerability observations. The heat product focuses on neighbourhood and asset conditions, service continuity and protective action.

Across all six adapters, the workbench produces comparable provenance and uncertainty while the underlying physics remain peril-specific.

10. Operational validation and model promotion

SCRI evaluates a model as part of a workflow. The starting point is a transparent baseline: human review, threshold rule, established remote-sensing classifier or current production model. A challenger earns promotion through event- and geography-held-out evidence, operational usefulness and deployability.

The evaluation catalogue contains four linked groups:

1. **Perception performance:** precision, recall, average precision, overlap, boundary error, count error, tracking association, depth error and calibration.
2. **Operating performance:** end-to-end latency, throughput, energy, memory, transmission volume, review workload, failure recovery and availability in degraded conditions.
3. **Catastrophe-state value:** change in state calibration, footprint accuracy, lead time, false and missed event states, and uncertainty reduction after dependence is accounted for.
4. **Decision value:** change in inspection yield, claims readiness, route selection, analyst time, response lead time and actuarial loss calibration.

Let k index error types, s evaluation slices and $c_{k,s}$ the owned consequence of an error. A decision-weighted perception risk is:

$$R_{\text{perception}}(f) = \sum_s w_s \sum_k c_{k,s} \Pr_f(\text{Error}_k \mid \text{Slice}_s), \quad (2.3)$$

where w_s represents the agreed evaluation population. A rescue-screening false negative can receive a different consequence from an extra review cue. A flood-boundary error near dense exposure can receive a different priority from the same pixel error in open water. The weights are published with the result so ranking remains understandable.

Calibration is assessed before and after score mapping. Thresholds are selected on validation data and frozen for the holdout. Event-block bootstrap intervals reflect within-event correlation. A model can be promoted for one class, county, sensor or device and remain a challenger elsewhere. This scoped promotion builds a portfolio that improves by evidence rather than by a single global launch.

The quality gate requires stated uses, a frozen origin-aware dataset split, identical baseline and challenger holdouts, uncertainty and subgroup results, exported-artefact equivalence, measured review capacity, replayed state and actuarial impact, and a ready rollback path.

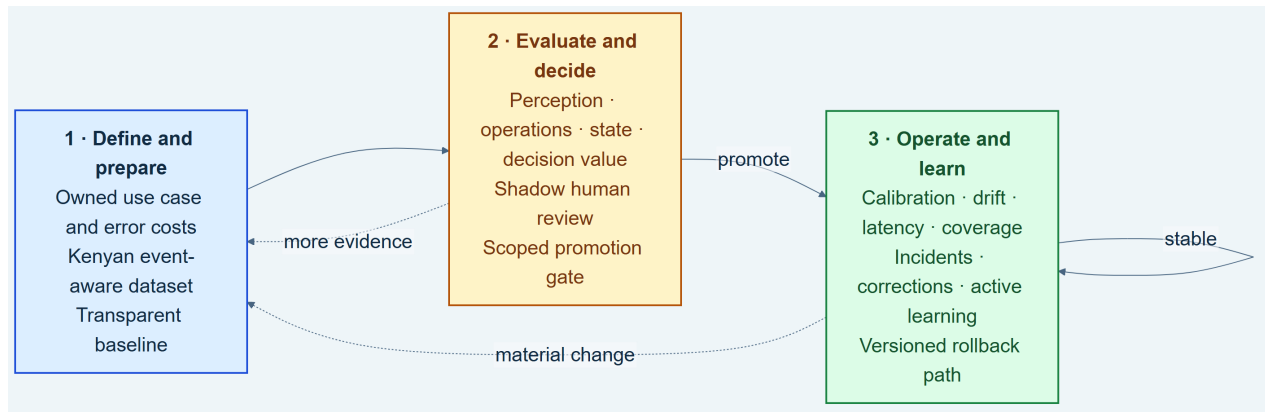


Figure 2.6: Machine-perception promotion links offline metrics to shadow operation and monitored product value

10.1. Shadow operation, drift and learning

In shadow mode, the model produces observations beside the existing workflow. Analysts record whether they accepted, corrected or ignored each result and why. SCRI measures review time, queue

ageing, error discovery, operational action and downstream state movement. Stratified sampling concentrates review on high-consequence, low-confidence, novel-domain and apparent-disagreement cases.

Drift monitoring distinguishes input change from performance change. Sensor, compression, brightness, ground sample distance, geography, season and class prevalence can all move. Feature-distribution diagnostics identify change for investigation. Delayed verified labels, analyst corrections and event reconstructions measure actual performance. A drift alert increases sampling or narrows scope; demonstrated performance deterioration triggers a threshold, model or workflow response.

The registry keeps candidate, shadow, production, degraded and retired states. Every result retains its model and weights version, so replay can show how a later model would have behaved without changing the historical decision record. Rollback restores the last validated artefact and thresholds. Capture-and-forward, human review and authoritative-source workflows continue during correction.

11. Deployment tiers and service design

National deployment uses several compute tiers because each has a different combination of latency, energy, privacy and imagery scale.

Tier	Typical work	Product priority	Promotion evidence
Phone or assisted field device	Image-quality guidance and bounded inference	Offline usability, battery, privacy and correction	Device matrix, thermal test, upload recovery and user test
Fixed camera or edge accelerator	Smoke/flame screening, road-water onset and counts	Low latency, continuous operation and compact evidence	Long-duration false-alert rate, recall, power and network-failure test
Drone computer	Flight-time quality check and selected inference	Throughput, motion handling and geospatial lineage	Altitude/GSD slices, stabilisation, battery and mosaic reconciliation
County or institutional server	Local batch inference, review and degraded service	Data control, resilience and team workflow	Capacity, failover, access controls and synchronisation replay
Cloud geospatial platform	Satellite processing, large segmentation, ensembles and replay	Scale, lineage and multi-source fusion	Cost, latency, reproducibility, outage drill and vendor-exit test

Every tier publishes a service envelope. A smoke-screening edge model states a tested frame rate and evidence-package latency under named hardware and visibility. A satellite flood product states acquisition time, processing time, native resolution and update interval. A field application states how long it queues evidence offline and what happens when storage fills. These envelopes let the state engine model freshness and operations plan around the actual service.

A deployment utility can combine performance and resources:

$$\begin{aligned}
 U(f, d) = & \textit{Benefit}_{\textit{decision}}(f, d) - \lambda_L \textit{Latency}(f, d) - \lambda_E \textit{Energy}(f, d) \\
 & - \lambda_C \textit{Cost}(f, d) - \lambda_R \textit{ReviewLoad}(f, d).
 \end{aligned} \tag{2.4}$$

The weights express the owned use case. Life-safety screening may value latency and recall strongly; retrospective claims mapping can favour boundary quality and throughput. The utility supports an explicit decision record.

The perception service publishes an immediate observation and any later correction. Each contains acquisition and processing times, geometry, score type, uncertainty, correlation group and review state. The event engine subscribes to admissible evidence; response users receive purpose-qualified cues; portfolio users receive hazard-state changes by default; claims users receive inspection rationale; model operations receives drift and device telemetry; authorised audit users can traverse lineage.

Latency is measured from acquisition, not merely the start of inference. Availability includes sensor, network, queue, inference, geospatial transform and delivery. Coverage reports where no eligible sensor exists. This aligns the service level with the observation process in Part 3.

12. Human imagery, participation and narrow purpose

The ingestion workflow identifies whether an image may contain faces, plates, homes, documents, children, injured people or precise private locations. It applies purpose-based access, redaction where appropriate, retention and review before wider use.

Community contribution states what is requested, why it helps, how it will be checked, who may use it, how long it is retained and how a contributor can correct or withdraw eligible data. Assisted and off-line channels support people without a modern smartphone. A contributor can report an effect without uploading identifiable imagery. Trusted field networks receive task-specific training and safety guidance.

Dataset creation records the lawful and contractual basis for training, evaluation and publication examples separately. Customer data, community evidence, open imagery and research datasets can have different permitted uses. Derived embeddings and annotations retain lineage because derivation does not erase underlying rights or sensitivity. Published examples pass through an editorial and redaction process.

The narrow-purpose design improves model quality. A rescue cue, road-passability observation and insurance damage indicator use different taxonomies, thresholds and retention. Each community and institution can therefore understand the precise service it supports.

13. A perception-to-actuarial worked example

Consider a road camera overlooking a market corridor during a flood. At 07:05, the quality check confirms the camera identity, clock and visibility. A semantic segmenter produces a water probability surface; a detector identifies three vehicles; a waterline routine finds a calibrated wall reference. Geometry processing projects the surface into the local coordinate system and returns an estimated visible inundated area with boundary samples. The depth workflow combines the waterline, terrain and an upstream gauge to produce a depth likelihood rather than a point estimate.

The observation contract sends four correlated products to the state engine: visible-water surface, provisional depth, vehicle observations and passability evidence. They share one camera sequence and therefore one correlation group. A second road report and later radar acquisition provide independent evidence. Part 3 updates the flood state across the drainage network.

Part 4 intersects posterior flood samples with shops, dwellings, vehicles and road-dependent businesses. For posterior sample m :

$$L^{(m)} = \sum_i V_i D_i(I^{(m)}(g_i), M_i, \varepsilon_i^{(m)}), \quad I^{(m)} \sim p(I \mid Y_{1:t}). \quad (2.5)$$

The perception model has not calculated a claim. It has improved the evidence distribution that generates depth and footprint samples. The claims workspace can show why expected loss changed: new water evidence, stronger depth information and corrected passability. The actuary retains the exposure, vulnerability and policy transformation.

Product interfaces express the same chain differently:

- **Community and field interface:** what was observed, the practical effect, recommended authoritative action and a correction route.
- **County or response interface:** verified obstruction, accessible alternatives, observation age and review priority.
- **Insurer portfolio interface:** affected exposure by probability and intensity band, with image evidence accessible under permitted use.
- **Claims interface:** candidate affected policies and inspection priority, without using perception as automatic coverage proof.
- **Actuarial interface:** posterior hazard samples, uncertainty attribution and the observation/model versions responsible for movement.
- **Model-operations interface:** drift, calibration, runtime, data coverage and rollback status.

This completes the perception product. Pixels become calibrated, georeferenced observations; observations update a catastrophe state; catastrophe states generate loss distributions; and institution-owned workflows turn those distributions into action.

14. Two product-interface journeys

14.1. A Mount Kenya landscape fire seen by several users

At 13:42, a camera overlooking a forest-agriculture boundary produces a smoke cue. The edge device has already checked clock synchronisation, lens obstruction and the proportion of sky in the frame. It packages six seconds of imagery before and after the cue, the camera bearing, the calibrated smoke probability, a broad angular uncertainty and the device/model versions. Rather than transmitting a continuous identifiable video stream, it sends the bounded evidence package required by the approved fire-screening purpose.

The event interface creates a candidate fire observation. A contextual classifier shows that bright cloud is a plausible alternative, so the cue appears as amber rather than red. Wind direction and a later thermal observation support the same general location. A ranger report supplies an independently described smoke bearing. The state engine intersects those uncertain bearings and updates ignition probability across a set of cells. The screen shows which evidence generated the movement and how much uncertainty remains.

For the response analyst, the interface leads with practical questions: where is the likely origin, which roads and water points remain accessible, how old is the evidence, and which observation can confirm

the state most efficiently? It proposes two camera views and a field verification point. The analyst chooses the next action and records the result. If the report confirms a fire, the perimeter product begins a time-indexed series using thermal, optical and field observations. Wind, fuel, slope and suppression feed the propagation model.

For an insurer, the same evidence appears later through a portfolio lens. The user sees insured farms, buildings, tourism assets and equipment intersecting posterior intensity bands. Raw frames containing people remain in the restricted evidence store; the portfolio view receives the governed state and permitted thumbnails. The loss distribution changes through the fire model, exposure and vulnerability---not through a direct multiplication of smoke confidence by insured value.

For a claims lead, post-event drone imagery creates building and crop inspection candidates. Pre-event scenes help distinguish existing roof condition from new loss. Each mask or oriented box carries geometry, model version, probability and review status. The interface groups multiple drone tiles by flight so they do not appear as independent confirmation. Adjusters can accept, correct or deprioritise a candidate, and their findings return to the evidence and dataset programmes under the agreed use.

For model operations, the event becomes a structured learning record. The team reviews whether the smoke cue arrived earlier than other evidence, whether the contextual classifier reduced false escalation, whether the edge package met latency and energy targets, how the perimeter masks drifted, and whether the downstream state was calibrated. The finding may promote a challenger for bright-cloud conditions, add a new negative example or refine the camera's service envelope. One real event improves the product without erasing what users knew at each earlier decision time.

14.2. A drought month in a pastoral livelihood system

The drought product moves at a different rhythm. At the start of the month, the county view already contains rainfall, vegetation condition, water-point reports, livestock movement, market access and previous phase history. Field officers use an offline-first application to collect a stratified sample of water and livestock observations. The application guides image angle, distance, reference scale and minimum metadata. It also accepts a structured non-image report where photography is unsafe, intrusive or unhelpful.

At a water point, an image-quality model confirms that the scene is usable. Segmentation proposes the visible water surface and identifies shadows as uncertain. The officer reviews the mask and records whether the point is functioning, estimated queue, access condition and the source of the water. A nearby livestock assessment uses a species- and protocol-specific workflow. The interface presents the model suggestion as a measurement aid and records the trained assessor's result. Neither observation is asked to stand for the entire livelihood zone.

The county analyst receives a coverage map before a drought-phase map. It shows which strata were sampled, which roads prevented access, where connectivity delayed upload and which communities have no current field observation. The analyst can direct the next visits toward information gaps rather than repeatedly collecting evidence near easy roads. Satellite and authoritative indicators continue to update while field packages wait offline; later synchronisation preserves their event time and knowledge time.

The probabilistic engine asks whether the field evidence is consistent with the evolving vegetation, water and livelihood state. A low water surface at one point becomes more influential when independent borehole, rainfall and usage evidence agree and when the sampling design supports representation.

Several forwarded photographs remain one evidence origin. Silence in a disconnected area increases uncertainty through the reporting model rather than creating an apparent absence of stress.

For an insurer or livelihood-protection partner, the interface shows the contractual index, observed livelihood effects and basis-risk evidence separately. For a lender, it shows possible revenue and repayment pressure through livestock condition, market access and customer exposure, while the lender's credit model owns PD and LGD. For a county or anticipatory-action programme, it shows the populations, water systems and livelihood assets associated with each credible state and the lead time for available interventions.

At month end, the product closes a versioned assessment rather than declaring the event finished. Drought duration continues, evidence can be corrected and the next period starts from the posterior state. The model-operations team reviews sampling balance, assessor agreement, device performance, uncertainty coverage and whether recommended information-gathering improved decisions. This is continuous intelligence adapted to a slow catastrophe: the product preserves persistence, livelihoods and uneven observation rather than forcing drought into a rapid-alert template.

These two journeys show why the perception architecture is both shared and plural. The evidence contract, calibration, lineage, review and deployment disciplines are common. Fire and drought create different clocks, physical states, field practices, product users and actuarial consequences.

15. Inclusive crowd intelligence

The defining statistical problem is dependence among crowd reports. Fifty forwarded copies of one image share a common information origin. SCRI first creates a **duplication and propagation cluster** using perceptual image hashes, text similarity, shared URLs, timestamps, geometry, device metadata where lawfully available and known forwarding relationships. The cluster becomes one correlated evidence object with an internal report count. Independent corroboration is then assessed across distinct evidence origins.

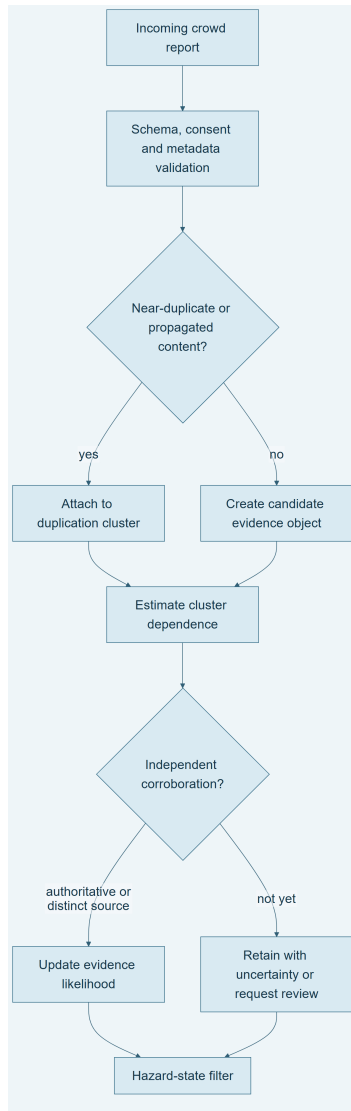


Figure 2.7: Separating correlated repetition from independent corroboration

Contributor reliability is contextual. The platform evaluates location precision, timestamp quality, report completeness, sensor calibration, agreement with independent sources and known operating conditions. Historical performance can contribute as a capped, regularly re-estimated input, while first-time reporters, shared-device users and remote communities remain eligible sources. Community coverage becomes a monitored variable, and sparse-reporting areas receive wider uncertainty and targeted outreach.

A practical evidence update can be written as:

$$p(Z_t \mid Y_{1:t}) \propto p(Z_t \mid Y_{1:t-1}, X_t) \prod_{c=1}^{C_t} p(Y_{c,t} \mid Z_t, q_{c,t}, \rho_c). \quad (2.6)$$

where Z_t is the hazard state, $Y_{c,t}$ is an evidence cluster rather than an individual forwarded message, $q_{c,t}$ represents observable quality and ρ_c represents residual dependence inside the cluster. The predictive prior marginalises uncertainty in the preceding state rather than conditioning on an unknown

Z_{t-1} . The interface supports a Kalman-family filter, particle filter, variational approximation, ensemble update or deterministic rule-based baseline according to hazard and latency. Offline MCMC remains useful for calibration and model comparison where computation permits. Appendix D derives Equation (2.6) from Bayes' rule, shows how duplication dependence changes effective information, and develops source-reliability and geolocation-error updates.

Privacy uses several reinforcing controls. Geotagged images, phone numbers, faces, vehicle plates and property details may be personal data. Kenya's Data Protection Act requires lawful, fair and transparent processing; specified purposes; data minimisation; accuracy; retention discipline; safeguards for transfer and enforceable data-subject rights [5]. SCRI separates the identifiable contribution store from the hazard-evidence store, applies encryption and role-based access, redacts visible identifiers where practical and records permitted uses. Pseudonymisation reduces direct identifiability, while access and retention controls address the residual identifiability of rich geospatial records.

Safe degradation is designed before scale. If the crowd channel fails, the state engine widens uncertainty and uses authoritative and earth-observation sources. A stale gauge receives a freshness flag and decaying evidential weight. Delayed satellite acquisition appears as footprint age. During cloud outages, edge devices retain evidence and authorised local warnings follow the responsible agency's manual route. Missing evidence produces an explicit coverage gap and wider confidence interval.

The output of Part 2 is a versioned collection of admissible evidence with uncertainty, correction history and provenance intact. Part 3 can therefore estimate hazard state while Part 4's actuary can trace every financial update back to its observational basis.

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Part 3

Spatiotemporal Hazard-State Modelling

Spatial Catastrophe Risk Intelligence

1. From evidence to an evolving physical state

At 08:00, an insurer may know that rain has fallen upstream, one gauge is rising and three community reports describe water on a road. At 11:00, a radar acquisition reveals a wider inundated area. At 15:00, a photograph corrects the assumed location of the road closure. The event has changed, but so has knowledge of the event. A catastrophe-state engine must represent both.

Spatial Catastrophe Risk Intelligence (SCRI) treats the latest polygon as one observation within a posterior distribution over the physical variables relevant to a peril and forecast horizon:

$$p(Z_{g,t} \mid Y_{1:t}). \quad (3.1)$$

where $Z_{g,t}$ is the latent or partially observed hazard state for spatial unit g at time t , and $Y_{1:t}$ is the evidence that was available by that valuation time. A map presents one view of this distribution, such as median extent, probability of inundation or a credible interval around a perimeter. The underlying model retains the complete probability representation.

The state evolves through a hazard-specific process:

$$p(Z_{g,t} \mid Z_{g,t-1}, Z_{\mathcal{N}_h(g),t-1}, X_{g,t}, \theta_h). \quad (3.2)$$

where $\mathcal{N}_h(g)$ defines the relevant neighbourhood for hazard h , $X_{g,t}$ contains forcing variables and θ_h contains model parameters. For flood, the neighbourhood may be directed upstream and downstream through a river or drainage network. For wildfire, it follows adjacency, wind, slope and fuel. For locusts, it is a movement corridor. For drought, temporal persistence and livelihood zones may dominate. For landslide, the state may be local to a slope unit. For heat, it may follow a gridded temperature and humidity field combined with the built environment. Appendix E derives the prediction-update recursion behind Equations (3.1)-(3.2) and states the conditional-independence assumptions needed for spatial factorisation.

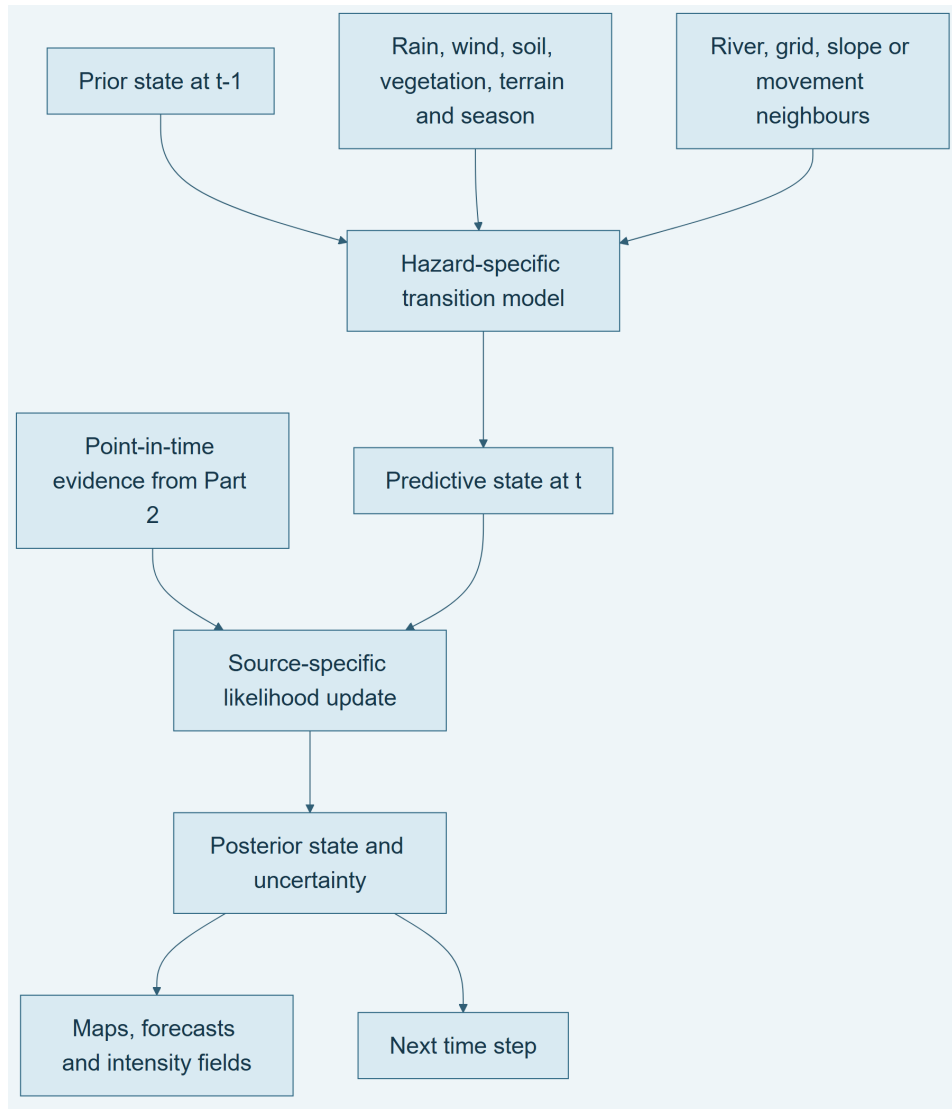


Figure 3.1: Prediction and evidence update are separate operations

Every advanced model has a transparent baseline. A flood baseline may combine gauge thresholds, terrain and a precomputed inundation library. A drought baseline may reproduce the existing indicator-and-phase logic used in public bulletins. A fire baseline may use confirmed ignition, wind buffers and simple rate-of-spread rules. A locust baseline may advect the last confirmed location with forecast wind and wide uncertainty. A landslide baseline may combine susceptibility with rainfall thresholds. A heat baseline may use station or gridded temperature thresholds and duration. These baselines establish an understandable, fast and operationally resilient benchmark.

The challenger may be a hierarchical Bayesian state-space model, finite hidden Markov model, hidden semi-Markov model, switching process model or physics-informed ensemble. An infinite hidden Markov model becomes a research candidate when discovering an unknown number of regimes is useful. Model choice follows calibration, lead time, transfer, decision value, operating latency and compute budget.

2. Six hazard adapters

2.1. Flood: routed dependence from extent to depth

A Sentinel-1 flood mask can reveal extent under cloud. SCRI combines that extent with rainfall, river levels, terrain, drainage and hydraulic or hydrological relationships to estimate water depth at exposed properties. WRA's basin structure and monitoring network provide the institutional and physical starting point [1]. The state vector may include:

$$Z_{g,t}^{flood} = \{p_{g,t}^{wet}, d_{g,t}, v_{g,t}, \tau_{g,t}^{on}, \tau_{g,t}^{rec}\}. \quad (3.3)$$

representing inundation probability, depth, velocity, onset and recession. Each component has its own uncertainty. The product reports spatial and depth precision at the support justified by the radar, terrain, gauges and model validation [2]. Appendix E expands Equation (3.3) into a joint state with marginal and cross-component uncertainty.

2.2. Drought: duration and livelihood condition

Drought requires memory. A finite HMM can represent phases; a hidden semi-Markov or continuous state-space model can represent explicit duration where livelihood deterioration persists, while a transparent composite indicator remains the baseline. NDMA's four indicator families - biophysical, production, access and utilisation - connect rainfall and vegetation to water access, livestock condition, markets and household utilisation [3]. The insurer's model consumes the authoritative drought phase as one observation or contextual variable, while the responsible public institution continues to issue that phase.

2.3. Wildfire: detection, spread and response

The fire state separates ignition from detection:

$$T^{ign} \leq T^{det} \leq T^{dispatch} \leq T^{response}. \quad (3.4)$$

The spread model conditions on wind, fuel, moisture, terrain and suppression. The observed perimeter anchors a forecast ensemble. Community knowledge remains operationally relevant around Mount Kenya, where access, land use, fire practice and preparedness affect both observation and response [4]. SCRI reports the probability that an exposure lies within future fire footprints at selected horizons, conditional on explicit response assumptions. Appendix E derives the associated latency decomposition and shows why occurrence probability and detection probability remain separate.

2.4. Locust: a mobile biological process

Locust state includes location, density, lifecycle and movement. Wind and vegetation affect the transition, while surveillance and control actions change what happens next. The 2019-2020 Kenya response relied on surveillance, identification of breeding grounds, ground and aerial control and livelihood restoration [5]. SCRI records control operations as state-changing interventions and widens uncertainty where surveillance coverage is sparse.

2.5. Landslide: slope units and threshold exceedance

Landslide susceptibility answers *where failure is more plausible*; rainfall and soil conditions help answer *when*. State estimation operates over slope units or other geotechnically defensible geometries. Reports of fissures or blocked drainage contribute local evidence with measured false-alarm potential. The output is a conditional failure probability and access-impact scenario with uncertainty suited to inspection and response planning. Kenyan synthesis evidence connects rainfall, steep terrain, deformation and drainage with regional patterns including Murang'a and West Pokot [6].

2.6. Severe storm and heat: fields, persistence and compound effects

Storm state includes rainfall intensity, wind, hail where observed, lightning and duration. Heat state includes temperature, humidity or an approved heat-stress measure, persistence, time of day and spatial differences created by altitude, coast and urban form. A storm can trigger flood and landslide; drought and heat can increase fire-conductive conditions; heat and power failure can compound health and business interruption. The state engine records those dependencies while preserving the identity and physics of each hazard.

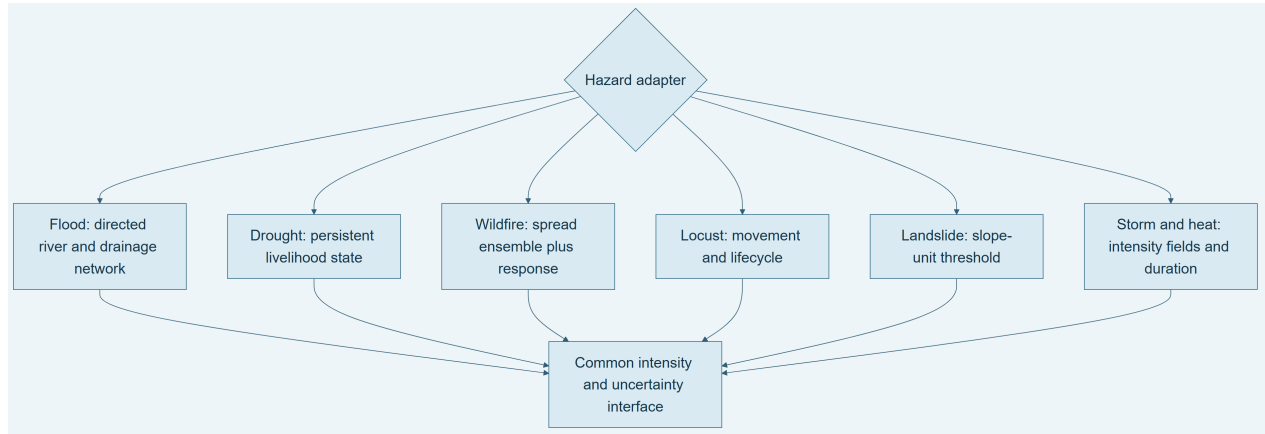


Figure 3.2: Common interface, different physics

3. Constructing the operational posterior

The state engine becomes operational when each source enters through a likelihood that describes what the source could have produced under alternative physical states. A confidence score alone cannot do that work. The system needs the probability of the measurement, mask, detection or report conditional on the catastrophe state, its spatial support and the conditions under which it was created.

The complete filtering step is:

$$p(Z_t | Y_{1:t}, X_{1:t}) \propto p(Y_t | Z_t, \Psi_t) \int p(Z_t | Z_{t-1}, X_t, A_t, \theta) p(Z_{t-1} | Y_{1:t-1}, X_{1:t-1}) dZ_{t-1}, \quad (3.5)$$

where Ψ_t contains source, reporting, spatial-support and dependence information, and A_t contains interventions that can change the physical process. The integral is the predictive prior. It carries

forward the full uncertainty from the preceding state rather than substituting one estimated state. Linear-Gaussian cases admit Kalman-family solutions; nonlinear or non-Gaussian cases may use ensembles, particles, variational approximations or other validated filters. Standard Bayesian filtering and smoothing provide the algorithmic foundation [7].

3.1. Source-specific likelihoods

A source adapter publishes the statistical relationship between the observation and the state. The first operational set is:

Source	Candidate observation model	Material uncertainty
River gauge	Gaussian or robust measurement error around stage; rating-curve model for discharge	Sensor calibration, datum, rating-curve parameter, freshness and failure state
Rainfall station	Error model conditional on intensity and accumulation interval	Wind undercatch, missing interval and spatial representativeness
Segmentation surface	Pixel or region probability model with spatially correlated boundary error	Calibration, sensor artefact, co-registration, class confusion and resolution
Object detector	Class-specific sensitivity and false-positive model conditional on quality	Threshold, object scale, occlusion, domain shift and duplicated frames
Crowd report	Truthfulness and geolocation model conditional on reporting access and wording	Reporting selection, contributor context, copied origin, memory and location error
Satellite product	Coverage-conditioned retrieval likelihood	Acquisition time, cloud or radar geometry, processing version and unavailable coverage
Institutional record	Revision and delay model around the recorded variable	Administrative cut-off, definition change and reporting latency

For a gauge reading $y_{s,t}$ measuring latent river stage h_t , a robust baseline is:

$$y_{s,t} \mid h_t, b_s, \sigma_s, \nu \sim t_\nu(h_t + b_s, \sigma_s), \quad (3.6)$$

where b_s is station bias, σ_s is observation scale and the Student- t likelihood reduces the influence of an isolated extreme residual. A separate health state models a stuck or stale gauge. A discharge estimate derived from stage carries rating-curve parameter uncertainty rather than being treated as a direct measurement.

For a calibrated detector with binary class output $D_{s,t}$ and physical class-presence indicator O_t :

$$P(D_{s,t} = 1 \mid O_t, Q_t) = \begin{cases} Se_s(Q_t), & O_t = 1, \\ 1 - Sp_s(Q_t), & O_t = 0, \end{cases} \quad (3.7)$$

where sensitivity Se_s and specificity Sp_s vary with the quality context Q_t , such as object scale, rain,

smoke, darkness or sensor. A segmentation likelihood operates over a spatial field and must preserve local correlation; treating every pixel as independent would create implausibly concentrated posteriors.

3.2. Reporting is part of the model

Silence in a crowd channel has several explanations. The hazard may be absent, connectivity may have failed, a community may lack access, people may prioritise safety, or the event may have damaged communications. SCRI models whether a report could have been observed:

$$R_{s,g,t} \sim \text{Bernoulli}(\pi_{s,g,t}), \quad \text{logit}(\pi_{s,g,t}) = \alpha_s + \beta_Z Z_{g,t} + \beta_A A_{g,t}^{\text{access}} + \beta_C C_{g,t} + \beta_T T_{g,t}. \quad (3.8)$$

Here $R = 1$ means a report is observed, A^{access} describes access to the reporting channel, C describes connectivity and T describes timing and contributor availability. The content likelihood is evaluated conditional on a report. The model can therefore distinguish a verified negative report from no report and can widen uncertainty where observation access is weak.

This structure also exposes data inequality. A remote area with limited reporting cannot appear safer merely because the reporting probability is lower. Coverage indicators and posterior uncertainty become visible product outputs, while field partnerships and offline channels can improve the observation process itself.

3.3. Contextual reliability through partial pooling

A contributor, sensor or model does not have one timeless reliability score. Reliability varies by hazard, location, variable, device, visibility, report age and verification method. The Beta-Binomial update in Appendix D remains the transparent baseline. The production challenger uses a hierarchical model:

$$\text{logit}(r_{s,k}) = \mu + u_s + v_{\text{hazard}(k)} + w_{\text{region}(k)} + \gamma^\top Q_k, \quad (3.9)$$

where $r_{s,k}$ is reliability for source s in context k , u_s , v and w are partially pooled effects and Q_k contains observable quality. A new source begins near the context-level distribution rather than at zero credibility. A source with strong flood-road reporting history does not automatically inherit the same reliability for wildfire smoke or crop-stage assessment.

Reliability is attached to a source-task context and capped as one component of the likelihood. It does not become a social score. Contributors can challenge corrections and remain eligible even when they share a device or operate in a previously sparse area.

3.4. Common origins and correlated manifestations

Copying and platform propagation create many reports from one origin. Video frames, overlapping satellite tiles and multiple predictions from one model run create similar dependence. SCRI introduces a latent evidence origin O_j :

$$p(Y_{1:n_j} | Z_t) = \int p(O_j | Z_t) \prod_{i=1}^{n_j} p(Y_{j,i} | O_j, \kappa_{j,i}) dO_j, \quad (3.10)$$

where κ describes copying, frame or processing relationships. This factorisation allows several manifestations to sharpen understanding of the origin without pretending they are independent observations of the physical state.

The duplication cluster from Part 2 supplies candidate origin links. The probabilistic layer can retain uncertainty when provenance is incomplete. One independent gauge, one satellite acquisition and one genuinely separate community photograph can provide stronger corroboration than hundreds of forwarded copies.

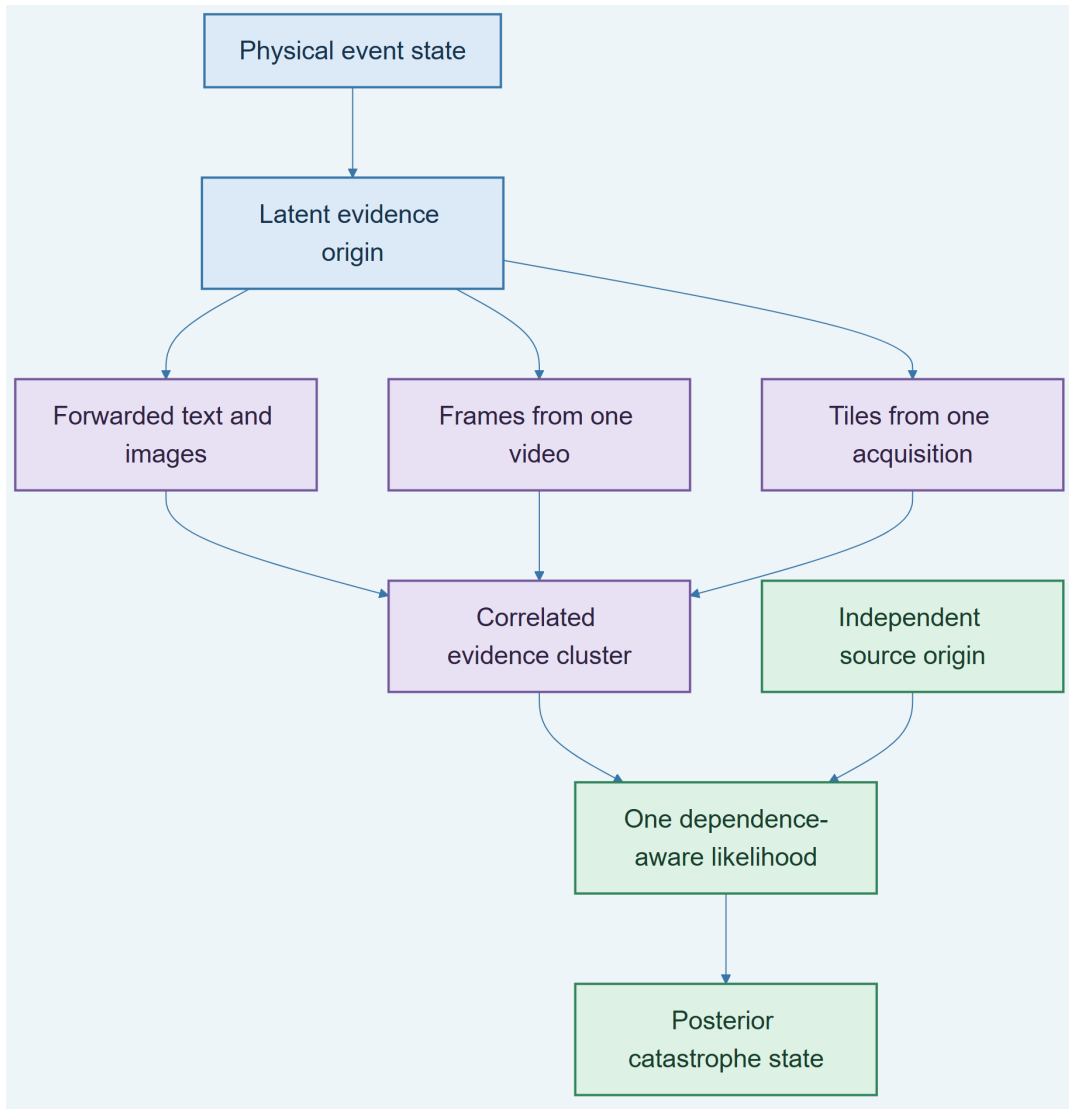


Figure 3.3: Visible report count is separated from independent evidence origins

3.5. Asynchronous evidence, correction and replay

Operational filtering processes evidence as it arrives. Catastrophe reconstruction must also handle a photograph uploaded hours late, a gauge correction, a revised satellite mask or a policy coordinate fixed after a claim. SCRI maintains three inference products:

1. **Online filter:** the best state using evidence available at the operational cut-off.
2. **Fixed-lag smoother:** a revised recent trajectory after delayed evidence arrives.
3. **Retrospective smoother:** a complete event reconstruction for validation and learning.

An out-of-sequence observation with event time $\tau < t$ updates the stored state at τ and re-propagates affected states to the current time. The system retains the previous version, correction reason and knowledge time. Operational decisions remain auditable against what was known then, while the retrospective record can become more accurate.

The replay engine is part of validation. A model is back-tested with the same ingestion lags, missing feeds and corrections that occurred historically. This prevents a retrospective reconstruction from being presented as if it had been available during the emergency.

3.6. Change of spatial support

SCRI combines point gauges, road lines, camera fields of view, raster pixels, polygons, basin totals and livelihood-zone indicators. Each observation is linked to the latent field through a support operator H_s :

$$Y_{s,t} = H_s[Z_t] + \epsilon_{s,t}, \quad H_s[Z_t] = \int_{B_s} w_s(g) Z(g, t) dg, \quad (3.11)$$

where B_s is the observation footprint and w_s is a normalised weighting or physical response. A point gauge has highly local support; a satellite pixel integrates a ground footprint; a drought bulletin aggregates a larger administrative or livelihood area. Spatial hierarchical modelling provides the broader framework for linking latent processes and misaligned observations [8].

The operator prevents false precision. A county-level drought phase cannot be copied into each building as if it were a property measurement. A small camera view cannot define a basin-wide flood surface. Downscaling, interpolation or hydrological routing becomes an explicit model with uncertainty.

3.7. Interventions change transitions

Information can change loss only when it changes action and the action changes the physical or social process. Drainage operation can alter urban water depth; suppression can alter fire spread; pest control can alter swarm survival; water allocation can alter drought impact; road closure can alter human exposure. The transition therefore includes A_t in Equation (3.5).

For two intervention scenarios a and a' , SCRI can simulate:

$$Z_{t+1}^{(m)}(a) \sim p(Z_{t+1} \mid Z_t^{(m)}, X_{t+1}, A_{t+1} = a), \quad (3.12)$$

then pass both state paths to the loss engine. The intervention effect remains conditional on implementation quality and causal evidence. Observed action time, location, capacity and completion are required inputs, not narrative annotations.

3.8. Nonstationarity and structural change

Seasonality belongs in the transition process, while long-run change can enter parameters, covariates or alternative scenarios. Relevant changes include climate forcing, land cover, urban drainage, irrigation, vegetation, building and infrastructure condition, observation coverage and response capacity. A dynamic parameter may follow:

$$\theta_{g,t} = \theta_{g,t-1} + Bx_{g,t}^{trend} + \eta_{g,t}, \quad (3.13)$$

with shrinkage or structural-break alternatives where gradual drift is implausible. The model separates changing hazard, changing exposure, changing vulnerability and changing observation access. Otherwise, improved reporting can be mistaken for increasing event frequency, or urban growth can be incorrectly absorbed into a vulnerability parameter.

Transportability is tested by county, basin, livelihood system and sensor. Local effects are partially pooled where mechanisms are shared. Scenario outputs state their climate, land-use, exposure and response assumptions and avoid unsupported precision beyond the evidence horizon.

3.9. Posterior samples are the actuarial interface

The principal interface to Part 4 is a set of weighted state paths:

$$\{Z_{1:T}^{(m)}, w_m\}_{m=1}^M \sim p(Z_{1:T} | Y_{1:t}, X_{1:T}), \quad \sum_{m=1}^M w_m = 1. \quad (3.14)$$

Each sample carries extent, intensity, duration, scenario assumptions and provenance. The actuarial engine calculates damage and financial terms for every sample. This preserves thresholds, limits, deductibles, network failure and nonlinear vulnerability. Posterior means and covariance summaries remain useful for display and reconciliation, but do not replace samples where the financial transform is nonlinear.

Model-form uncertainty can be represented through stacking or Bayesian model averaging. Predictive stacking selects weights from out-of-sample predictive performance rather than rewarding model complexity in-sample [9]. The ensemble retains a transparent baseline and names each component. Posterior predictive checks compare simulated observations and events with real ones, including false alarms, missed events, duration, footprint, reporting coverage and extremes.

The operational posterior is therefore more than a coloured map. It is a versioned probability distribution that explains which sources entered, how their observation processes were modelled, which dependencies were retained, how interventions changed transitions and how uncertainty reaches the loss engine.

4. Resolution, validation and operational promotion

Resolution is chosen from the decision backwards. Property claims triage may need building or parcel-level intersection, while the underlying hazard may support a coarser estimate. Drought may be credible at a livelihood-zone or subcounty scale while a gauge is a point measurement and a slope unit is

highly local. SCRI stores native resolution, display resolution and any downscaling method separately, then communicates uncertainty at the support justified by the data.

Each operational state output includes:

- event_id and hazard_type
- valuation_time and forecast_horizon
- state_variable and unit
- native_geometry and resolution
- posterior mean or median
- credible or prediction interval
- probability thresholds
- observation cut-off time
- model and parameter version
- forcing-data versions
- quality and degradation status
- scenario assumptions, including response

Validation follows the event through time. Hindcasting freezes the evidence available at each historical cut-off. Event holdouts test temporal generalisation; spatial holdouts test transfer to untrained areas; extreme-event holdouts test the tail. Metrics are matched to output: Brier or log score and reliability diagrams for probabilities; intersection-over-union for extent; continuous ranked probability score for distributions; absolute or relative error for depth, intensity or onset; lead time at a stated false-alarm rate; and duration error for slow states.

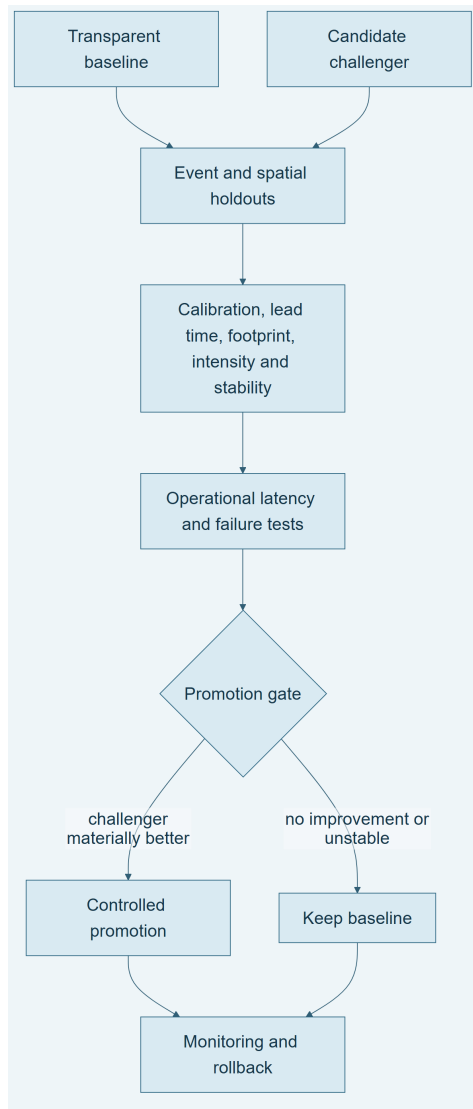


Figure 3.4: Promotion follows demonstrated operational value

The state engine passes distributions with provenance to the actuarial layer. A flood footprint may narrow as new radar arrives; a drought posterior may remain broad because household data are sparse; a fire forecast may branch under alternative suppression scenarios. Part 4 converts those uncertain states into damage and insured loss while preserving where the uncertainty came from.

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Part 4

Actuarial Catastrophe Loss Intelligence

Spatial Catastrophe Risk Intelligence

1. The loss engine follows the policy from damage to recovery

When floodwater reaches a warehouse, depth and duration influence damage; construction and contents influence vulnerability; and replacement values establish the ground-up financial scale. Deductibles, limits, sublimits, exclusions, waiting periods, coinsurance and coverage interpretation then turn covered damage into an insured claim. Reinsurance terms determine how much of the portfolio loss remains with the insurer.

This sequence is the actuarial centre of Spatial Catastrophe Risk Intelligence (SCRI). Catastrophe modelling practice separates event, hazard, exposure, vulnerability and financial modules for a reason [1]. The state engine from Part 3 estimates the peril. The actuarial engine answers a different question: *given what is currently known about this event and this exact exposure snapshot, what distribution of ground-up, gross insured and net retained loss follows?*

For asset or coverage unit i affected by event e and hazard h :

$$L_{e,i}^{GU} = V_i D_{h,c(i)}(I_{e,h}(g_i), M_i, \varepsilon_{e,i}). \quad (4.1)$$

where V_i is the relevant value measure, $I_{e,h}(g_i)$ is the uncertain intensity at location g_i , $c(i)$ is the exposure class, M_i contains secondary vulnerability modifiers and $\varepsilon_{e,i}$ represents residual damage uncertainty. The damage function D is a bounded distribution where the physical quantity requires it, carrying both a mean damage ratio and variability. For a crop it may depend on inundation duration and crop stage; for a building, depth and construction; for livestock drought, the insured index or mortality mechanism; for business interruption, physical damage, access and restoration time.

A simplified occurrence-level insurance transform is:

$$L_{e,i}^{INS} = q_i \min[\max(L_{e,i}^{COV} - d_i, 0), \ell_i]. \quad (4.2)$$

where $L_{e,i}^{COV}$ is the covered portion of ground-up loss after exclusions and sublimits, d_i is the deductible, ℓ_i is the applicable limit and q_i is the insured or coinsurance share. Real implementations also preserve policy aggregation, hours clauses, waiting periods, reinstatements and the order in which terms apply. The Oasis financial module illustrates why policy and reinsurance terms must be applied explicitly to ground-up samples to obtain gross and net perspectives [2]. Appendix F derives the moments of Equation (4.1), proves the piecewise policy transform in Equation (4.2), and shows how dependence enters portfolio aggregation.

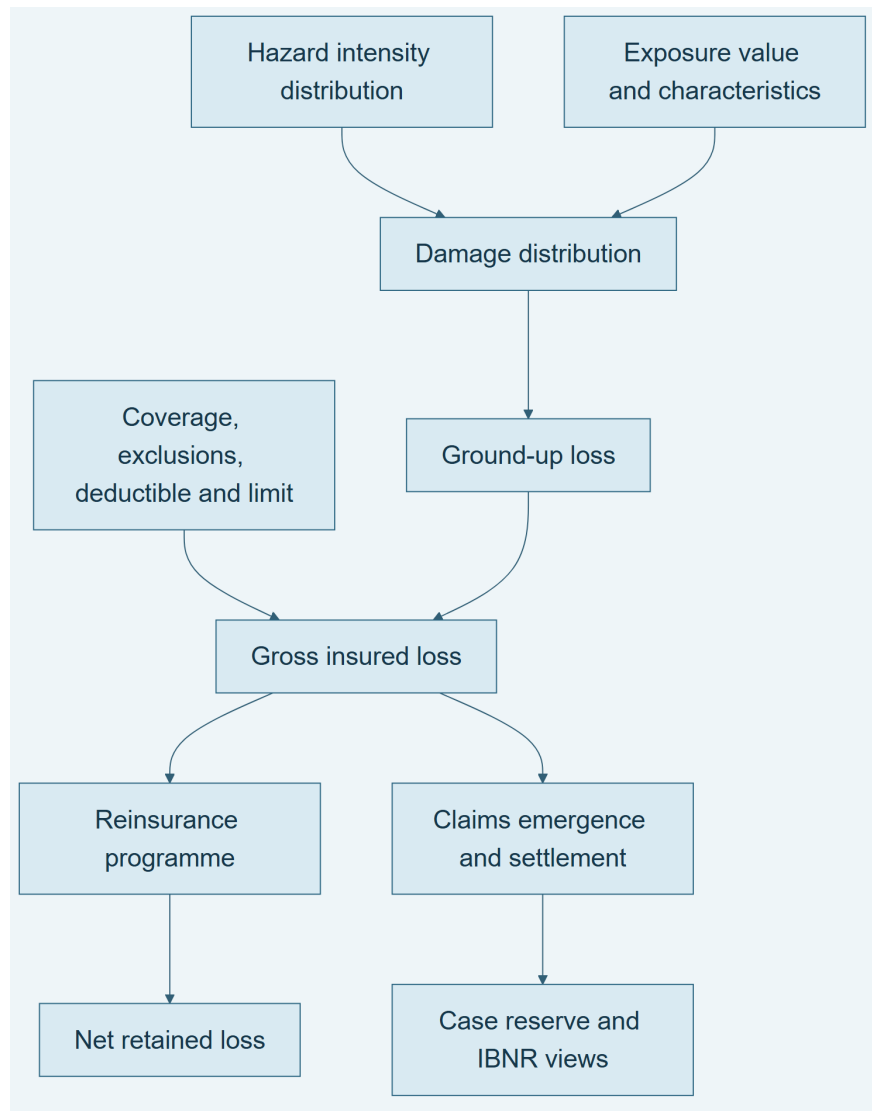


Figure 4.1: Physical damage, insured loss and accounting views are distinct

The insured event total is $L_e^{INS} = \sum_i L_{e,i}^{INS}$, but the sum must retain dependence. Two adjacent buildings can share flood depth, construction practice, utility failure and access interruption. Crop policies may share the same weather and planting calendar. Motor, property and business interruption can accumulate around one blocked transport corridor. Independent asset simulations would understate the tail.

2. A worked flood portfolio: from footprint to retained loss

The following **synthetic, reproducible Nzoia flood portfolio** makes every financial transformation inspectable before real exposure, claims and vulnerability evidence are introduced. Its counts, values, damage ratios, policy terms, event frequency and curve points are demonstration assumptions rather than Kenya market estimates.

The illustrative insurer has four groups exposed to the same flood footprint. The central scenario translates the posterior footprint into affected counts and one mean damage ratio for each group.

Values and all losses are in KES millions.

Segment	Policies	Value per unit	Total insured value	Central affected count	Mean damage ratio	Deductible	Limit	Insured share
Dwellings	120	4.00	480.00	72	18%	0.10	2.50	100%
Small businesses	80	8.00	640.00	36	22%	0.25	5.00	100%
Crop plots	300	0.30	90.00	210	35%	0.03	0.27	80%
Vehicles	50	2.00	100.00	15	28%	0.10	1.50	100%
Portfolio	550	---	1,310.00	333	---	---	---	---

For dwellings, the central ground-up loss is $72 \times 4.00 \times 0.18 = 51.84$. The insured amount per affected dwelling is $\min[\max(4.00 \times 0.18 - 0.10, 0), 2.50] = 0.62$, giving KES 44.64 million. The same order of operation is repeated for each segment; applying a portfolio-average deductible after aggregation would be wrong.

Segment	Ground-up loss	Gross insured loss	Difference created by terms
Dwellings	51.84	44.64	7.20
Small businesses	63.36	54.36	9.00
Crop plots	22.05	12.60	9.45
Vehicles	8.40	6.90	1.50
Central scenario	145.65	118.50	27.15

Suppose, for this demonstration, that a catastrophe excess-of-loss layer pays losses above KES 75 million up to KES 100 million. At the central gross loss of KES 118.50 million, the illustrative recovery is $\min[\max(118.50 - 75, 0), 100] = 43.50$ million and the insurer's event retention is KES 75.00 million. A production extension would add reinstatement premium, aggregate terms, exclusions, collectability, loss-adjustment expense and the timing of cash recovery.

The simulation expands the central scenario into a loss distribution. It treats affected share and mean damage ratio as uncertain beta variables, applies policy terms at unit level and preserves one common event draw within each segment. Across 60,000 event simulations, gross insured loss has a P10 of KES 83.51 million, a median of KES 115.86 million, a P90 of KES 155.58 million and a P95 of KES 168.83 million. The deterministic central case differs from the median because deductibles, limits and stochastic affected counts make the transform nonlinear.

Here **P10** means the tenth percentile of the simulated event-loss distribution: 10% of simulated losses are at or below that value. It is not a "1-in-10-year" loss. A 1-in-10-year annual loss corresponds to a 10% annual exceedance probability on a stated OEP or AEP curve. SCRI displays percentile, exceedance probability, return period, horizon and gross/net basis together.

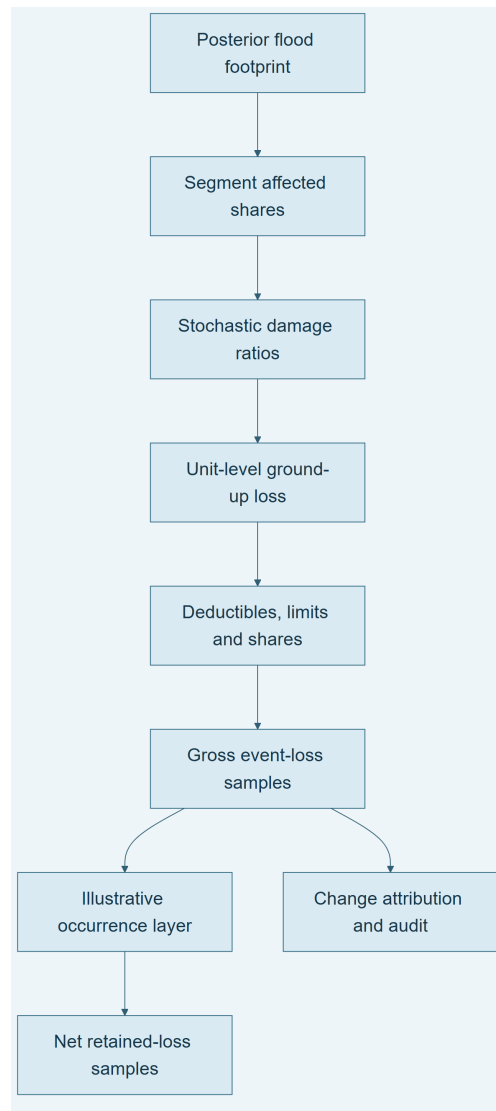


Figure 4.2: Calculation path for the synthetic worked example

To demonstrate the annual clock, a second simulation samples 100,000 synthetic years with a Poisson mean of 0.65 events per year and the event-loss distribution above. It produces an illustrative annual average gross insured loss of KES 76.36 million. The model's three-year AEP and OEP points are KES 110.36 million and KES 107.06 million respectively; at 100 years they are KES 386.83 million and KES 188.17 million. The widening gap is expected: AEP includes every event in a year, whereas OEP uses only the largest event.

Synthetic flood portfolio exceedance curves

Reproducible demonstration - not a tariff, reserve, capital figure or Kenya market estimate

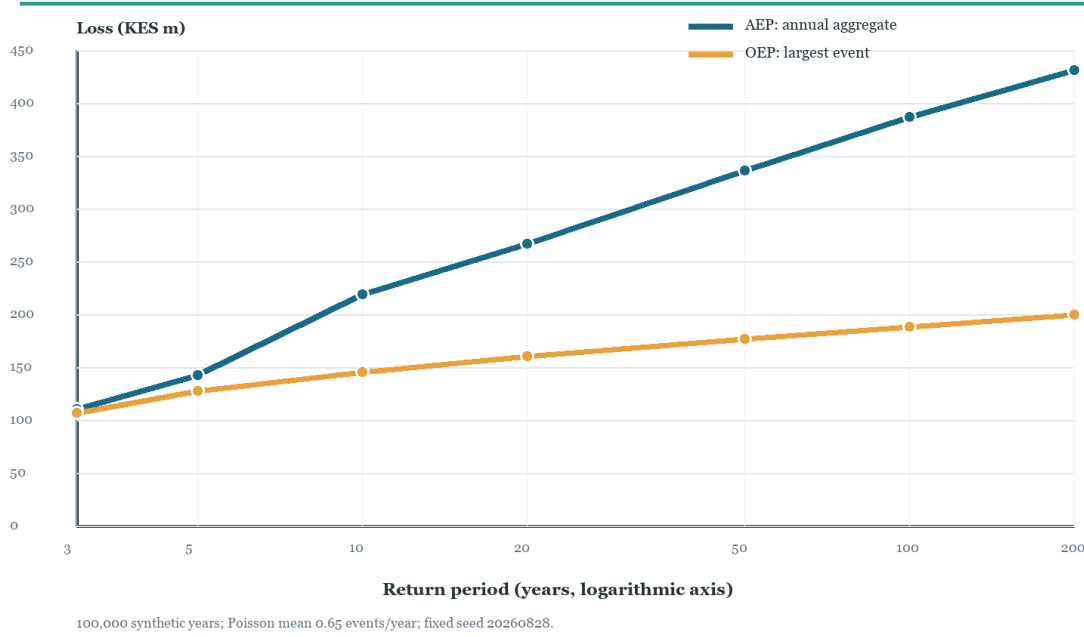


Figure 4.3: Synthetic aggregate and occurrence exceedance curves. The fixed seed, assumptions and generated values are stored with the manuscript. These curves demonstrate the interface; production pricing, reserving, reinsurance and capital use begins after empirical calibration and independent review.

The next evidence layer upgrades the demonstration into an empirically calibrated model. Policy records establish location, coverage, values and terms. Claims and engineering evidence calibrate vulnerability by depth, duration and asset type. Historical event catalogues and climate assumptions support frequency. Reinsurance contracts are encoded and independently reconciled. Until those steps are complete, the product label remains **demonstration**.

3. Current-event loss and long-term catastrophe risk

SCRI operates on two clocks. The **event clock** updates the unfolding catastrophe. New gauges, imagery and reports change the posterior intensity and footprint, which change the event-loss distribution. The **portfolio clock** governs annual expected loss, tariff support, risk appetite, reinsurance purchase and capital. It changes only through a controlled model version using an event catalogue, exposure assumptions, vulnerability calibration and financial terms appropriate to that horizon.

For the long-term catalogue, event count in hazard h , region g and year y may use a Poisson, Negative Binomial, point-process, renewal or state-duration model. A Negative Binomial is a candidate when conditional counts remain overdispersed after relevant covariates:

$$N_{h,g,y} \sim \text{NegBin}(\mu_{h,g,y}, \phi_h). \quad (4.3)$$

$$\log \mu_{h,g,y} = \log E_{h,g,y} + \eta_{h,g,y}. \quad (4.4)$$

where $E_{h,g,y}$ is an exposure-to-occurrence measure such as years, area-time or another definition established for that hazard. When μ contains the exposure offset, it already represents the expected count for that observation cell and enters the pure-premium calculation once. Appendix F derives the mean-variance relationship for Equation (4.3) and the rate interpretation of the exposure offset in Equation (4.4).

Frequency modelling follows each hazard. A drought episode may be represented by a persistent regime and duration. A wildfire catalogue may distinguish ignitions from escaped fires. Locust outbreaks may be conditioned on regional biological and climate regimes. A Poisson or empirical-frequency baseline makes the challenger's improvement measurable. Real-time reports update the current event record, while the annual catalogue changes through a separate controlled event-definition process.

Annual aggregate loss is generated from the catalogue:

$$S_y = \sum_{e=1}^{N_y} L_{e,y}^{NET}. \quad (4.5)$$

with event clustering, cross-peril relationships and policy/reinsurance terms applied in the correct order. Simulated year-loss tables or equivalent analytical outputs support:

- annual average loss, $\mathbb{E}[S_y]$;
- occurrence exceedance probability, based on the largest event loss in a year;
- aggregate exceedance probability, based on total annual loss;
- selected quantiles such as VaR at a stated probability;
- TVaR or expected shortfall beyond a stated quantile;
- scenario losses and uncertainty ranges.

“Probable maximum loss” is reported with its definition. It may refer to a quantile of occurrence loss, aggregate annual loss or a deterministic scenario; the probability, horizon, gross/net basis, currency, exposure date and model version accompany it.

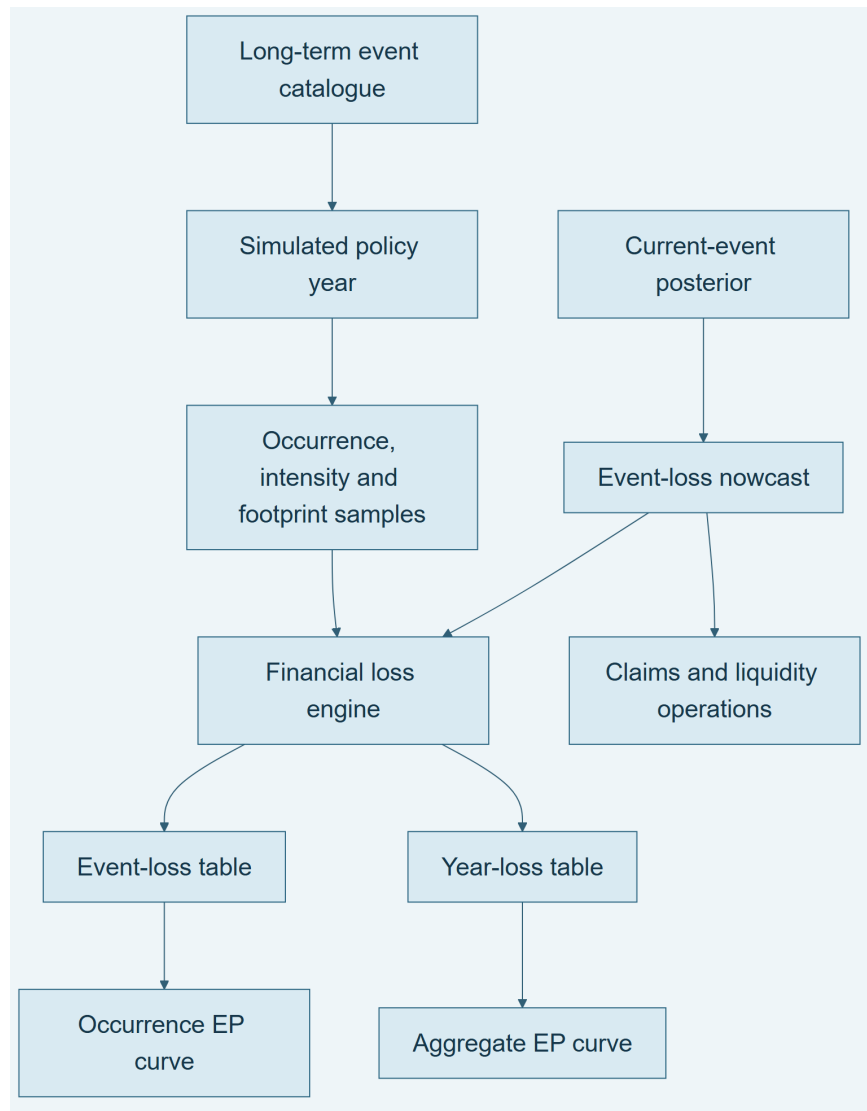


Figure 4.4: Event nowcasting and annual risk use related but different clocks

4. Severity, tails and dependence are empirical choices

SCRI evaluates candidate bulk distributions such as Gamma, Lognormal, Weibull or mixtures; asset-level loss may instead emerge from stochastic vulnerability functions. Selection uses exposure- and inflation-normalised data, censored and truncated likelihoods where required, diagnostic plots, proper scoring rules and out-of-sample validation.

If a peaks-over-threshold model is used, exceedances above threshold u may follow a Generalized Pareto distribution:

$$L - u \mid L > u \sim \text{GPD}(\xi, \beta_u). \quad (4.6)$$

Threshold choice is a model decision with a bias-variance trade-off. The analysis states the exceedance count, threshold diagnostics, parameter stability, continuity or splicing method, treatment of limits

and inflation, and uncertainty. Research on insurance-loss modelling likewise treats threshold selection and bootstrap uncertainty as substantive parts of the method [3], while standard quantitative-risk texts place tail estimates inside a wider framework of model and parameter uncertainty [4]. Where the Kenyan tail sample is sparse, documented external data, engineering scenarios and expert judgement support wider and more transparent uncertainty ranges. Appendix F derives compound annual-loss moments, OEP and AEP relationships and the tail-quantile expression implied by Equations (4.5)-(4.6).

Dependence receives the same discipline. A Gaussian copula is symmetric and has no asymptotic tail dependence; a conventional Student-*t* copula has symmetric tail dependence. Where upper- and lower-tail behaviour differs, candidates may include appropriate Archimedean families, vines, factor models, max-stable or spatial-process approaches [5]. Model selection considers dimensionality, data volume, physical mechanism and tail diagnostics. For many catastrophe portfolios, shared event and intensity fields already induce dependence, so any residual copula is calibrated to the dependence remaining after the hazard simulation.

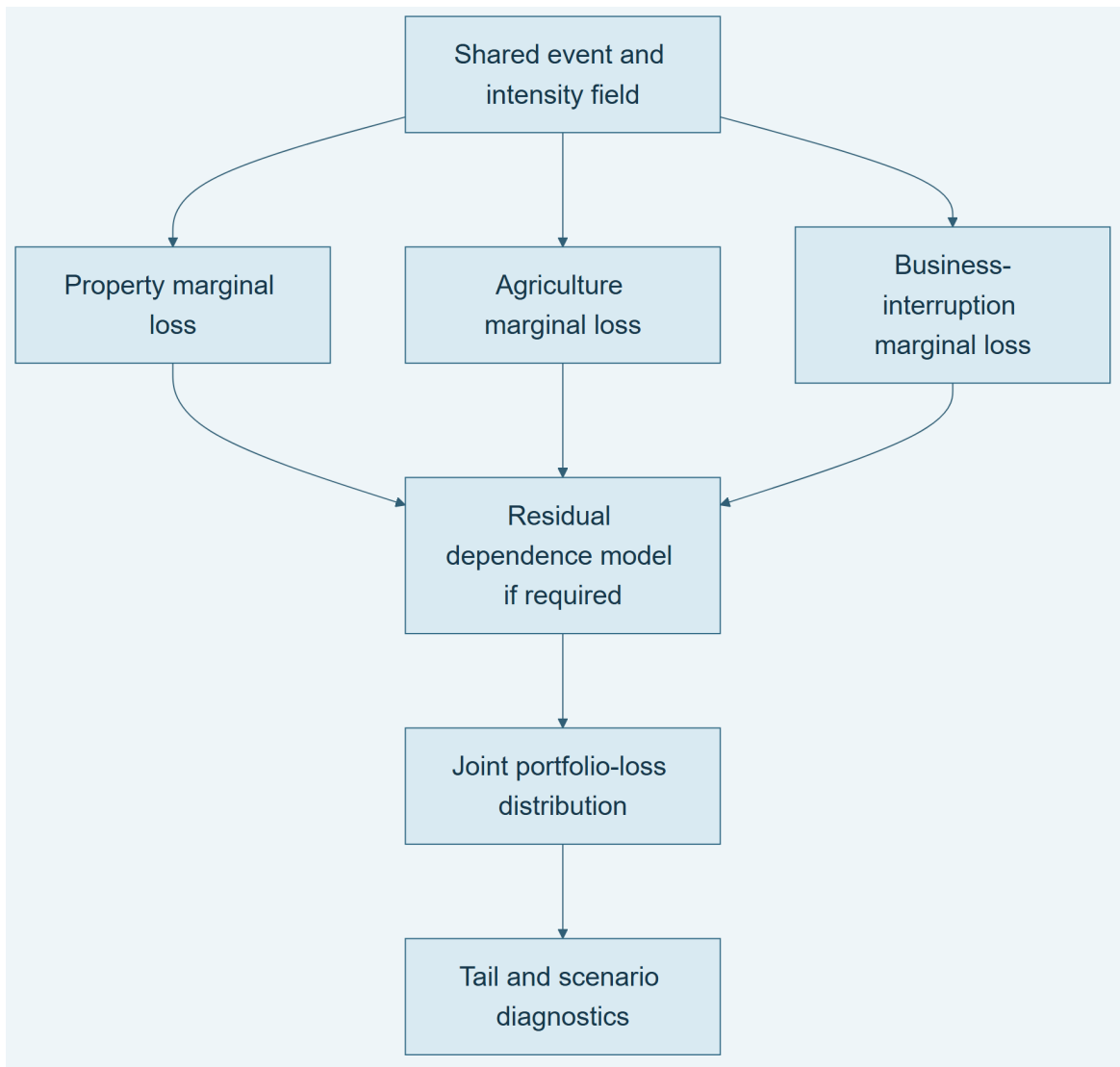


Figure 4.5: Dependence begins with the common physical event

5. Completing the event-loss distribution

The event-loss distribution must carry uncertainty from every major transformation, not only hazard intensity. A building may be geocoded to a parcel centroid when the insured structure sits at a different elevation. Replacement value may be indexed from an old declaration. Construction, occupancy, crop stage or asset condition may be misclassified. A vulnerability curve may be estimated from a small, selectively reported claims sample. SCRI represents these as model inputs with distributions and provenance.

For posterior hazard sample m , an expanded unit loss is:

$$L_{e,i}^{GU,(m)} = V_i^{(m)} D_{h,C_i^{(m)}} \left(I_{e,h}^{(m)}(G_i^{(m)}), M_i^{(m)}, \beta_h^{(m)}, \varepsilon_{e,i}^{(m)} \right), \quad (4.7)$$

where value V , geolocation G , class C , modifiers M and vulnerability parameter β can all be uncertain. Shared draws preserve dependence: one post-event inflation factor may affect many repairs; one location-quality process may affect a block of legacy policies; one vulnerability parameter draw affects all assets in its class. The output can then attribute variance to hazard, exposure, vulnerability and financial terms.

5.1. Damage distributions with boundary mass

A mean damage ratio alone rarely describes claims adequately. Many exposed assets experience no damage, while a smaller group reaches constructive or actual total loss. A flexible bounded model can combine a zero-damage mass, a continuous damage ratio and a total-loss mass:

$$D_i \sim \begin{cases} 0, & \text{with probability } \pi_{0,i}, \\ \text{Beta}(a_i, b_i), & \text{with probability } 1 - \pi_{0,i} - \pi_{1,i}, \\ 1, & \text{with probability } \pi_{1,i}. \end{cases} \quad (4.8)$$

Both boundary probabilities can depend on intensity and vulnerability. For drought or crop loss, the boundary may represent no yield effect and complete insured yield loss. For business interruption, a separate occurrence indicator and restoration-time distribution are often more interpretable than forcing the outcome into one damage ratio.

Claims used for calibration can be censored by policy limits, truncated by reporting thresholds or altered by deductibles. The likelihood must follow the observed quantity. A policy-limit payment indicates that covered loss was at least the exhaustion point; it is not evidence that ground-up loss equalled the limit. Closed-without-payment claims, partial inspections and missing uninsured damage also carry selection information.

5.2. Demand surge, inflation and settlement economics

Catastrophes change the cost of repair. Labour, materials, transport, accommodation and specialist capacity may become scarce. SCRI separates ordinary valuation inflation from event demand surge and from claims-development inflation. A scenario factor can depend on affected construction volume, access and elapsed time, with common draws across the affected region. The model reports the price date of values and the currency basis of every loss.

Claims leakage, fraud, salvage and subrogation are separate transformations. Leakage and fraud require claims evidence and operational controls; they are not inferred from a hazard footprint. Salvage and subrogation reduce ultimate cost with their own timing and recovery uncertainty. Direct and indirect loss-adjustment expense are modelled explicitly so operational workload and external-adjuster cost do not disappear inside indemnity loss.

5.3. Business interruption and infrastructure networks

Business interruption depends on more than physical damage. Restoration time can include inspection, repair, utility recovery, access, supplier recovery and policy waiting periods. A simple representation is:

$$L_i^{BI} = m_i [T_i^{restore} - w_i]^+, \quad T_i^{restore} = \max(T_i^{asset}, T_i^{access}, T_i^{utility}, T_i^{supplier}), \quad (4.9)$$

where m_i is the insured contribution margin or other contractually defined rate and w_i is the waiting period. Contingent business interruption uses supplier or customer dependencies even when the insured site has limited physical damage.

Infrastructure networks create correlated interruption. A bridge, substation, water plant or communication link can connect many exposures. SCRI represents critical nodes and service areas, then simulates component failure and restoration. The same network view supports uninsured livelihood and public-service loss, while insured coverage remains subject to policy wording.

5.4. Reinsurance, event definition and collectability

Reinsurance recovery depends on contract terms, event aggregation and timing. Hours clauses and event definitions determine which losses may be grouped. Event clustering can create several occurrences close together or one compound event affecting several perils. The event catalogue therefore stores physical lineage, contractual occurrence assignment and alternative interpretations where wording is unresolved.

Expected recovery also carries counterparty and dispute uncertainty. A recoverable calculated from treaty terms is different from cash collected. Collectability scenarios consider reinsurer credit, collateral, documentation, dispute, currency and settlement timing. The net-loss record presents contractual recovery, expected collectable recovery and cash timing separately.

Multi-peril annual catalogues preserve the possibility that drought, heat and fire share a climate regime; storm and flood occur in one system; or flooding and landslide affect the same road network. Correlation is calibrated through physical and statistical evidence, with scenario stress where data are sparse.

6. Claims nowcasting, reserves and the value of intervention

An event-loss nowcast supports operations before all claims are reported. It estimates a range of ultimate insured loss from the current footprint, exposure, vulnerability and financial terms. A claims emergence model then connects that ultimate view to reported counts and paid amounts:

$$N_{e,k}^{rep} \mid N_e^{ult} \sim \text{Binomial}(N_e^{ult}, F_N(k \mid x_e)). \quad (4.10)$$

where k is elapsed development time and F_N is a reporting curve conditional on event and operational characteristics. Severity development, reopening, salvage, loss-adjustment expense and claims inflation require separate treatment. The nowcast enters the reserving process as evidence, while the insurer's actuarial and finance functions own case reserves, IBNR and accounting entries under the applicable basis.

Detection latency enters the model only where an intervention pathway exists. Let A_e record whether a response was available and implemented, and τ_e the effective response delay. Conditional severity may include:

$$\mathbb{E}[L_e^{GU} \mid I_e, X_e, A_e, \tau_e] = m_h(I_e, X_e) + \delta_h(A_e, \tau_e). \quad (4.11)$$

Here δ_h is a causal effect estimated from evidence. Confounding is substantial: severe events may be detected earlier because they are visible, while easier-to-reach events may receive faster intervention and also cause less loss. Credible estimation may require matched events, quasi-experimental designs, randomised operational trials where ethical, or structured scenario analysis. The output is an avoided-loss range conditional on assumptions. Appendix F derives the reporting-count moments in Equation (4.10), an ultimate-count estimator, and the causal contrast associated with Equation (4.11).

7. Catastrophe-informed reserving

Claims nowcasting and reserving answer related but different questions. The nowcast asks what ultimate insured loss follows from the current catastrophe state and exposure. The reserve combines that prior view with reported, paid, case-estimate and settlement evidence under the insurer's reserving basis.

Let U_e denote ultimate event loss and $C_{e,1:k}$ the complete claims-development evidence available through development time k . The challenger posterior is:

$$p(U_e \mid C_{e,1:k}, H_{e,t}) \propto p(C_{e,1:k} \mid U_e, \theta_C) p(U_e \mid H_{e,t}, E, V, F), \quad (4.12)$$

where the prior is generated by posterior hazard H , exposure E , vulnerability V and financial terms F . As claims mature, the likelihood increasingly informs the posterior. The system records how much movement came from changing catastrophe evidence, new claims, case revisions, payment, inflation or closure.

The event reserve view reconciles:

$$U_e = P_e + RBNS_e + IBNER_e + IBNR_e + REOPEN_e + LAE_e - SALV_e - SUBR_e, \quad (4.13)$$

where P is paid loss, $RBNS$ is reported but not settled, $IBNER$ is development on reported claims, $IBNR$ is unreported loss, $REOPEN$ covers reopened claims, LAE is loss-adjustment expense, and salvage and subrogation reduce ultimate cost. Reinsurance recoveries are developed in a separate gross-to-net reconciliation.

Count and severity development are modelled separately where useful. Reporting-time survival models condition on channel, accessibility, catastrophe severity and operational disruption. Settlement and closure models condition on claim type, damage, documentation, repair capacity and dispute. Paid and incurred development models incorporate calendar inflation and event-specific operational effects.

Transparent baselines remain mandatory:

- chain ladder for stable development triangles;
- Bornhuetter-Ferguson to combine an a priori expectation with emergence;
- Cape Cod where exposure and historical experience support the expected loss ratio;
- Mack distribution-free process uncertainty under its assumptions;
- bootstrap or GLM reserving for predictive distributions; and
- claim-level reporting, payment, settlement and reopening models.

The catastrophe-informed Bayesian reserve is promoted only when point-in-time back-testing shows improved ultimate prediction, interval coverage and operational interpretation. The test set reconstructs historical events at successive knowledge cut-offs. It measures bias, mean absolute or squared error, quantile loss, coverage, reserve variability and one-year deterioration. Established stochastic reserving literature provides the baseline framework for prediction and uncertainty [6], [7].

The reserve dashboard displays paid, case, IBNER, IBNR, LAE, recoveries and uncertainty by event, peril, county and line. Each figure states valuation date, currency and gross/net basis. The responsible actuary owns assumptions and booking; SCRI supplies reproducible evidence, posterior samples and diagnostics.

8. IFRS 17 is an accounting translation

IFRS 17 measurement is introduced after the reserve model because it has a different purpose. The standard describes fulfilment cash flows as estimates of amounts expected to be collected and paid, including adjustments for timing and risk, and defines a risk adjustment for non-financial risk [8]. SCRI can provide claim and expense cash-flow scenarios, timing evidence and uncertainty analysis, while the insurer's accounting policy determines grouping, measurement model, discounting, risk adjustment, presentation and disclosure.

For incurred claims, an illustrative fulfilment-cash-flow view is:

$$FCF_t = \mathbb{E}_t \left[\sum_{u>t} \frac{CF_u}{B(t, u)} \right] + RA_t, \quad (4.14)$$

where CF_u contains probability-weighted future claim, expense and recovery cash flows within the applicable boundary, $B(t, u)$ is the accumulation or discount factor and RA_t is the entity's risk adjustment for non-financial risk. This expression is explanatory; the insurer applies the complete standard and its accounting policy.

Four quantities remain distinct:

Quantity	Primary purpose	Owner
Event-loss nowcast	Operational estimate of ultimate loss from the unfolding event	Catastrophe analytics and claims
Actuarial reserve estimate	Best estimate and uncertainty for outstanding claims	Reserving actuary and finance
IFRS 17 liability and risk adjustment	Financial reporting under the entity's accounting policy	Finance, actuarial and audit
Internal economic capital	Management view of capital required for the risk profile	Risk, capital and board

9. One-year economic-capital modelling

Economic capital measures the adverse one-year change in available capital or net asset value under the institution's internal risk framework. Define:

$$X_{1y} = -\Delta NAV_{1y}. \quad (4.15)$$

An internal unexpected-loss measure at confidence level α is:

$$EC_{\alpha} = VaR_{\alpha}(X_{1y}) - \mathbb{E}[X_{1y}], \quad (4.16)$$

while tail value at risk is:

$$TVaR_{\alpha}(X_{1y}) = \mathbb{E}[X_{1y} \mid X_{1y} > VaR_{\alpha}(X_{1y})], \quad (4.17)$$

with an appropriate discrete-distribution definition where needed. VaR gives a quantile; TVaR describes the average beyond it and is often more informative about tail severity. Confidence level, horizon, balance-sheet perimeter, management actions and valuation basis accompany both.

The one-year simulation combines:

- catastrophe underwriting loss from the multi-peril year catalogue;
- attritional premium risk and ordinary claims variability;
- reserve deterioration on prior events and accident years;
- reinsurance recovery and counterparty-credit risk;
- market and asset-value risk;
- operational, cyber, vendor and model events where included in the framework;
- liquidity stress from claim and recovery timing; and
- concentration by county, basin, peril, line and counterparty.

Dependencies follow plausible mechanisms. A catastrophe can simultaneously increase claims, weaken investment assets, delay reinsurance cash and create operational disruption. Drought, heat and fire may share a regime. Flood can damage collateral and create credit stress for policyholders or counterparties. Alternative dependency and reverse-stress scenarios test model-form uncertainty.

Kenya's regulatory capital requirement remains the applicable statutory measure. The Insurance (Capital Adequacy) Guidelines require insurers to assess capital required and available and maintain

a capital adequacy level commensurate with their risk profile [9]. SCRI does not replace that calculation. It supports catastrophe scenarios, concentrations and evidence that may inform the insurer's regulatory and internal capital processes.

Capital or margin concept	Meaning
Kenya regulatory risk-based capital	Statutory solvency requirement under the applicable Kenyan framework
Internal economic capital	Management estimate of capital needed for the institution's chosen risk appetite and profile
IFRS 17 risk adjustment	Compensation the insurer requires for bearing non-financial risk in insurance-contract cash flows
Reinsurance or transaction risk margin	Commercial pricing component determined by the contract or transaction

9.1. Capital allocation and risk-adjusted return

If portfolio capital is a positively homogeneous risk measure $\rho(X)$, Euler allocation assigns component j :

$$EC_j = x_j \frac{\partial \rho(X)}{\partial x_j}, \quad \sum_j EC_j = \rho(X), \quad (4.18)$$

under the required differentiability conditions. Components can be peril, county, line, treaty or customer segment. Marginal and scenario contributions remain useful when the conditions are not satisfied. Allocation is a management lens, not a claim that each county holds a physically separate pot of capital.

Risk-adjusted return can be expressed as:

$$RAROC_j = \frac{\mathbb{E}[Profit_j] - Cost_j^{risk}}{EC_j}, \quad (4.19)$$

with numerator and capital definitions documented. The measure can reveal concentrations that earn attractive nominal premium but consume disproportionate tail capital. It can also recognise a validated resilience intervention where it changes the loss distribution rather than merely receiving a label.

9.2. Reinsurance optimisation and liquidity

Candidate reinsurance programmes are applied to identical year-loss samples. The optimisation considers expected premium and expenses, retained loss, regulatory and economic capital, counterparty risk, reinstatement, collateral, basis risk and claim/recovery timing:

$$\min_{r \in \mathcal{R}} \{Premium(r) + \lambda_C EC_\alpha(r) + \lambda_L LiquidityStress(r)\} \quad (4.20)$$

subject to risk-appetite, coverage, counterparty and operational constraints. The result is a decision

frontier rather than an automatic treaty purchase. Reverse stress testing asks which combination of catastrophe, reserve deterioration, asset loss, failed recovery and operational disruption would breach a capital or liquidity threshold.

Multi-year scenarios connect the one-year capital view to business planning and climate uncertainty. They do not convert a long-run climate scenario directly into a one-year probability without a documented method. The board can see current risk, plausible change, management actions and the conditions under which the strategy needs revision.

The canonical loss record makes every financial quantity auditable:

event_id and hazard_type
valuation_time and observation cut-off
scenario or posterior-state version
exposure snapshot and currency basis
ground-up loss distribution
gross insured loss distribution
net retained loss distribution
reported and ultimate claims views
mean, quantiles and exceedance probabilities
occurrence or aggregate horizon
policy and reinsurance terms versions
uncertainty decomposition
model version and data-quality status
accountable user and permitted use

Output	Meaning	Typical owner	Not interchangeable with
Event-loss nowcast	Current estimate of ultimate loss from one unfolding event	Catastrophe analytics / claims	Booked reserve
Gross insured loss	Covered loss after direct policy terms	Actuarial / underwriting	Ground-up economic loss
Net retained loss	Gross loss after reinsurance recoveries	Reinsurance / capital	Available cash
AAL	Mean annual loss over the model horizon	Pricing / portfolio / capital	Current-event loss
OEP / AEP	Annual occurrence or aggregate exceedance relationship	Reinsurance / capital	Deterministic forecast
Case reserve / IBNR	Accounting estimate under the insurer's reserving basis	Reserving actuary / finance	Parametric payout trigger
IFRS 17 fulfilment cash flows and risk adjustment	Accounting measurement under the entity's policies and the applicable standard	Finance / actuarial / audit	Internal economic capital

Output	Meaning	Typical owner	Not interchangeable with
Internal economic capital	One-year management view of unexpected loss in available capital or net asset value	Risk / capital / board	Regulatory capital requirement or booked reserve

The architecture shortens the distance between physical evidence and a financially defined, revisable distribution. It shows why that distribution changed, which policies drove it, which uncertainty remains and which institution owns the next decision. Speed and traceability together make the estimate operationally useful.

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Part 5

Insurance, Investment and Resilience Finance

Spatial Catastrophe Risk Intelligence

1. Continuous underwriting with a stable customer promise

When the Nzoia leaves its banks, an insurer has several urgent questions. Which active policies intersect the credible flood footprint? How much insured value sits within each depth band? Which claims channels, adjusters and repair networks should be prepared? What range of gross and net loss should be communicated internally and to reinsurers? SCRI answers these operational questions while the customer's in-force promise remains stable.

That operating model defines **continuous catastrophe underwriting** in Spatial Catastrophe Risk Intelligence (SCRI). The product continuously observes the portfolio and serves each insurance team on its own decision clock. During an event, it supports accumulation, claims readiness, event-loss nowcasting, reinsurance notification and liquidity planning. At a future quotation or renewal, governed experience informs rating, deductibles, limits, mitigation requirements and appetite. The Insurance Regulatory Authority's insurance-risk guideline treats pricing, underwriting, claims, reserving and reinsurance as distinct controlled processes [1]; SCRI gives them a common evidence foundation.

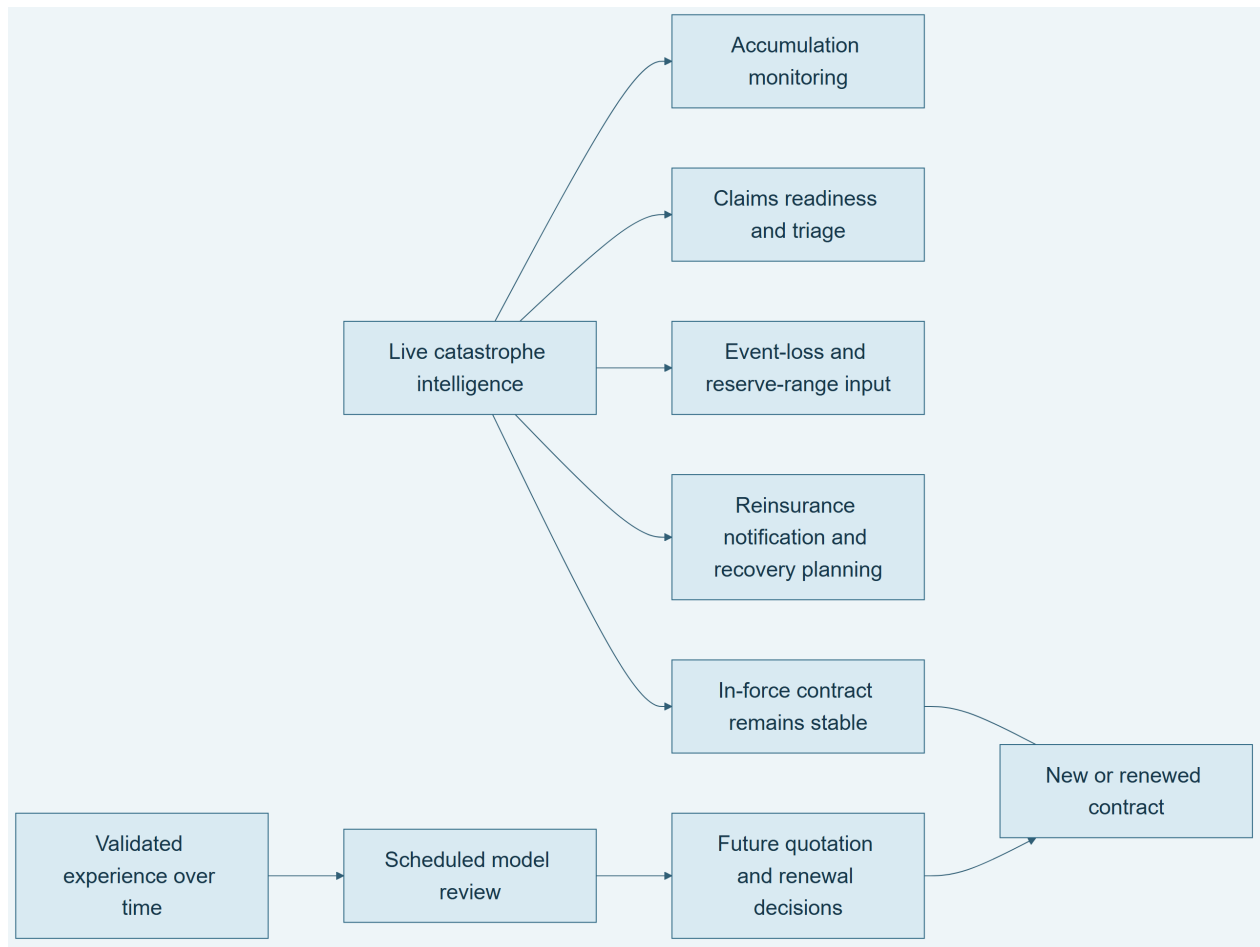


Figure 5.1: Continuous intelligence and contractual underwriting run on different clocks

The distinction changes the product interface. A portfolio manager sees exposure by hazard intensity, confidence and line of business. A claims manager sees likely affected policies, contact status, access constraints and triage priority. A reserving actuary sees a versioned event-loss distribution, re-

ported emergence and uncertainty. A reinsurance manager sees estimated loss by treaty, attachment proximity and data cut-off. An underwriter's prospective workspace presents long-term hazard and mitigation evidence at quotation or renewal.

Decision moment	SCRI may support	SCRI does not authorise
Active event, in-force policy	Exposure matching, welfare contact, claims triage, reserve-range input, reinsurance notice	Premium change, cancellation, new exclusion or retroactive deductible
Quote already issued	Flag for authorised human review under documented rules	Silent withdrawal or discriminatory treatment
New quotation	Prospective risk assessment using approved model and data	Unreviewed automated refusal based on a crowd report
Renewal	Experience analysis, mitigation recognition, filed or approved pricing workflow as applicable	Treating one event nowcast as a new annual tariff
Claims decision	Evidence retrieval and prioritisation	Automatic coverage or causation determination from a hazard polygon

More precise geography can improve risk differentiation by replacing county-wide assumptions with local exposure mechanisms. It also reveals affordability challenges for highly exposed households. SCRI therefore reports affordability, take-up, declination and subgroup outcomes alongside discrimination and calibration. Partial pooling stabilises sparse estimates, while explicit product choices govern cross-subsidy, public support and risk-pool design.

2. One flood, four desks, one evidence trail

At 06:40, a basin gauge and rainfall nowcast push the lower-Nzoia state above the insurer's accumulation-watch threshold. The screen shows how the probability of specified depth bands has changed, names the observations responsible, marks one gauge as twelve minutes stale and displays active exposure within each credible footprint. The accumulation analyst acknowledges the watch and requests an update in thirty minutes.

At 07:15, two independently sourced road reports and radar-derived water extent increase confidence in one area while a copied image cluster is consolidated. The claims operations lead receives likely affected policies, last verified contact channel, accessibility, policy type and the reason for each priority. Welfare contact and claim-intake preparation begin under an approved playbook. Customers can respond through smartphone, contact-centre or field channels, and sparse digital reporting appears as uncertainty rather than apparent safety.

At 09:00, the reserving actuary opens the same event at the same knowledge cut-off. The interface displays ground-up, gross insured and net retained distributions separately. A change panel attributes the movement since 07:15 to four factors: expanded footprint, revised flood depth, corrected policy coordinates and new reported claims. The actuary can download unit-level samples, reproduce the financial transform and interpret the nowcast within the insurer's reserve process.

At 10:30, the reinsurance manager sees the occurrence layer approaching attachment under the central estimate, with a probability distribution around that outcome. The communication pack includes the treaty version, exposure cut-off, model version, outstanding data failures and a link to supporting evidence. Recovery status develops as the contract and event facts mature. The prospective underwriting workspace remains scheduled around quotation and renewal.

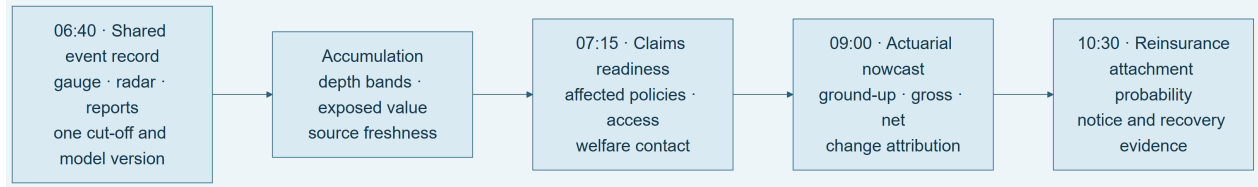


Figure 5.2: Product-interface narrative during one flood morning

The interface is organised around questions, not model names:

Workspace	First question	Minimum visible evidence	Human action
Event overview	What changed, where and how certain is it?	Cut-off time, footprint bands, source freshness, contradictions	Acknowledge or challenge the state
Portfolio	Which active exposures drive the range?	Policy snapshot, value, line, location quality, depth band	Investigate concentration and data defects
Claims readiness	Who may need contact or field support?	Triage reason, access, contact status, uncertainty	Contact, assign or override
Actuarial loss	What are the ground-up, gross and net distributions?	Samples, terms, version, change attribution	Interpret for reserving and risk management
Reinsurance	Which contractual layers may be affected?	Treaty encoding, attachment probability, exclusions, evidence pack	Notify and prepare recovery documentation

Every workspace carries the same event identifier, observation cut-off, exposure snapshot and model version. That continuity prevents a polished dashboard from hiding mismatched vintages. The user can move from a portfolio total to the physical evidence and back to the exact policy transform. A decision log records what the user saw, what action was taken and whether the recommendation was overridden. This is the product proposition in operational form: earlier shared intelligence, different owned decisions.

3. Reinsurance, parametric products and financial protection

An unfolding event estimate can improve the conversation with reinsurers. It provides earlier notice, likely affected layers, recovery documentation and a consistent view of uncertainty. Attachment, limit, hours clauses, reinstatements, exclusions and reporting obligations continue to govern the executed

treaty. Over time, validated event intelligence strengthens future placement and capital planning by giving both parties a more transparent risk record.

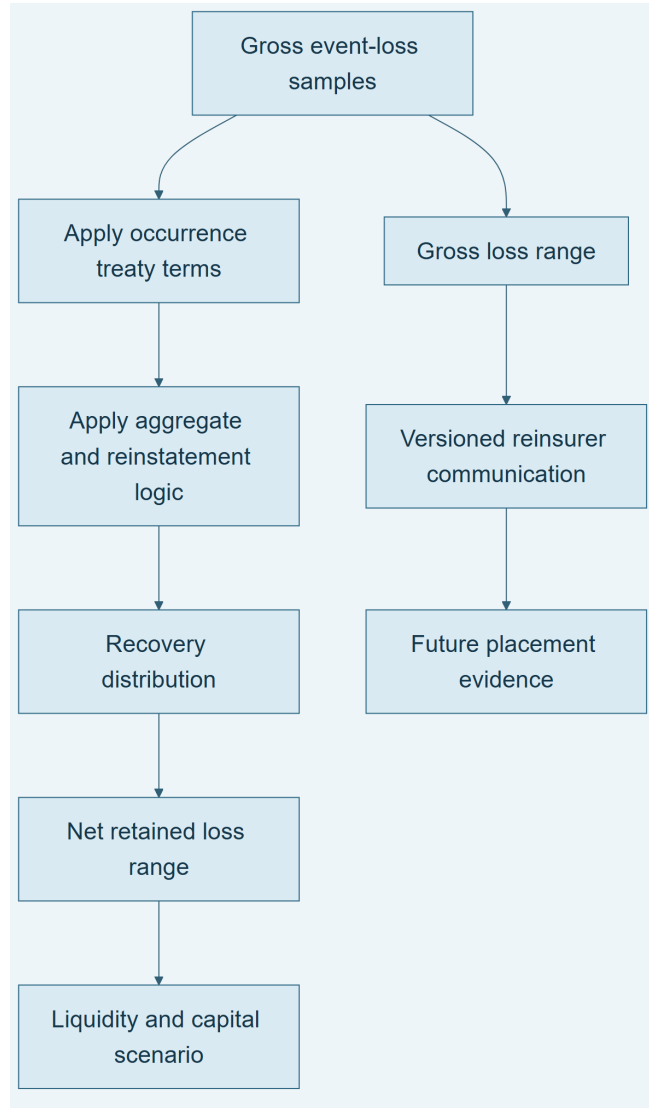


Figure 5.3: Reinsurance intelligence applies existing terms before informing future strategy

Parametric insurance defines its promise in advance. The contract names the trigger variable, observation source, calculation agent, threshold, geographic unit, data-continuity rule and payout formula. Crowd evidence can improve product design, validate the historical relationship between index and loss, support customer communication and activate a predefined fallback where the contract provides one.

Let T_e denote trigger status and L_e experienced loss. Basis risk remains:

$$BR_e = P(T_e) - L_e. \quad (5.1)$$

under a common monetary or utility scale. A payout with limited local loss is one direction; severe local loss without payout is the other. Higher-resolution evidence can reduce expected basis error,

while trigger testing, layered indices, contractually defined fallbacks, dispute processes and complementary indemnity or social-protection arrangements manage the remaining difference. Spatial tail dependence and local heterogeneity are recognised challenges in rainfall insurance [2]. Appendix G derives the expected payout, mean basis error and mean-squared basis-risk decomposition associated with Equation (5.1).

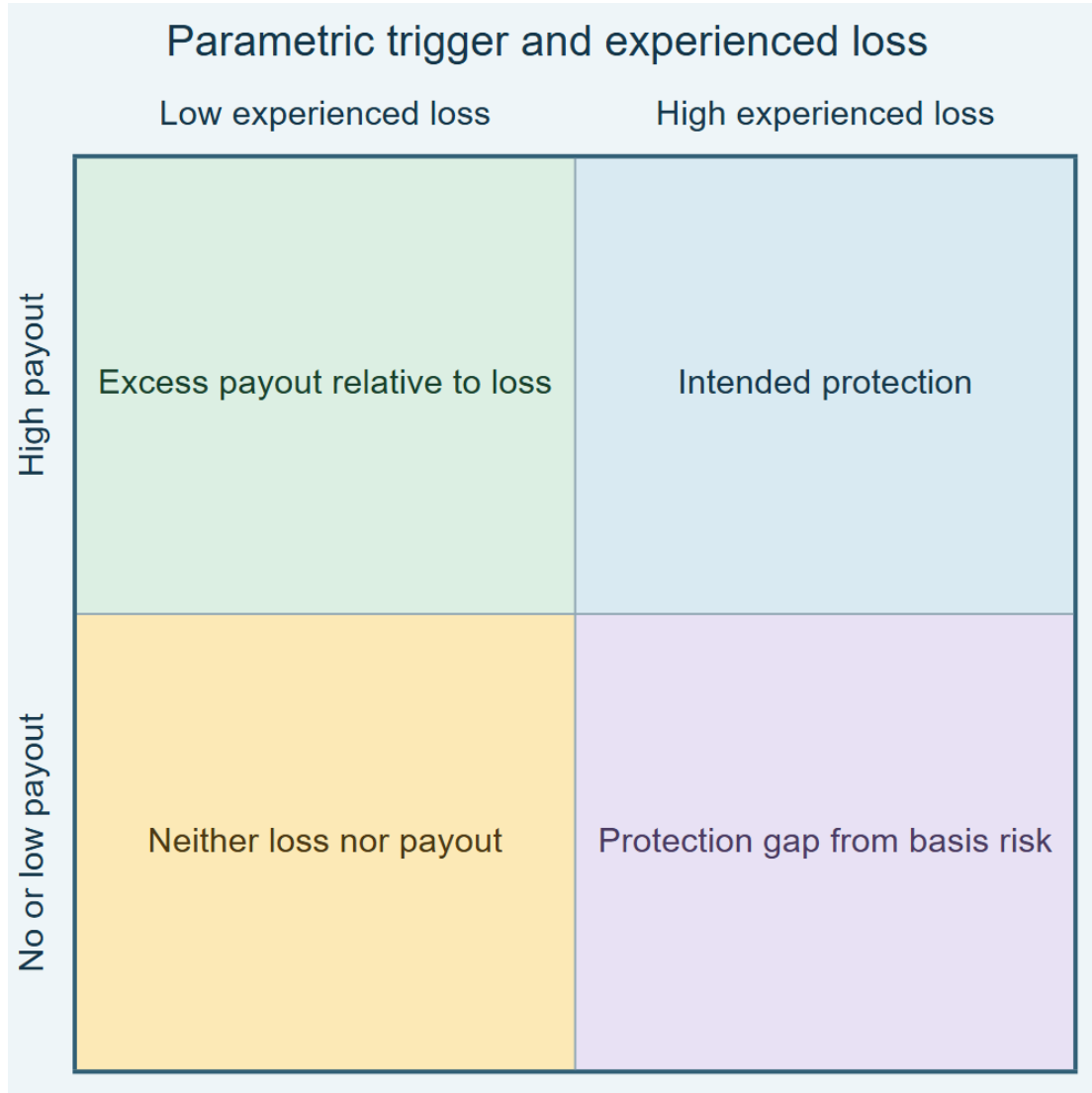


Figure 5.4: Parametric accuracy is tested against experienced loss

At sovereign and county level, catastrophe intelligence sits inside a broader financial-protection strategy. Kenya has used a layered approach including contingency mechanisms, contingent credit, agricultural and livestock insurance and social-protection instruments. The World Bank's account of Kenya's Cat DDO describes a US\$200 million contingent line and the role of prearranged finance in faster response [3]. The wider disaster-risk-finance framework distinguishes retention for frequent or moderate losses from risk transfer for less frequent, severe losses [4], while established guidance emphasises that public intervention responds to the market and affordability constraints of catastrophe finance [5]. SCRI estimates the loss and funding need associated with each layer, giving the authorised institution earlier evidence for drawdown, allocation or trigger decisions.

4. Resilience investment requires a cash-flow bridge

Resilience finance begins by connecting physical performance to an investable cash-flow structure. A financeable structure needs an issuer or borrower, a lawful source of repayment, a term, covenants, performance measurement, a verification agent, risk allocation and a credible counterfactual. SCRI supplies risk analytics and monitoring evidence; the transaction sponsor supplies the repayment source and financial structure.

For a proposed intervention a , the analytical starting point is:

$$NPV(a) = -C_0 + \sum_{t=1}^T \frac{\mathbb{E}[\Delta L_t(a)] + B_t(a) - O_t(a)}{(1+r)^t}. \quad (5.2)$$

where C_0 is initial capital expenditure, ΔL_t is avoided loss under an explicit counterfactual, B_t is other measurable benefit, O_t is operating cost and r is a discount rate appropriate to the decision owner. This is a benefit-cost assessment. It becomes a financing structure only after mapping benefits to a party able and willing to make payments. Appendix G derives Equation (5.2) from discounted cash flow, develops the benefit-cost ratio and shows how attachment and exhaustion define a financial risk layer.

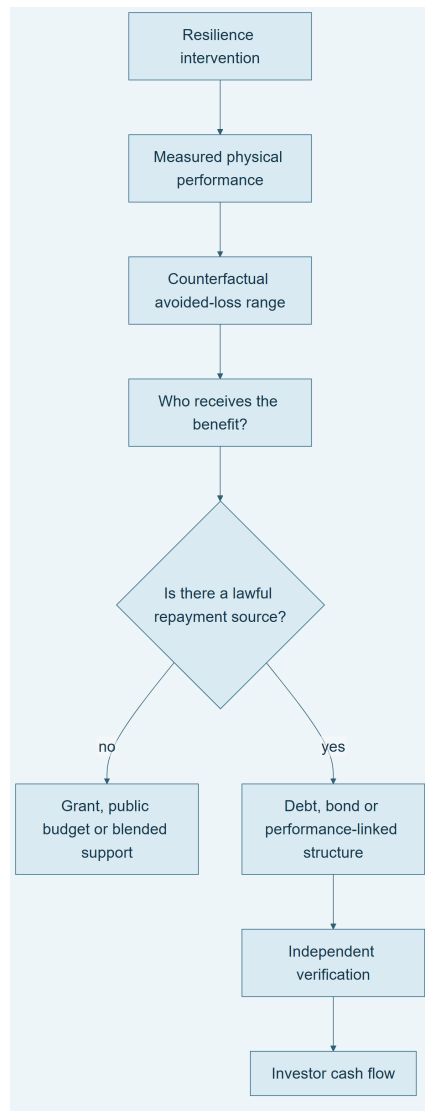


Figure 5.5: Avoided loss becomes financeable only through an identified payment mechanism

The intervention mechanism varies by hazard. Flood investment may improve drainage, protect a road or restore a wetland. Drought investment may improve water access, fodder systems or anticipatory livestock support. Wildfire investment may combine observation with access, fuel management, equipment and trained response. Locust investment may strengthen surveillance and timely control. Landslide investment may address drainage, slope stabilisation and road redundancy. Heat investment may support shade, ventilation, work practices, cooling and resilient power. Each requires different physical evidence and beneficiaries.

The platform therefore generates separate decision products:

- **Insurer:** event-loss range, affected policies, claims workload, accumulation, gross/net loss and future risk evidence.
- **Reinsurer:** versioned event estimate, treaty-layer impact, recovery documentation and uncertainty.
- **Lender or investor:** site hazard, downtime, revenue stress, resilience expenditure and resid-

ual risk.

- **Government:** affected population, infrastructure and fiscal-loss scenarios, emergency liquidity need and financing-layer exhaustion.
- **Development financier:** additionality, beneficiary reach, avoided-loss range, monitoring evidence and failure conditions.
- **Policyholder or community:** warning source, expected effects, protective action, coverage, trigger and claims information in accessible language.

These outputs share the catastrophe state while each institution applies its own decision rule. A lender's probability of default model, an insurer's loss model and a county's emergency-allocation rule receive separate calibration, governance and legal authority. That separation allows one product platform to serve several markets with precision.

5. The four-ledger finance engine

SCRI turns the cash-flow bridge into a working product through four linked ledgers. Each ledger answers a different diligence question, and their shared identifiers allow a financier to follow one intervention from the original catastrophe problem to verified performance.

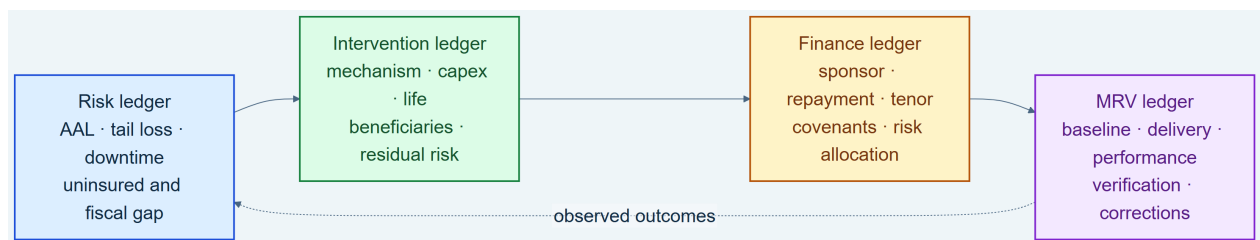


Figure 5.6: Four linked ledgers preserve the chain from catastrophe risk to financed and verified resilience

The **risk ledger** contains the exposure snapshot, hazard-state samples, annual average loss, occurrence and aggregate exceedance curves, expected downtime, uninsured loss, fiscal exposure and uncertainty. It records whose risk is being measured, at which valuation date and under which climate and development assumptions.

The **intervention ledger** describes the physical or social change: its location, cost, engineering or behavioural mechanism, construction schedule, useful life, maintenance obligation, beneficiaries, expected avoided-loss distribution, residual risk and possible spillovers. Flood drainage, for example, is represented through changed depth, duration or vulnerability---not through an untested resilience label.

The **finance ledger** records the sponsor, borrower or issuer, source and use of funds, repayment source, tenor, pricing, guarantees, insurance, triggers, covenants, waterfall and allocation of construction, performance, market, credit and catastrophe risk. It explains how a socially valuable avoided loss becomes a viable cash flow, public expenditure commitment or grant-supported outcome.

The **monitoring, reporting and verification ledger** preserves the baseline, implementation milestones, physical-performance observations, beneficiary evidence, event outcomes, verifier findings and correction history. Its version links back to the observations and models used at appraisal.

A later model revision therefore enriches the record rather than silently rewriting the basis on which capital was committed.

Together, the ledgers support origination, due diligence, approval, drawdown, covenant monitoring, post-event review and refinancing. They also make productive disagreement visible: an engineer may support the physical mechanism while a lender finds the repayment source weak; an actuary may validate loss reduction while a verifier finds incomplete implementation. The product keeps those conclusions distinct and traceable.

6. Investment modelling from hazard to debt service

An infrastructure owner experiences catastrophe through service interruption, restoration expenditure, revenue volatility and asset impairment. A lender experiences those effects through borrower cash flow, debt service, collateral and recovery. SCRI therefore translates posterior hazard samples into operating and financing scenarios without treating physical loss as credit loss.

For asset or project j , hazard-adjusted operating cash flow in scenario m is:

$$OCF_{j,t}^{(m)} = Revenue_{j,t}^{(m)} - OperatingCost_{j,t}^{(m)} - RestorationCost_{j,t}^{(m)}, \quad (5.3)$$

where revenue reflects downtime, customer disruption or production loss and restoration cost reflects the damage and recovery pathway. A project debt-service coverage ratio is then:

$$DSCR_{j,t}^{(m)} = \frac{CFADS_{j,t}^{(m)}}{DebtService_{j,t}}, \quad (5.4)$$

where $CFADS$ is cash flow available for debt service under the financing definition. The distribution of minimum DSCR, time below covenant and recovery to covenant provide more operational information than a single deterministic stress.

Where a bank has separately validated credit models, catastrophe evidence can enter through conditional probability of default, loss given default and exposure at default:

$$EL_{j,t}^{credit} = PD_{j,t}(H, E, V, R) \times LGD_{j,t}(H, C, T) \times EAD_{j,t}. \quad (5.5)$$

C represents collateral and T recovery and legal conditions. SCRI supplies physical-risk, interruption, collateral-condition and recovery-time features; the bank owns model calibration, credit policy and decision. The linkage is validated against borrower outcomes because a damaged asset may be well insured, a lightly damaged business may face prolonged access failure, and public support may change recovery.

Project comparison retains the NPV in Equation (5.2) and adds internal rate of return as the rate r^* satisfying:

$$0 = -C_0 + \sum_{t=1}^T \frac{CF_t(a)}{(1 + r^*)^t}. \quad (5.6)$$

For resilience projects, $CF_t(a)$ states whose cash flow is counted and how avoided loss enters it. Distributional NPV, probability of negative NPV, covenant breach and tail loss are reported alongside the mean. A staged intervention also has option value: a sponsor can fund a reversible first phase, observe performance and climate evidence, then expand when the expected benefit of learning exceeds the cost of waiting. That real-option view is valuable for interventions whose design can be modular, while urgent life-safety works retain their own decision criteria.

Portfolio screens aggregate assets through shared hazards and dependencies. Roads, substations, water systems and telecommunications may fail together, so apparent geographic diversification can disappear during one event. SCRI reports concentration by basin, county, service network, peril and critical dependency, then runs present-climate, plausible-change and reverse-stress scenarios. The product asks which combination of outage, restoration delay, insurance recovery and refinancing condition challenges the investment thesis.

7. Eight finance products, each with a complete product journey

The shared evidence spine supports several financial structures. Each product journey begins with a user, a financing need and a defined decision.

Structure	Product user and decision	SCRI evidence	Financial completion test
Resilience loan	Bank and borrower choose capex and loan terms	Baseline loss, intervention effect, DSCR stress, residual risk and covenants	Repayment source remains credible under declared stresses
Green or adaptation bond	Issuer and investors select eligible projects and reporting terms	Physical-risk rationale, project screening, use-of-proceeds mapping and impact evidence	Eligible use, governance, reporting and external review are established
Sustainability-linked finance	Borrower and lender set performance targets	KPI baseline, ambition analysis, observation method and verification trail	KPI is material, measurable, time-bound and financing terms are explicit
Parametric insurance	Policyholder, insurer and reinsurer define the trigger	Historical trigger-loss fit, basis-risk distribution, data continuity and fallback	Contract names index, source, agent, geography, thresholds and payout
Contingent credit	Public or institutional borrower prepares liquidity	Financing-layer exhaustion, trigger evidence, drawdown need and timing	Eligibility, drawdown authority and repayment terms are prearranged
Resilience bond	Sponsor connects risk reduction to insurance and capital	Pre/post loss distributions, premium or risk-cost pathway, MRV and residual risk	A credible source pays investors; modelled savings alone are insufficient

Structure	Product user and decision	SCRI evidence	Financial completion test
Blended or results-based finance	Public, concessional and commercial funders allocate risk	Additionality, beneficiaries, performance states and verification	Capital stack, loss absorption, payment conditions and verifier are agreed
County or sovereign risk finance	Treasury and county leaders select retention, contingency and transfer layers	Fiscal AAL, tail loss, vulnerable populations, liquidity timing and layer exhaustion	Budget, credit and transfer instruments align with mandates and response plans

7.1. Resilience loan: a water system that keeps earning through drought

A water utility or community enterprise proposes borehole rehabilitation, storage, leakage reduction and solar pumping. SCRI reconstructs drought duration, water-point reliability, demand, household and livestock dependence, operating cost and revenue collection. The intervention model produces operating-cash-flow and service-continuity distributions with and without the investment. The lender tests DSCR, reserve accounts and insurance under those scenarios. Covenants may track storage availability, preventive maintenance and service continuity. The MRV ledger records commissioning and performance. The loan is supported by identifiable water-service or public-payment cash flows; the avoided livelihood loss strengthens the public-value case and can justify concessional participation.

7.2. Green or adaptation bond: a portfolio with traceable physical rationale

A county, infrastructure issuer or financial institution assembles drainage, water, energy and transport investments. SCRI maps each asset to a climate hazard, documents how the activity improves adaptation, estimates beneficiaries and residual risk, and maintains use-of-proceeds and outcome evidence. Kenya's Green Finance Taxonomy and Climate Risk Disclosure Framework, issued by the Central Bank of Kenya in April 2025, provide a current banking-sector classification and disclosure context [6]. CMA's Policy Guidance Note supplies the Kenyan capital-market pathway for green bonds [7]. SCRI supports the evidence; the issuer, arranger, external reviewer and relevant authority complete the transaction.

7.3. Sustainability-linked finance: catastrophe resilience as a measured operating target

An infrastructure or agricultural borrower selects a material performance indicator such as percentage of critical assets meeting an approved resilience standard, verified restoration time, or continuity of service under defined stress. SCRI establishes the baseline, tests ambition against the asset plan, monitors the observation method and preserves verification evidence. Financing terms specify the target dates and economic consequence. The KPI reflects a change the borrower can influence and is accompanied by safeguards against maintaining performance by shifting risk to vulnerable communities.

7.4. Parametric protection: a transparent trigger and a visible basis-risk record

Before placement, SCRI replays the proposed rainfall, river, vegetation or heat index against observed loss and livelihood outcomes. It quantifies missed-loss and excess-payout states, tests alternative

thresholds and geographic units, and evaluates source outage and revision. During the contract, the platform displays the governed trigger calculation and local impact evidence side by side. After the event, it updates the basis-risk record. This creates a learning product around the contractual promise in Equation (5.1).

7.5. Contingent credit: liquidity ready before the emergency

A public borrower estimates the probability that contingency budgets and reserves will be exhausted under current and future scenarios. SCRI links the event state, affected services, fiscal-loss distribution and cash timing to the pre-agreed drawdown conditions. The authorised institution chooses whether the conditions are met and executes the facility. The product's value is preparation: data definitions, responsible users and the evidence pack exist before roads, power and communications are under stress.

7.6. Resilience bond: connecting reduced risk to an investor payment

A resilience bond requires a mechanism that turns improved risk into cash. SCRI estimates pre- and post-intervention loss distributions and tracks performance. An insurer may recognise validated reduction through a prospective premium or capacity arrangement; a public sponsor may make availability payments for service continuity; or project revenue may fund debt service. The finance ledger records the chosen mechanism. Sensitivity analysis shows how construction failure, maintenance lapse, changing hazard, model uncertainty and insurance-market conditions affect investor cash flow.

7.7. Blended and results-based finance: allocating risks to the capital best able to carry them

A drought, landscape or community-resilience programme may produce diffuse benefits and limited direct revenue. Concessional or grant capital can fund public goods, absorb first loss, pay for verified outcomes or lower the cost of capital. SCRI defines the baseline, additionality, beneficiary distribution, performance states and verification protocol. Commercial capital enters where a dependable payment or revenue stream exists. Results-based payments follow verified delivery or outcomes specified in advance rather than a retrospective claim of success.

7.8. County and sovereign finance: one multi-hazard liquidity strategy

The National Treasury's *Kenya Disaster Risk Financing Strategy 2026--2030* frames risk reduction, retention, contingency finance and transfer within a multi-hazard national approach [8]. SCRI can provide its operating evidence layer: county and national fiscal-risk views, financing-layer exhaustion, vulnerable populations, event timing, response needs and post-event reconciliation. The public authority retains budget, borrowing, allocation and emergency powers. The platform helps each layer activate with a common, versioned account of the event and funding need.

8. Sustainable insurance and climate-risk disclosure inside the product

Sustainability becomes concrete when it changes the data and workflow. The insurer-facing interface maps portfolio concentration, protection gaps, customer resilience and claims outcomes into the institution's sustainable-insurance process. IRA's 2024 Principles of Sustainable Insurance provide a Kenyan supervisory reference for embedding environmental, social and governance considerations

[9]. The bank-facing interface maps physical-risk exposure, scenario results, client adaptation and data limitations into the CBK disclosure framework [6].

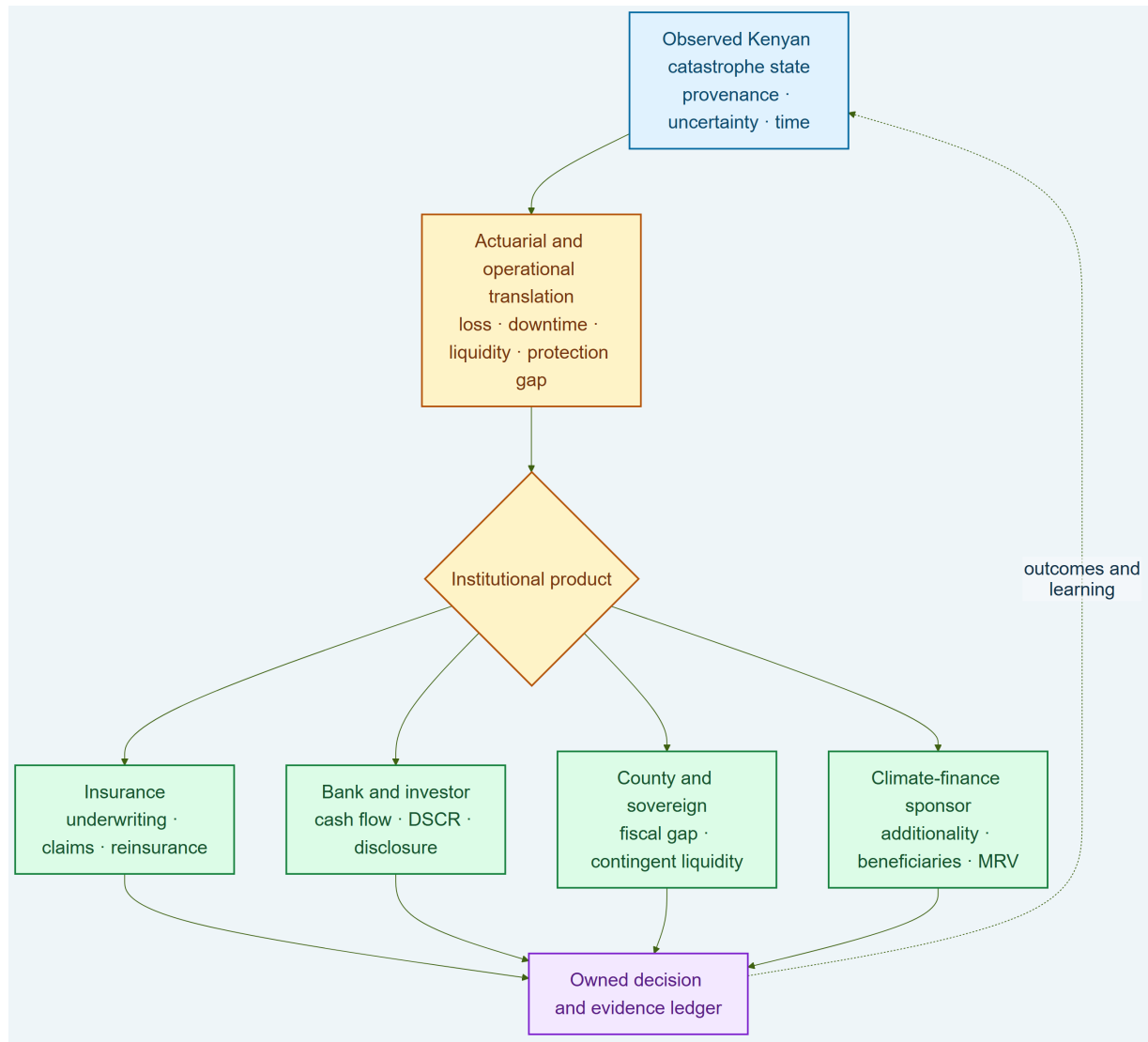


Figure 5.7: One catastrophe state becomes distinct insurance, investment, public-finance and climate-finance products

The finance engine measures distributional outcomes as well as aggregate benefit: people reached, affordability, service continuity, geographic inclusion, livelihood effects and the allocation of residual risk. This allows capital committees to see whether a project improves resilience for the intended beneficiaries and whether another group inherits displaced risk. It also creates evidence for learning across financed interventions.

The commercial thesis is focused and testable. SCRI can help capital arrive earlier by reducing uncertainty, preparing operations and providing traceable evidence. It can improve the cost of capital where validated information changes risk selection, mitigation or reinsurance confidence. Success is measured through lead time, operational readiness, basis-risk performance, customer outcomes and finance structures with real repayment mechanisms.

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Part 6

Governance, Validation and Implementation

Spatial Catastrophe Risk Intelligence

1. Turning evidence into accountable decisions

In the first minutes of a catastrophe, speed and accountability matter together. A community reporter may describe water across a road. Spatial Catastrophe Risk Intelligence (SCRI) may infer that insured properties are likely to be affected. A public authority may issue a warning. A claims officer may contact policyholders. A reserving actuary may revise an estimate. One evidence trail supports all four users, while each acts through a defined institutional mandate.

The operating model therefore assigns ownership before technology:

Role	Core responsibility
Data controller or joint controller	Determines lawful purposes and essential means of personal-data processing
Data processor	Processes personal data under documented controller instructions
Data steward	Owns schema quality, lineage, access classification and retention execution
Hazard-model owner	Defines intended use, physical assumptions, limitations and monitoring
Actuarial-model owner	Owns loss definitions, calibration, financial terms and actuarial interpretation
Product owner	Connects validated outputs to a bounded insurance workflow and commercial outcome
Decision owner	Has legal and institutional authority for the consequential action
Independent validator	Challenges data, method, code, performance, limitations and use
Community-governance forum	Reviews contribution terms, harms, accessibility, redress and distributional effects

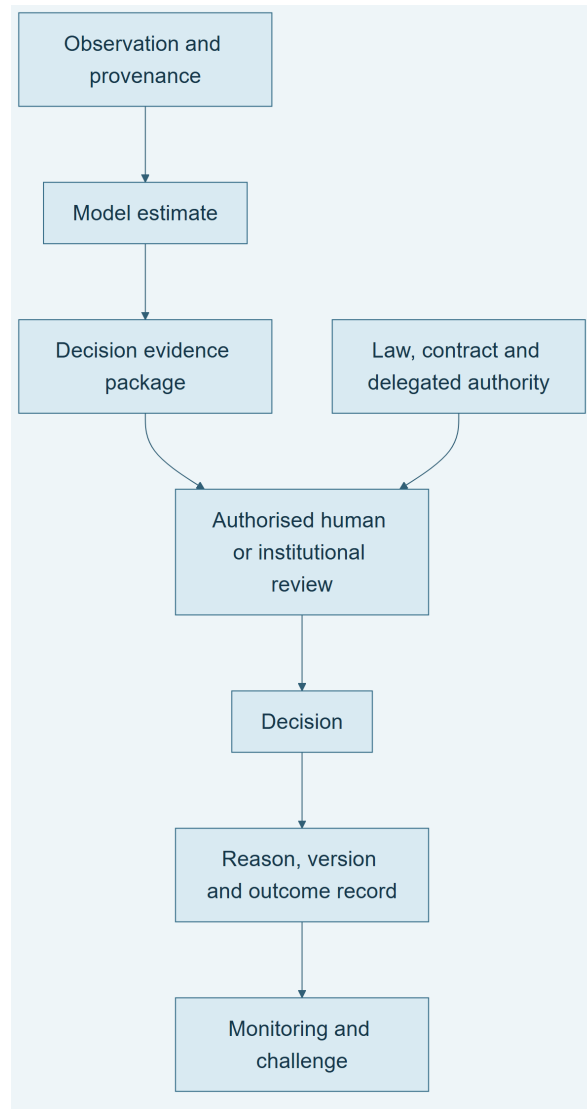


Figure 6.1: Evidence flows into accountable institutional decisions

For insurance, the governing starting point is the Kenyan framework. The Insurance Regulatory Authority’s guideline on insurance risk covers product design, pricing, underwriting, claims management, reserving and reinsurance [1]. SCRI maps its evidence and workflows to those existing responsibilities and to any later applicable instruments confirmed through legal review. Deployment proceeds through the appropriate institutional and regulatory approvals.

2. Privacy, AI risk and community legitimacy

Geotagged photographs, phone numbers, voices, property details and contributor histories may identify a person directly or indirectly. Encryption protects confidentiality in transit and at rest. Data minimisation, redaction, pseudonymisation, access control and retention rules address the additional identifiability created by a distinctive home, face, vehicle plate, device trail or precise location.

Kenya’s Data Protection Act requires lawful, fair and transparent processing; explicit purposes; necessity and data minimisation; accuracy; retention limits; safeguards for transfer; and mechanisms

through which data subjects can exercise their rights [2]. The General Regulations identify several high-risk activities directly relevant to SCRI, including large-scale processing, systematic monitoring of publicly accessible areas, combining datasets, innovative technologies, sensitive or vulnerable-group data and automated decisions with significant effects. Those conditions can require a data-protection impact assessment before processing [3].

SCRI's privacy design therefore follows the data lifecycle:

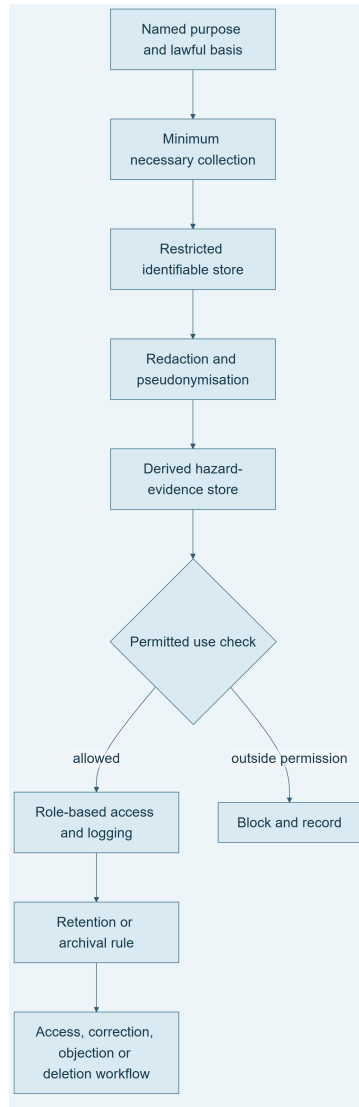


Figure 6.2: Privacy controls operate across the complete data lifecycle

Before pilot collection, the operator documents the purpose and lawful basis for every source class; controller and processor roles; data flows and transfers; retention; access roles; security controls; incident response; image-redaction rules; treatment of children and vulnerable persons; and routes for access, correction, withdrawal or deletion where applicable. The Office of the Data Protection Commissioner's DPIA guidance supplies the relevant Kenyan process [4]. Proportionality guides both emergency value and privacy protection.

Community legitimacy is a product requirement. Reporting channels include low-bandwidth and assisted options; contribution terms explain whether data may be used for warnings, insurance ana-

lytics, research or claims. Those purposes are separated. Any contributor payment follows a defined programme with clear terms. Reporting gaps are monitored by geography, language, gender, disability, connectivity and other relevant dimensions. Sparse-reporting areas receive wider uncertainty and targeted outreach.

Machine perception is consequential because its outputs can change hazard state and claims priority. Explainability therefore runs end to end: source image, model version, output mask, georeferencing, state update, exposure intersection, policy transform and human decision. Layer separation, common identifiers and versioned lineage make that traceability operational.

SCRI maintains a model inventory and one model card for each material component. Each card states intended and prohibited uses, owner, training and validation data, performance by relevant slice, calibration, uncertainty, dependencies, human-review point, monitoring, rollback and retirement. The NIST AI Risk Management Framework is comparative rather than binding in Kenya, but its govern-map-measure-manage structure is useful for organising continuous AI risk management [5].

3. From shadow mode to bounded production

The first deployment is a focused test of whether better catastrophe intelligence improves one insurer workflow with acceptable customer and community outcomes. Flood and drought are valuable contrasting pilots because one can evolve rapidly through a directed basin network while the other persists across biophysical and livelihood indicators. Location selection follows evidence and readiness: hazard relevance, authoritative data, exposure quality, claims history, institutional mandate, community participation, connectivity, response pathway and validation feasibility.

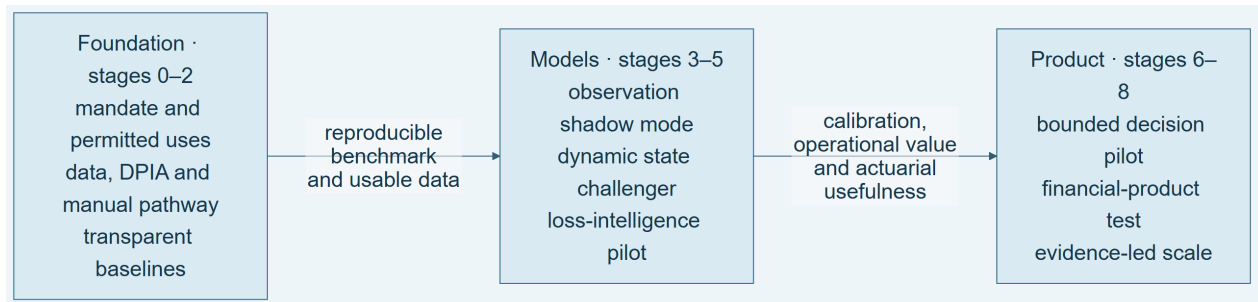


Figure 6.3: Research-to-operation stage gates

3.1. Stage 0 - mandate and permitted uses

The insurer names the product owner, hazard and actuarial owners, decision owners and independent reviewers. It selects one use case, such as flood accumulation and claims readiness. The approved-use statement covers accumulation, claims readiness and event-loss nowcasting; it reserves coverage determination, in-force contract changes and public warnings for their established authorities.

Evidence to pass: approved problem statement, user workflow, authority map, harm assessment and measurable baseline.

Stop condition: no accountable decision owner or no lawful/contractual path from output to action.

3.2. Stage 1 — data, permissions and manual fallback

The team completes source agreements, the source-to-use register, privacy assessment, observation schema, identity and access design, retention and incident procedures. A manual channel allows authorised staff to receive and review critical evidence during outages.

Evidence to pass: data lineage tests, access tests, consent or notice materials, correction workflow and continuity exercise.

Stop condition: source rights are ambiguous, high privacy risk remains untreated, or remote communities cannot participate safely.

3.3. Stage 2 — transparent baselines

The team implements the hazard-specific baseline and a manual event-loss workflow using a frozen exposure snapshot and explicit policy terms. Historical replay respects event time, knowledge time and ingestion time. The baseline establishes the standard a challenger must beat.

Evidence to pass: reproducible replay, dimensional checks, signed actuarial definitions and documented limitations.

Stop condition: claims, exposure or hazard data cannot be reconciled sufficiently to define ground truth.

3.4. Stages 3 and 4 — observation and state challengers

Crowd and AI components run in shadow mode, where outputs remain advisory while the team tests duplicate attacks, recycled photographs, incorrect coordinates, stale gauges, darkness, rain, low connectivity and sparse-reporting areas. The dynamic state challenger is compared with the baseline on event and spatial holdouts.

Evidence to pass: calibrated probabilities, acceptable false-alarm and missed-event trade-offs, stable latency, documented subgroup performance and operational rollback.

Stop condition: improved average discrimination masks worse subgroup calibration; the challenger cannot outperform the baseline; or the system fails during degraded inputs.

3.5. Stage 5 — loss intelligence

The product intersects posterior hazard samples with exposure, vulnerability and policy terms. It reports a range, contributors to change and uncertainty decomposition. Outputs remain advisory.

Evidence to pass: historical-event loss capture, stable gross-to-net reconciliation, claims emergence performance, user comprehension and independent actuarial review.

Stop condition: exposure double-counting, unstable tail estimates, untraceable financial transforms or repeated user misinterpretation.

3.6. Stages 6 and 7 — bounded decisions and financial products

A live pilot may allow predefined low-risk actions such as claims staffing, policyholder welfare contact or reinsurance notification. Consequential actions retain human review and a recorded reason. Parametric or financing structures are tested only after trigger independence, continuity, basis risk, dispute routes, repayment mechanics and authority are established.

Evidence to pass: operational benefit relative to baseline, acceptable customer outcomes, no material control breach and independent approval.

Stop condition: the pilot encourages opportunistic underwriting, unapproved financial action, customer harm or evidence use outside its permitted purpose.

4. Validation, monitoring and safe degradation

Validation measures more than predictive accuracy. Before the pilot, owners set thresholds from decision costs and baseline performance rather than inventing universal numbers. The scorecard includes:

- probability calibration using reliability plots, Brier score or log score;
- event detection recall, false-alarm rate and lead time;
- footprint overlap and intensity/depth error;
- loss-distribution calibration and interval coverage;
- claims-count, claims-severity and triage performance;
- OEP/AEP and tail sensitivity where long-term models are in scope;
- performance and coverage by relevant community and portfolio groups;
- data age, missingness, source failure and correction rate;
- system latency, uptime, recovery-time and recovery-point performance;
- user override, complaint, redress and decision-usefulness outcomes;
- intervention uptake and avoided-loss evidence where claimed.

For a model m , a cost-sensitive operational validation score can summarise scenario-specific false-negative and false-positive consequences:

$$R(m) = \sum_{s \in \mathcal{S}} \pi_s [c_s^{FN} \Pr(FN \mid s, m) + c_s^{FP} \Pr(FP \mid s, m)], \quad (6.1)$$

where $\mathcal{S}$ is the predeclared scenario set, π_s is its evaluation weight, and the costs are owned by the relevant product and decision functions. Accuracy and calibration constraints remain visible alongside this score. A challenger m_c is eligible for promotion over baseline m_b when the evidence supports a material improvement δ at confidence level $1 - \alpha$:

$$\Pr(R(m_c) < R(m_b) - \delta \mid \mathcal{D}_{val}) \geq 1 - \alpha. \quad (6.2)$$

Appendix H derives Equations (6.1)-(6.2), connects them to expected decision loss, and expands calibration, drift, subgroup and safe-degradation tests. Promotion remains a multi-criterion institutional decision: the equations make the evidential threshold reproducible.

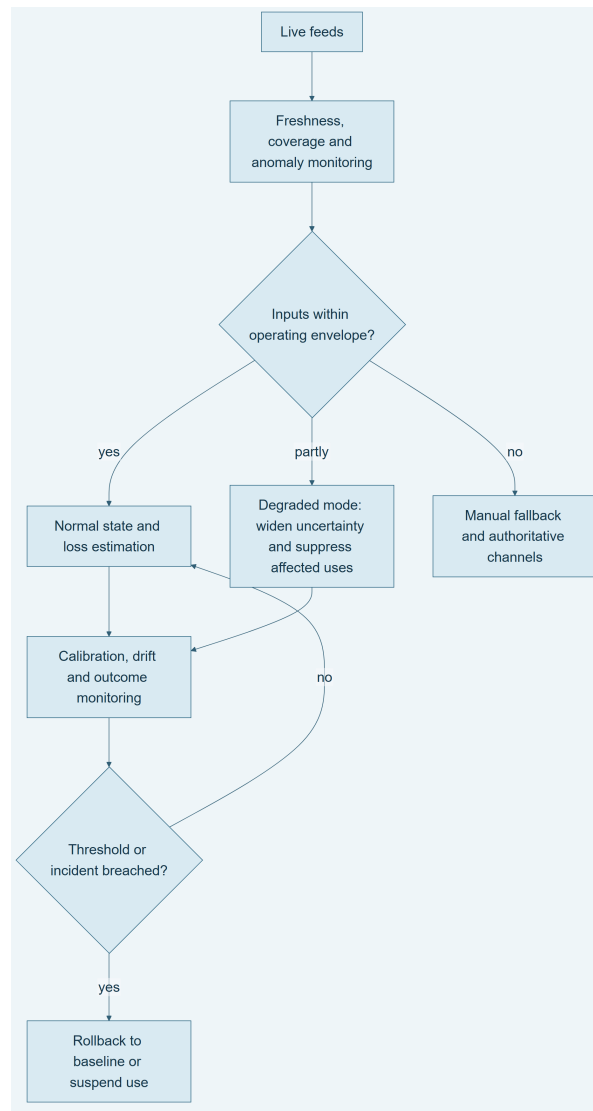


Figure 6.4: Degradation changes the product’s claims and permissions

Technology choices follow these requirements. The architecture needs durable event ingestion, spatial processing, versioned storage, model execution, access control, observability and offline recovery. Kubernetes is one possible implementation within a functional plan centred on portability, export rights, reproducible model artefacts, tested backups, vendor-exit procedures and the ability to run the transparent baseline when specialist services fail.

The definition of done is deliberately institutional. The product is ready for bounded use when a contributor can correct evidence, an actuary can reproduce a loss, an underwriter can explain a permitted action, a claims manager can override a triage recommendation, an independent validator can challenge the model and the system enters an honest degraded mode during failure. Scale follows demonstrated value and community legitimacy.

5. Independent review before rebuilding the production case

This sample edition has completed internal consistency and rendering checks. The next release gate is independent hydrological, actuarial, legal, community and insurance-operations review. Until those reviews are recorded, the edition remains a design demonstration of how the book integrates narrative, data, equations, maps, curves and interfaces.

Independent review begins with a reproducibility package: frozen manuscript, source and claim ledgers, event cut-offs, exposure schema, model cards, code, seeds, generated data, policy and treaty transforms, figure sources and a list of known limitations. Reviewers receive questions tied to their authority rather than a request to endorse the work in general.

Review lane	Questions that must be answered	Evidence required to close	Current sample status
Flood and hydrology	Are basin topology, depth, duration and gauge-failure assumptions physically credible?	Event replay, gauge and radar comparison, routing diagnostics	Pending independent review
Drought and livelihoods	Do indicators preserve NDMA meaning and livelihood sequencing?	Bulletin reconciliation, county and pastoral expertise	Pending independent review
Locust and agriculture	Are swarm identity, movement, control and crop-stage effects represented correctly?	Surveillance replay, intervention records, agronomic loss evidence	Pending independent review
Fire, landslide and heat	Do detection and susceptibility claims respect local mechanisms and sparse evidence?	Peril-specific holdouts, field and response review	Pending independent review
Actuarial modelling	Are exposure, vulnerability, terms, dependence, OEP/AEP and uncertainty coherent?	Reproducible calculations, claims back-test, gross-to-net reconciliation	Synthetic arithmetic checked; empirical calibration pending
Insurance operations	Does each screen improve a named workflow without creating conduct risk?	User testing, overrides, customer outcomes, incident scenarios	Narrative prototype only
Legal and data protection	Are purposes, authority, controller roles and consequential uses lawful?	Kenyan legal opinion, DPIA, contracts and notices	Pending independent review
Community and accessibility	Can contributors understand, contest and safely use the system?	Participatory review, language/accessibility tests, redress exercise	Pending independent review

Review lane	Questions that must be answered	Evidence required to close	Current sample status
Security and continuity	Does degraded mode work under peak-event failure?	Penetration test, outage exercise, recovery evidence	Design requirement only

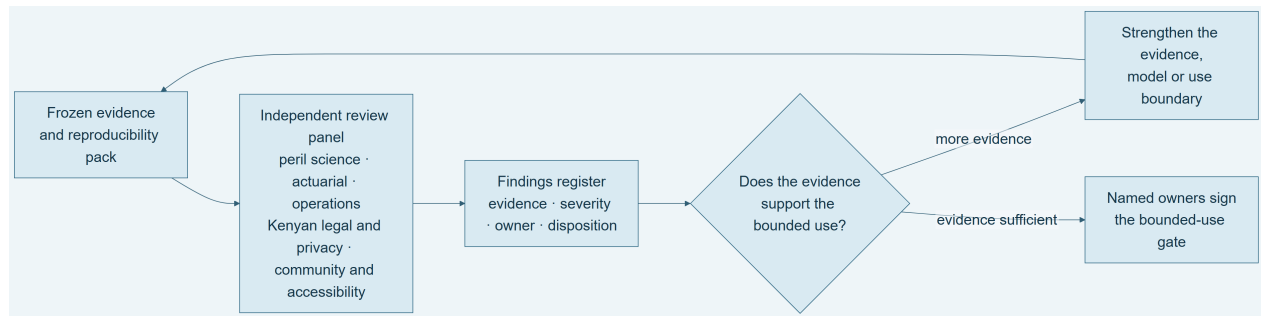


Figure 6.5: Independent review turns specialist challenge into release evidence

An independent reviewer works outside ownership of the component being validated and has freedom to record a negative conclusion. Each finding receives an identifier, severity, evidence, accountable owner, due date and disposition. “Accepted risk” requires the named decision owner and a stated reason; legal, actuarial and community-harm objections close through evidence from the responsible discipline.

Five release questions govern the next rebuild. First, can each event reconstruction be replayed at historical knowledge cut-offs using only the evidence then available? Second, do empirical Kenyan claims and exposure data support the vulnerability and tail assumptions? Third, can users reproduce every change from observation to net loss? Fourth, does the workflow remain useful when crowd, gauge, satellite or cloud inputs fail? Fifth, do affected communities and policyholders have meaningful information, correction and redress routes? A research edition can continue exploring open questions; production readiness follows affirmative evidence across all five.

The review sequence also strengthens the narrative. Technical depth makes the livelihood story more precise. Peril reviewers challenge the physical account; actuaries challenge the financial translation; operations teams test whether the interface supports real work; legal and community reviewers establish whether the proposed use is legitimate. Their findings and evidenced closures determine what the production white paper can confidently say.

References

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- [2] Republic of Kenya, *Data Protection Act*, Cap. 411C. [Online]. Available: <https://new.kenyalaw.org/akn/ke/act/2019/24/eng@2022-12-31>. Accessed: Aug. 28, 2026.
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[5] E. Tabassi, *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*, NIST AI 100-1. Gaithersburg, MD, USA: National Institute of Standards and Technology, 2023, doi: 10.6028/NIST.AI.100-1. [Online]. Available: <https://doi.org/10.6028/NIST.AI.100-1>. Accessed: Aug. 29, 2026.

Appendices

Spatial Catastrophe Risk Intelligence

Appendix A: Shared notation and definitions

The notation preserves the transformations established in the six parts. Observations inform hazard state; hazard and exposure generate damage; policy terms generate insured loss; and authorised actuarial, claims or contractual processes translate loss evidence into reserves or payouts.

Indices, space and time

Symbol	Definition
e	Catastrophe event under a stated event-definition rule
i	Exposure, asset, coverage or policy unit, stated by context
h	Hazard type
g	Spatial unit in the hazard model
$c(i)$	Exposure or vulnerability class of unit i
s	Observation-source class or individual source, stated by context
t	Event-state or valuation time
y	Annual catalogue or portfolio year
k	Claims-development period or other local sequence index
$\mathcal{N}_h(g)$	Hazard-specific neighbours of spatial unit g

Observation and state layer

Symbol	Definition and unit requirement
$Y_{s,g,t}$	Source-specific observation; its physical unit and geometry are stored with the record
$q_{s,g,t}$	Observable source-quality vector, not hazard probability
ρ_c	Dependence parameter or descriptor for duplication cluster c
$Z_{g,t}$	Latent or partially observed physical hazard state
$X_{g,t}$	Exogenous forcing and contextual covariates
$I_{e,h}(g)$	Hazard intensity for event e , hazard h and geometry g , with a peril-specific unit
$T^{ign}, T^{det}, T^{response}$	Ignition or onset, detection and effective response times where those concepts apply

Exposure, damage and finance layer

Symbol	Definition
V_i	Replacement value, sum insured, crop value or another explicitly named exposure measure
M_i	Secondary vulnerability modifiers for exposure i
$D_{h,c(i)}(\cdot)$	Stochastic damage-ratio function for hazard and exposure class
$L_{e,i}^{GU}$	Ground-up physical/economic loss before insurance terms
$L_{e,i}^{COV}$	Covered portion of ground-up loss after exclusions and coverage definition
$L_{e,i}^{INS}$	Gross insured loss after direct policy terms
L_e^{NET}	Insurer-retained event loss after reinsurance terms
d_i, ℓ_i, q_i	Deductible, applicable limit and insured/coinsurance share
$N_{h,g,y}$	Long-term event count for a stated hazard, region, year and event definition
S_y	Aggregate annual loss in year y on a stated gross or net basis

Canonical meanings

- **Observation confidence** describes support for an observation or model-derived feature under its source model.
- **Hazard probability** describes uncertainty about the physical state or event.
- **Damage ratio** describes physical/economic damage relative to a stated value measure.
- **Event-loss nowcast** is a versioned estimate of ultimate loss for an unfolding event.
- **Reserve** is an accounting estimate owned under the insurer's reserving basis.
- **Parametric payout** is calculated under a contractually specified trigger and formula.
- **Avoided loss** is a causal comparison requiring an intervention record and counterfactual.

Appendix B: Canonical systems architecture

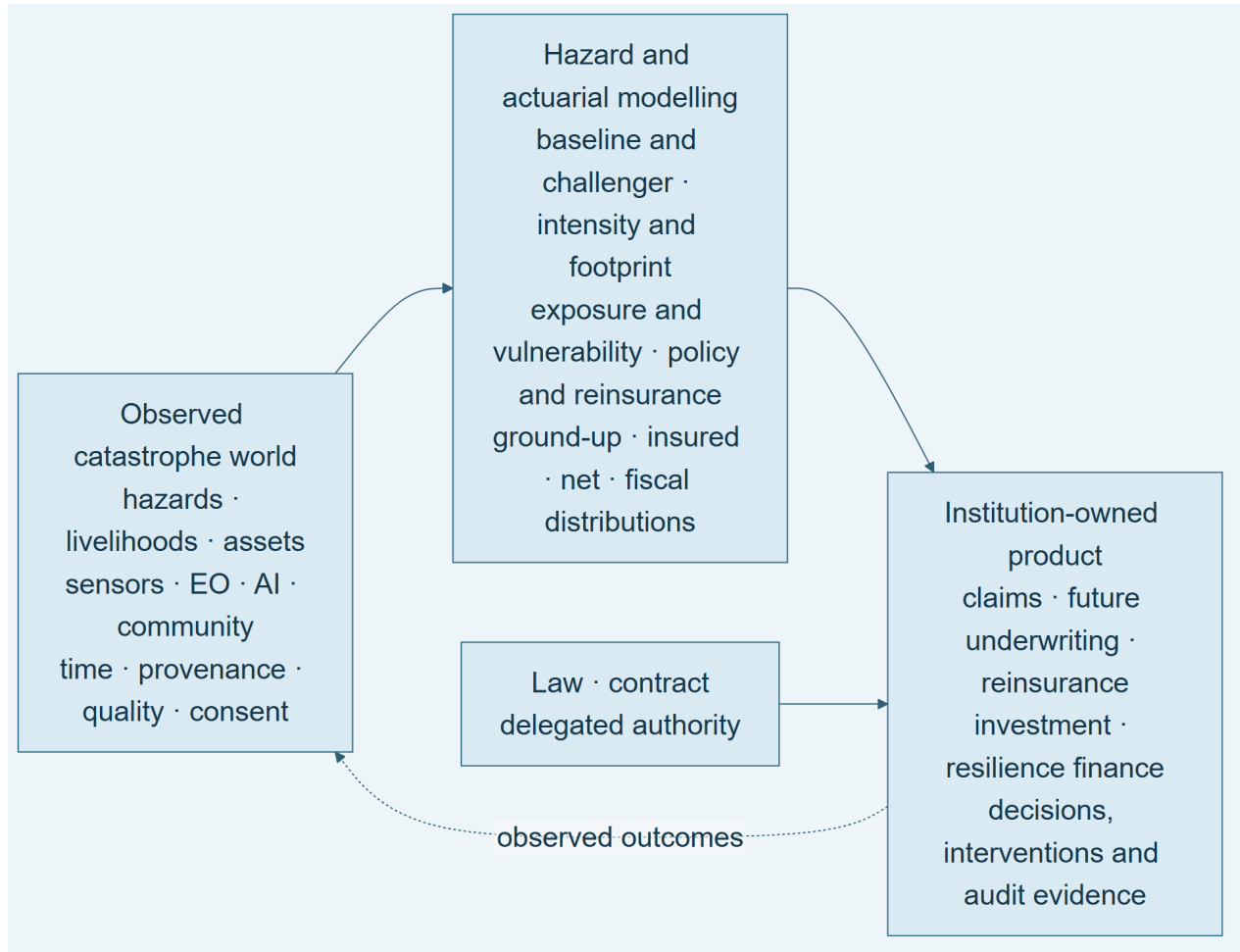


Figure B.1: Canonical SCRI architecture

Appendix C: Mathematical supplement to Part 1 - product thesis and value creation

C.1 Numbering convention

Main-text equations use the part number and their order of appearance, such as Equation (4.5). Appendix derivations use the form $(Ap.n)$, where p is the associated part. Thus $(A4.3)$ is the third appendix equation supporting Part 4. Inline symbol definitions remain inline; every displayed mathematical statement is numbered.

C.2 Ground-up loss: bounds, expectation and variance

Part 1 begins with the asset-level relationship in Equation (1.1). For this derivation, condition on hazard intensity $I_i = I_{e,h}(g_i)$ and vulnerability modifiers M_i . Write the random damage ratio as $D_i = D_{h,c(i)}(I_i, M_i)$. Then:

$$L_{e,i}^{GU} = V_i D_i, \quad 0 \leq D_i \leq 1. \quad (\text{A1.1})$$

If $V_i \geq 0$, multiplication preserves the bounds, which proves:

$$0 \leq L_{e,i}^{GU} \leq V_i. \quad (\text{A1.2})$$

The upper bound changes when the value measure includes business interruption, debris removal or another loss component that can exceed physical replacement value; the value basis and admissible damage-ratio range must then be restated. Under the bounded physical-damage definition, conditional expectation follows from linearity because V_i is fixed at the valuation snapshot:

$$\mathbb{E}[L_{e,i}^{GU} \mid I_i, M_i] = V_i \mathbb{E}[D_i \mid I_i, M_i]. \quad (\text{A1.3})$$

Variance scales with the square of value:

$$\text{Var}(L_{e,i}^{GU} \mid I_i, M_i) = V_i^2 \text{Var}(D_i \mid I_i, M_i). \quad (\text{A1.4})$$

Equations (A1.3)-(A1.4) show why exposure error matters multiplicatively. A 10 percent upward value error raises the conditional mean by 10 percent and, holding the damage-ratio distribution fixed, raises the conditional variance by 21 percent.

For a portfolio of n assets, $L_e^{GU} = \sum_i L_{e,i}^{GU}$. The variance expansion is:

$$\text{Var}(L_e^{GU}) = \sum_{i=1}^n \text{Var}(L_{e,i}^{GU}) + 2 \sum_{i < j} \text{Cov}(L_{e,i}^{GU}, L_{e,j}^{GU}). \quad (\text{A1.5})$$

The covariance term proves why catastrophe accumulation cannot be represented by adding independent asset variances. Shared depth, wind, fire perimeter, drought regime, construction practice and response conditions create dependence.

Dimensional check. D_i is dimensionless and V_i is measured in currency or another declared value unit. Therefore $L_{e,i}^{GU}$ has the same unit as V_i . Variance has squared currency units, while standard deviation returns to currency units.

C.3 Avoided expected loss as a causal contrast

Let $L_e(a)$ denote the potential loss for event e under intervention a , and let a_0 denote the reference response. Equation (1.2) is equivalently:

$$\Delta EL(a) = \mathbb{E}[L_e(a_0) - L_e(a)]. \quad (\text{A1.6})$$

The equality between Equation (1.2) and Equation (A1.6) follows directly from linearity of expectation:

$$\mathbb{E}[L_e(a_0) - L_e(a)] = \mathbb{E}[L_e(a_0)] - \mathbb{E}[L_e(a)]. \quad (\text{A1.7})$$

Observed data reveal only one potential outcome for each event. Identification from non-random operational data therefore requires a defensible version of four conditions:

1. **Consistency:** the recorded response corresponds to the defined intervention.
2. **Conditional exchangeability:** after conditioning on measured confounders X , response assignment is independent of the potential losses.
3. **Positivity:** each relevant event type has a non-zero probability of receiving the compared responses.
4. **Stable treatment definition:** the intervention does not conceal materially different response intensities, and spillovers are represented where one intervention changes neighbouring losses.

Under consistency, conditional exchangeability and positivity, the adjustment identity is:

$$\mathbb{E}[L_e(a)] = \int \mathbb{E}[L_e \mid A = a, X = x], dF_X(x). \quad (\text{A1.8})$$

Equation (A1.8) is obtained by the law of total expectation, followed by exchangeability to replace the unobserved conditional potential-outcome mean with the observed conditional mean. This is an identification result, not a guarantee that the conditions hold. The manuscript therefore treats avoided loss as an estimated range with stated assumptions.

C.4 Value of information and value of action

Let $U(a, Z)$ be the utility of action a when catastrophe state Z occurs, and let Y be new information. Before observing Y , the best expected utility is $\max_a \mathbb{E}[U(a, Z)]$. With Y , the action may depend on the observation. The expected value of sample information is:

$$EVSI = \mathbb{E}_Y \left[\max_a \mathbb{E}[U(a, Z) \mid Y] \right] - \max_a \mathbb{E}[U(a, Z)]. \quad (\text{A1.9})$$

To prove $EVSI \geq 0$, choose the pre-information optimal action a^* after every possible observation. Because the conditional maximum is at least the value of this feasible constant rule:

$$\mathbb{E}_Y \left[\max_a \mathbb{E}[U(a, Z) \mid Y] \right] \geq \mathbb{E}_Y [\mathbb{E}[U(a^*, Z) \mid Y]] = \mathbb{E}[U(a^*, Z)]. \quad (\text{A1.10})$$

The last equality is the tower property. Information therefore has weakly non-negative decision value when it is free and the decision-maker may ignore it. Operational cost, delay and harmful action can make the **net** value negative, which is why SCRI measures value after implementation cost and actual response behaviour.

Appendix D: Mathematical supplement to Part 2 - evidence, dependence and reliability

D.1 Bayesian evidence update

Let Z_t be the current hazard state and $Y_{1:C_t,t}$ the set of evidence clusters available at time t . Bayes' rule gives:

$$p(Z_t | Y_{1:C_t,t}) = \frac{p(Y_{1:C_t,t} | Z_t)p(Z_t)}{\int p(Y_{1:C_t,t} | z)p(z)dz}. \quad (\text{A2.1})$$

The denominator does not depend on the candidate value of Z_t , so the posterior is proportional to likelihood times prior. If evidence clusters are conditionally independent given Z_t and their quality vectors, the joint likelihood factorises:

$$p(Y_{1:C_t,t} | Z_t, q_{1:C_t,t}) = \prod_{c=1}^{C_t} p(Y_{c,t} | Z_t, q_{c,t}). \quad (\text{A2.2})$$

Substituting the full predictive distribution $p(Z_t | Y_{1:t-1}, X_t)$ for the prior in Equation (A2.1) and using Equation (A2.2) yields Equation (2.6). The factorisation applies to independent evidence origins, not to every forwarded copy or adjacent video frame.

D.2 Dependence and effective information

Suppose a duplication cluster contains n_c reports with common pairwise correlation ρ_c and equal marginal variance σ^2 . The variance of their mean is:

$$\text{Var}(\bar{Y}_c) = \frac{\sigma^2}{n_c} [1 + (n_c - 1)\rho_c]. \quad (\text{A2.3})$$

Proof follows by expanding the variance of a sum: n_c variance terms contribute $n_c\sigma^2$, while $n_c(n_c - 1)$ ordered covariance terms contribute $n_c(n_c - 1)\rho_c\sigma^2$, after which division by n_c^2 gives Equation (A2.3).

Equating Equation (A2.3) to the variance σ^2/n_c^{eff} of an independent sample defines the effective sample size:

$$n_c^{eff} = \frac{n_c}{1 + (n_c - 1)\rho_c}. \quad (\text{A2.4})$$

When $\rho_c = 0$, all reports contribute independent information and $n_c^{eff} = n_c$. When $\rho_c = 1$, identical copies have $n_c^{eff} = 1$. This derivation supplies a transparent diagnostic; the production likelihood can model dependence more directly through a common latent origin.

D.3 Source reliability as a sequential posterior

For a narrowly defined report class, let r_s be the probability that source s produces a verified report under stated conditions. A Beta prior and Bernoulli verification outcomes provide a simple baseline:

$$r_s \sim \text{Beta}(a_0, b_0), \quad K_s | r_s \sim \text{Binomial}(n_s, r_s). \quad (\text{A2.5})$$

Multiplying the Beta density by the Binomial likelihood gives the conjugate posterior:

$$r_s | K_s, n_s \sim \text{Beta}(a_0 + K_s, b_0 + n_s - K_s). \quad (\text{A2.6})$$

The posterior mean is:

$$\mathbb{E}[r_s \mid K_s, n_s] = \frac{a_0 + K_s}{a_0 + b_0 + n_s}. \quad (\text{A2.7})$$

Equation (A2.7) shrinks sparse histories toward the prior rather than assigning extreme scores to first-time reporters. In SCRI, reliability is contextual and capped; it complements present-report quality and cannot substitute for coverage monitoring or independent corroboration.

D.4 Geolocation uncertainty

Let G be the unknown true location and G^{obs} the reported coordinate. The evidential likelihood must average over plausible true locations:

$$p(Y \mid Z, G^{obs}) = \int p(Y \mid Z, G = g) p(g \mid G^{obs}), dg. \quad (\text{A2.8})$$

Equation (A2.8) follows from marginalisation over the latent location. A narrow accuracy radius concentrates the integral near the reported point; a broad or displaced distribution spreads evidential weight across neighbouring cells. This prevents false precision when a report names the correct event but supplies an inaccurate coordinate.

Dimensional check. Probabilities and reliability parameters are dimensionless. A likelihood for a continuous measured variable is a density and therefore carries reciprocal measurement units; integration over the observation or latent location restores a dimensionless probability. Geometry retains the coordinate reference system and accuracy unit from the observation contract.

D.5 Calibration of perception scores

If a perception model emits score $S \in [0, 1]$, calibration asks whether the empirical event frequency matches the score:

$$\Pr(Y = 1 \mid S = s) = s. \quad (\text{A2.9})$$

A calibration mapping g estimated on held-out Kenyan data produces $\hat{p} = g(S)$. The calibrated score remains a probability about the perception label, not the full hazard-state probability. Equation (2.6) performs the later evidence fusion.

D.6 Contextual and partially pooled reliability

Reliability varies by task and operating condition. For verification outcome $K_j \in \{0, 1\}$ from source $s(j)$, hazard $h(j)$ and county $c(j)$, a hierarchical logistic model is:

$$K_j \sim \text{Bernoulli}(r_j), \quad \text{logit}(r_j) = \beta_0 + u_{s(j)} + v_{h(j)} + w_{c(j)} + \beta^\top \mathbf{x}_j, \quad (\text{A2.10})$$

with $u_s \sim N(0, \sigma_s^2)$ and analogous priors for hazard and county effects. The condition vector $\mathbf{x}_j$ can include darkness, smoke, rain, device type, observation age, geolocation quality and verification

method. Partial pooling follows because each source effect is estimated jointly with a population distribution: sparse sources shrink toward the relevant contextual mean, while repeated verified evidence supports source-specific learning. Reliability is attached to a source-task-context combination rather than treated as a permanent character judgement.

D.7 Missing-not-at-random reporting

Let $R_{s,g,t} = 1$ when a source has an opportunity and submits a report. A reporting model can be written:

$$\Pr(R_{s,g,t} = 1 \mid Z_{g,t}, A_{g,t}, C_{g,t}, Q_s) = \text{logit}^{-1}(\gamma_0 + \gamma_Z Z_{g,t} + \gamma_A A_{g,t} + \gamma_C C_{g,t} + \gamma_Q Q_s), \quad (\text{A2.11})$$

where A is access and C connectivity. The observed-data likelihood sums over unobserved states and reporting opportunities:

$$p(Y^{obs}, R \mid Z) = p(R \mid Z, A, C, Q) [p(Y^{obs} \mid Z, R = 1)]^R. \quad (\text{A2.12})$$

Equation (A2.12) shows why $R = 0$ is informative only through the reporting model. If flood damage destroys connectivity, silence can become more likely as hazard intensity rises. Sensitivity analysis varies the unidentifiable parts of this mechanism and compares them with representative field or authoritative observations.

D.8 Common evidence origins

Let O_c be a latent original observation and $Y_{c1}, \dots, Y_{cn_c}$ its visible copies, frames or derived detections. The cluster likelihood is:

$$p(Y_{c1:n_c} \mid Z) = \int p(O_c \mid Z) \prod_{j=1}^{n_c} p(Y_{cj} \mid O_c, \psi_{cj}) dO_c. \quad (\text{A2.13})$$

Conditional independence applies after the shared origin is introduced. If all copies reproduce the origin nearly deterministically, they sharpen knowledge about O_c but provide little additional evidence about Z . Independent observations have different origin variables and contribute separate likelihood factors. The same construction applies to adjacent video frames, tiles from one satellite acquisition and repeated inference over one source image.

D.9 Conformal uncertainty for perception outputs

For exchangeable calibration examples with nonconformity scores $a_1, \dots, a_n$, define the finite-sample quantile:

$$q_{1-\alpha} = \text{Quantile}_{\lceil (n+1)(1-\alpha) \rceil / n} \{a_1, \dots, a_n\}. \quad (\text{A2.14})$$

The prediction set for new input x is:

$$\Gamma_\alpha(x) = \{y : a(x, y) \leq q_{1-\alpha}\}. \quad (\text{A2.15})$$

Under exchangeability of the calibration and test examples, their score ranks are uniform, giving the marginal coverage result:

$$\Pr\{Y_{n+1} \in \Gamma_\alpha(X_{n+1})\} \geq 1 - \alpha. \quad (\text{A2.16})$$

Coverage is evaluated by peril, county, season, device and visibility condition because marginal coverage can conceal weak subgroups. Spatially or temporally correlated event data require event-blocked calibration or methods whose assumptions match that dependence. Conformal sets express perception uncertainty; the hazard engine subsequently combines them with other evidence.

Appendix E: Mathematical supplement to Part 3 - dynamic hazard state

E.1 Prediction-update recursion

Let Z_t be a Markov state and $Y_{1:t-1}$ the evidence available before the current observation. The predictive distribution is obtained by marginalising the previous state:

$$p(Z_t | Y_{1:t-1}) = \int p(Z_t | Z_{t-1}, X_t) p(Z_{t-1} | Y_{1:t-1}) dZ_{t-1}. \quad (\text{A3.1})$$

The derivation uses the law of total probability and the Markov property $Z_t \perp Y_{1:t-1} | Z_{t-1}, X_t$. On receipt of Y_t , Bayes' rule gives the filtering update:

$$p(Z_t | Y_{1:t}) = \frac{p(Y_t | Z_t) p(Z_t | Y_{1:t-1})}{\int p(Y_t | z) p(z | Y_{1:t-1}) dz}. \quad (\text{A3.2})$$

Equations (A3.1)-(A3.2) derive the posterior in Equation (3.1). Kalman, ensemble, particle, variational and discrete-state filters are alternative numerical realisations of this recursion.

E.2 Spatial dependency graph

For grid cells or hazard units $g = 1, \dots, G$, a local Markov assumption yields the transition factorisation:

$$p(Z_{1:G,t} | Z_{1:G,t-1}, X_t) = \prod_{g=1}^G p(Z_{g,t} | Z_{g,t-1}, Z_{\mathcal{N}_h(g),t-1}, X_{g,t}). \quad (\text{A3.3})$$

Equation (A3.3) follows from the assumption that each current cell is conditionally independent of non-neighbouring prior cells once its own prior state, hazard-specific neighbours and forcing variables are known. For a river network, $\mathcal{N}_h(g)$ is directed; for fire, it is anisotropic under wind and slope; for locusts, it changes with movement conditions. The factorisation is therefore hazard-specific rather than a universal square-grid rule.

E.3 Joint flood state and uncertainty propagation

Equation (3.3) is a state vector rather than five independent estimates. Let:

$$\mathbf{Z}_{g,t}^{flood} = (p_{g,t}^{wet}, d_{g,t}, v_{g,t}, \tau_{g,t}^{on}, \tau_{g,t}^{rec})^\top. \quad (\text{A3.4})$$

Its posterior covariance matrix is:

$$\Sigma_{g,t} = \mathbb{E}[(\mathbf{Z}_{g,t} - \mu_{g,t})(\mathbf{Z}_{g,t} - \mu_{g,t})^\top \mid Y_{1:t}]. \quad (\text{A3.5})$$

Off-diagonal terms preserve relationships such as deeper water tending to recede later. If an asset-loss function is $f(\mathbf{Z})$, a first-order uncertainty approximation is:

$$\text{Var}[f(\mathbf{Z}) \mid Y_{1:t}] \approx \nabla f(\mu)^\top \Sigma \nabla f(\mu). \quad (\text{A3.6})$$

Equation (A3.6) is the multivariate delta method, obtained by a first-order Taylor expansion of f around μ . Simulation is preferred where the loss function is highly nonlinear or contains policy thresholds.

E.4 Occurrence, detection and false reports

Let O_t denote true event occurrence and R_t a report or detector alert. The probability of an alert is:

$$\Pr(R_t = 1) = \Pr(R_t = 1 \mid O_t = 1) \Pr(O_t = 1) + \Pr(R_t = 1 \mid O_t = 0) \Pr(O_t = 0). \quad (\text{A3.7})$$

This is the law of total probability. The first conditional probability is detection sensitivity; the second is the false-positive rate. Bayes' rule then gives the probability that an event occurred after an alert:

$$\Pr(O_t = 1 \mid R_t = 1) = \frac{\Pr(R_t = 1 \mid O_t = 1) \Pr(O_t = 1)}{\Pr(R_t = 1)}. \quad (\text{A3.8})$$

Equations (A3.7)-(A3.8) prove that detector confidence or sensitivity is not occurrence probability. Base rates and false alerts matter.

E.5 Detection and response latency

Equation (3.4) implies three non-negative operational intervals:

$$\tau^{det} = T^{det} - T^{ign}, \quad \tau^{dispatch} = T^{dispatch} - T^{det}, \quad \tau^{response} = T^{response} - T^{dispatch}. \quad (\text{A3.9})$$

Adding the intervals telescopes, proving total ignition-to-response latency:

$$T^{response} - T^{ign} = \tau^{det} + \tau^{dispatch} + \tau^{response}. \quad (\text{A3.10})$$

This decomposition separates value created by earlier detection from delay in decision, mobilisation or access.

E.6 Explicit duration for slow-onset states

For a semi-Markov drought regime k , let D_k be its duration. The probability of leaving at duration d is the discrete hazard:

$$h_k(d) = \Pr(D_k = d \mid D_k \geq d) = \frac{\Pr(D_k = d)}{\Pr(D_k \geq d)}. \quad (\text{A3.11})$$

The survival recursion follows from conditioning on survival through the previous period:

$$\Pr(D_k > d) = \prod_{j=1}^d [1 - h_k(j)]. \quad (\text{A3.12})$$

Equation (A3.12) allows persistence that differs from the geometric duration imposed by a conventional HMM.

E.7 Out-of-sequence evidence and fixed-lag smoothing

Suppose an observation Y_τ with event time $\tau < t$ arrives at knowledge time t . The corrected current posterior is obtained by inserting its likelihood at the event-time state and integrating the intervening trajectory:

$$p(Z_t \mid Y_{1:t}^{old}, Y_\tau^{late}) \propto \int p(Y_\tau^{late} \mid Z_\tau) p(Z_{\tau:t}, Y_{1:t}^{old}) dZ_{\tau:t-1}. \quad (\text{A3.13})$$

A fixed-lag smoother retains particles or sufficient state over $t - L : t$ so observations within lag L can be replayed operationally. Older evidence enters a versioned retrospective reconstruction. This preserves the difference between what was knowable at decision time and what was learned later.

E.8 Change of spatial support

Let $Z_t(s)$ be a continuous latent field and observation j represent spatial support A_j , such as a pixel, road segment, catchment or livelihood zone. A linear observation operator is:

$$Y_{j,t} = \frac{1}{|A_j|} \int_{A_j} h_j\{Z_t(s)\} ds + \varepsilon_{j,t}. \quad (\text{A3.14})$$

For point support, the integral becomes evaluation at a location; for a line, integration uses length; for a basin, area or hydrological weighting replaces a simple average. The operator h_j maps state variables to the quantity observed. Discretising the field gives $\mathbf{Y}_t = \mathbf{H}_t \mathbf{Z}_t + \varepsilon_t$, making the support transformation explicit and reproducible.

E.9 Controlled transition and value of action

Where action A_t changes the process, the predictive step is:

$$p(Z_t | Y_{1:t-1}, A_{1:t-1}) = \int p(Z_t | Z_{t-1}, X_t, A_{t-1}) p(Z_{t-1} | Y_{1:t-1}, A_{1:t-2}) dZ_{t-1}. \quad (\text{A3.15})$$

The contrast between action a and baseline a_0 propagates through the same transition model:

$$\Delta_t(a, a_0) = \mathbb{E}[f(Z_t) | do(A_{t-1} = a)] - \mathbb{E}[f(Z_t) | do(A_{t-1} = a_0)]. \quad (\text{A3.16})$$

The *do* notation signals that observed association alone is insufficient. Operational trials, process models or a defensible causal design identify the intervention effect. Fire suppression, pest control, drainage operation and water allocation each require their own action mechanism.

E.10 Posterior samples and the loss transform

Draw weighted state samples $Z_t^{(m)} \sim p(Z_t | Y_{1:t})$, exposure samples $E^{(m)}$ and vulnerability parameters $\theta_V^{(m)}$. The Monte Carlo loss sample is:

$$L_t^{(m)} = \mathcal{L}(Z_t^{(m)}, E^{(m)}, \theta_V^{(m)}, F^{(m)}), \quad m = 1, \dots, M. \quad (\text{A3.17})$$

For integrable loss, the sample mean converges by the law of large numbers:

$$\frac{1}{M} \sum_{m=1}^M L_t^{(m)} \xrightarrow[M \rightarrow \infty]{a.s.} \mathbb{E}[L_t | Y_{1:t}]. \quad (\text{A3.18})$$

Applying deductibles, limits, triggers and network thresholds to every sample preserves nonlinearities that a mean-state approximation can lose.

E.11 Nonstationary county effects and model stacking

A dynamic hierarchical parameter can evolve as:

$$\theta_{c,t} = \mu_t + u_c + \delta_{c,t}, \quad \delta_{c,t} = \rho \delta_{c,t-1} + \eta_{c,t}, \quad (\text{A3.19})$$

where μ_t captures shared temporal change, u_c persistent county variation and $\delta_{c,t}$ local drift. Climate, land-use, urbanisation and reporting covariates enter the mean. Structural-break challengers are activated when residual diagnostics show that gradual drift is insufficient.

For candidate predictive densities $p_k(\tilde{y} | y)$, stacking selects simplex weights:

$$\mathbf{w}^* = \arg \max_{\mathbf{w}: w_k \geq 0, \sum_k w_k = 1} \sum_{i=1}^n \log \left[\sum_{k=1}^K w_k p_k(y_i | y_{-i}) \right]. \quad (\text{A3.20})$$

This optimises held-out predictive performance rather than choosing a model by novelty. Event, spatial and temporal blocks replace pointwise leave-one-out terms where dependence requires them.

Dimensional check. Equation (A3.4) deliberately contains components with different units, so it is a typed state vector rather than a quantity to be added componentwise. Depth is a length, velocity is length per time, onset and recession are times, and inundation probability is dimensionless. Covariance entries in Equation (A3.5) carry the product of the corresponding component units.

Appendix F: Mathematical supplement to Part 4 - actuarial loss

F.1 Damage moments with uncertain hazard

Conditional on intensity and modifiers, Equation (4.1) has the moments derived in Appendix C. When intensity is uncertain, the tower property gives:

$$\mathbb{E}[L_{e,i}^{GU} \mid Y] = V_i \mathbb{E}_{I|Y}[\mathbb{E}[D_i \mid I, M_i]]. \quad (\text{A4.1})$$

The law of total variance separates vulnerability randomness from hazard-state uncertainty:

$$\text{Var}(L_{e,i}^{GU} \mid Y) = V_i^2 \mathbb{E}_{I|Y}[\text{Var}(D_i \mid I, M_i)] + V_i^2 \text{Var}_{I|Y}(\mathbb{E}[D_i \mid I, M_i]). \quad (\text{A4.2})$$

The first term is conditional damage uncertainty; the second is uncertainty about hazard intensity. This decomposition supports the product's explanation of why a loss range changed.

F.2 Piecewise policy transform

Let $x = L_{e,i}^{COV}$. Equation (4.2) is equivalent to:

$$L_{e,i}^{INS}(x) = q_i \begin{cases} 0, & x \leq d_i, \\ x - d_i, & d_i < x < d_i + \ell_i, \\ \ell_i, & x \geq d_i + \ell_i. \end{cases} \quad (\text{A4.3})$$

Proof follows by evaluating the nested maximum and minimum in each region. Below the deductible, $x - d_i \leq 0$ and the maximum is zero. Between deductible and exhaustion, $0 < x - d_i < \ell_i$ and the inner value passes through. Above exhaustion, the minimum fixes the result at ℓ_i . Consequently:

$$0 \leq L_{e,i}^{INS} \leq q_i \ell_i. \quad (\text{A4.4})$$

The result is monotone non-decreasing in covered loss, with slope 0, q_i and 0 across the three regions. Actual wording can introduce additional layers and aggregation rules, but the same sample-by-sample logic applies.

F.3 Negative Binomial frequency and exposure offset

Use the mean-dispersion parameterisation in which Equation (4.3) satisfies:

$$\mathbb{E}[N_{h,g,y}] = \mu_{h,g,y}, \quad \text{Var}(N_{h,g,y}) = \mu_{h,g,y} + \frac{\mu_{h,g,y}^2}{\phi_h}. \quad (\text{A4.5})$$

One derivation uses a Poisson-Gamma mixture. Let $N \mid \Lambda \sim \text{Poisson}(\Lambda)$, with $\mathbb{E}[\Lambda] = \mu$ and $\text{Var}(\Lambda) = \mu^2/\phi$. Then:

$$\text{Var}(N) = \mathbb{E}[\text{Var}(N \mid \Lambda)] + \text{Var}(\mathbb{E}[N \mid \Lambda]) = \mu + \frac{\mu^2}{\phi}. \quad (\text{A4.6})$$

Exponentiating Equation (4.4) gives:

$$\mu_{h,g,y} = E_{h,g,y} \exp(\eta_{h,g,y}) = E_{h,g,y} \lambda_{h,g,y}. \quad (\text{A4.7})$$

Thus $\lambda = \exp(\eta)$ is the unit-exposure event rate and μ is already the expected cell count. Equation (A4.7) proves why multiplying by exposure a second time would double-count it.

F.4 Compound annual loss moments

Let annual event count N be independent of identically distributed event severities $L_1, L_2, \dots$, with mean m_L and variance v_L . From Equation (4.5):

$$S = \sum_{e=1}^N L_e. \quad (\text{A4.8})$$

Conditioning on N gives $\mathbb{E}[S \mid N] = Nm_L$ and $\text{Var}(S \mid N) = Nv_L$. The tower property and total-variance identity therefore yield:

$$\mathbb{E}[S] = \mathbb{E}[N]m_L, \quad (\text{A4.9})$$

$$\text{Var}(S) = \mathbb{E}[N]v_L + \text{Var}(N)m_L^2. \quad (\text{A4.10})$$

Event clustering, shared climate regimes and cross-peril dependence invalidate the simple independence assumption. SCRI then simulates the joint annual process, while Equations (A4.9)-(A4.10) remain valuable reconciliation baselines.

F.5 Occurrence exceedance probability

Let $M = \max(L_1, \dots, L_N)$ and suppose $N \sim \text{Poisson}(\lambda)$. Conditional on $N = n$, $\Pr(M \leq x \mid N = n) = F_L(x)^n$. Averaging over the Poisson count gives:

$$\Pr(M \leq x) = \sum_{n=0}^{\infty} F_L(x)^n \frac{e^{-\lambda} \lambda^n}{n!} = \exp\{-\lambda[1 - F_L(x)]\}. \quad (\text{A4.11})$$

The occurrence exceedance probability is therefore:

$$OEP(x) = 1 - \exp\{-\lambda[1 - F_L(x)]\}. \quad (\text{A4.12})$$

The second equality in Equation (A4.11) uses the exponential power series.

F.6 Aggregate exceedance probability

For a compound Poisson sum, let $M_L(t) = \mathbb{E}[e^{tL}]$ be the severity moment-generating function where it exists. Conditioning on N and summing its Poisson distribution gives:

$$M_S(t) = \mathbb{E}[e^{tS}] = \exp\{\lambda[M_L(t) - 1]\}. \quad (\text{A4.13})$$

Equation (A4.13) characterises the annual aggregate distribution; numerical inversion or simulation yields:

$$AEP(x) = \Pr(S > x) = 1 - F_S(x). \quad (\text{A4.14})$$

OEP and AEP coincide only in special cases. AEP captures multiple events and is therefore generally the relevant annual aggregate measure for capital and reinsurance accumulation.

F.7 Pure premium

For exposure E_i , unit event rate λ_i and mean insured severity m_i , expected count is $\mu_i = E_i\lambda_i$. Applying Equation (A4.9) gives:

$$PP_i = \mathbb{E}[S_i] = E_i\lambda_i m_i = \mu_i m_i. \quad (\text{A4.15})$$

This derivation reconciles the rate and expected-count formulations and confirms that either $E_i\lambda_i$ or μ_i enters, not both.

F.8 Generalized Pareto tail quantile

For excess $X = L - u \mid L > u$ under Equation (4.6), the Generalized Pareto distribution for $\xi \neq 0$ has:

$$F_X(x) = 1 - \left(1 + \frac{\xi x}{\beta_u}\right)^{-1/\xi}, \quad x \geq 0, \quad 1 + \frac{\xi x}{\beta_u} > 0. \quad (\text{A4.16})$$

Set $F_X(x_p) = p$ and solve algebraically:

$$x_p = \frac{\beta_u}{\xi} [(1 - p)^{-\xi} - 1]. \quad (\text{A4.17})$$

As $\xi \rightarrow 0$, the limit is $x_p = -\beta_u \log(1 - p)$, obtained from the standard exponential limit. Tail estimates also require the probability of exceeding u and parameter uncertainty.

F.9 Claims emergence

Let $F_k = F_N(k \mid x_e)$. From Equation (4.10), conditional moments are:

$$\mathbb{E}[N_{e,k}^{rep} \mid N_e^{ult}] = N_e^{ult} F_k, \quad \text{Var}(N_{e,k}^{rep} \mid N_e^{ult}) = N_e^{ult} F_k (1 - F_k). \quad (\text{A4.18})$$

If F_k is known and positive, the chain-ladder-style count estimator is:

$$\widehat{N}_e^{ult} = \frac{N_{e,k}^{rep}}{F_k}. \quad (\text{A4.19})$$

Conditional unbiasedness follows immediately from Equation (A4.18): $\mathbb{E}[\widehat{N}_e^{ult} \mid N_e^{ult}] = N_e^{ult}$. In practice, uncertainty in F_k , event heterogeneity, reporting dependence and changing operations widen the prediction interval.

F.10 Intervention contrast

Equation (4.11) writes conditional mean loss as baseline mechanism plus intervention effect. For two response states a and a_0 at common intensity and covariates:

$$\mathbb{E}[L^{GU} \mid I, X, A = a, \tau_a] - \mathbb{E}[L^{GU} \mid I, X, A = a_0, \tau_{a_0}] = \delta_h(a, \tau_a) - \delta_h(a_0, \tau_{a_0}). \quad (\text{A4.20})$$

The baseline $m_h(I, X)$ cancels algebraically. Causal interpretation still requires the identification conditions in Appendix C.

F.11 Uncertain exposure, vulnerability and boundary masses

Let value V , hazard intensity I , class C and vulnerability parameter θ be uncertain. Total expectation gives:

$$\mathbb{E}[L^{GU} \mid Y] = \sum_c \iiint v d(i, c, \theta) p(v, i, \theta, C = c \mid Y) dv di d\theta. \quad (\text{A4.21})$$

Monte Carlo evaluation samples the joint distribution, preserving correlations between location, class, value and hazard. For damage ratio $D \in [0, 1]$ with no-damage mass π_0 , total-loss mass π_1 and continuous density f_B on $(0, 1)$:

$$p(D) = \pi_0 \delta_0(D) + \pi_1 \delta_1(D) + (1 - \pi_0 - \pi_1) f_B(D). \quad (\text{A4.22})$$

The conditional mean is:

$$\mathbb{E}[D] = \pi_1 + (1 - \pi_0 - \pi_1) \mathbb{E}_B[D]. \quad (\text{A4.23})$$

This structure represents undamaged and destroyed assets directly. Censored claims contribute tail probabilities to the likelihood; truncated samples divide the density by their observation probability. Treating either as an ordinary observed amount biases severity estimation.

F.12 Catastrophe-informed reserve posterior

Let U be ultimate event loss, H the catastrophe state and $C_{1:k}$ claims evidence. Bayes' rule gives Equation (4.12):

$$p(U \mid C_{1:k}, H) = \frac{p(C_{1:k} \mid U, \theta_C) p(U \mid H, E, V, F)}{\int p(C_{1:k} \mid u, \theta_C) p(u \mid H, E, V, F) du}. \quad (\text{A4.24})$$

A transparent Bornhuetter-Ferguson interpretation emerges at the mean level. If U_0 is the a priori ultimate and q_k is the expected reported proportion, the unreported expectation is $(1 - q_k)U_0$. With reported loss C_k :

$$\widehat{U}_k^{BF} = C_k + (1 - q_k)U_0. \quad (\text{A4.25})$$

The catastrophe model supplies a distribution for U_0 rather than a fixed number, while the claims likelihood reflects count, paid, incurred, closure, reopening and calendar effects. Posterior uncertainty narrows only as evidence becomes informative; it can widen when inflation, access failure or reporting delay introduces new uncertainty.

F.13 IFRS 17 cash-flow translation

For scenario m with probability weight w_m , future net claim and expense cash flow $CF_u^{(m)}$, and discount factor $D(t, u)$, the probability-weighted present value is:

$$PVFCF_t = \sum_{m=1}^M w_m \sum_{u>t} D(t, u) CF_u^{(m)}, \quad \sum_m w_m = 1. \quad (\text{A4.26})$$

Adding an entity-determined risk adjustment RA_t produces the simplified incurred-claims fulfilment-cash-flow expression in Equation (4.14). The scenario engine can supply cash-flow timing and non-financial uncertainty, while contract boundary, grouping, discount methodology, risk-adjustment method and presentation remain accounting-policy matters.

F.14 Economic capital and tail value at risk

For loss variable X with distribution function F_X , define the lower quantile:

$$VaR_\alpha(X) = \inf\{x : F_X(x) \geq \alpha\}. \quad (\text{A4.27})$$

For a continuous distribution, tail value at risk can be written as the average quantile above α :

$$TVaR_\alpha(X) = \frac{1}{1 - \alpha} \int_\alpha^1 VaR_u(X) du. \quad (\text{A4.28})$$

The identity with $\mathbb{E}[X \mid X > VaR_\alpha(X)]$ follows by the probability-integral transform under continuity. For discrete or mixed losses, Equation (A4.28) correctly handles probability mass at the quantile. Economic capital in Equation (4.16) subtracts expected loss when that expected loss is already allowed for in pricing, reserves or planning; the framework states the balance-sheet convention explicitly.

F.15 Euler allocation

Let $X(\mathbf{x}) = \sum_j x_j X_j$ and suppose risk measure ρ is differentiable and positively homogeneous of degree one: $\rho(c\mathbf{x}) = c\rho(\mathbf{x})$. Euler's homogeneous-function theorem gives:

$$\rho(X) = \sum_j x_j \frac{\partial \rho(X)}{\partial x_j}. \quad (\text{A4.29})$$

Defining $EC_j = x_j \partial \rho / \partial x_j$ proves the full-allocation result in Equation (4.18). For TVaR under regularity conditions, the marginal contribution has the conditional-expectation interpretation:

$$\frac{\partial TVaR_\alpha(X)}{\partial x_j} = \mathbb{E}[X_j \mid X > VaR_\alpha(X)]. \quad (\text{A4.30})$$

Simulation estimates this quantity using tail scenarios. County, peril and line allocations are reconciled to the same portfolio sample and used as management diagnostics.

F.16 Reinsurance optimisation on common samples

For gross annual sample $S^{(m)}$ and treaty programme r , let recovery be $R_r(S^{(m)}, \xi^{(m)})$, where ξ includes hours clauses, reinstatement, collectability and settlement timing. Net retained loss is:

$$S_r^{net,(m)} = S^{(m)} - R_r(S^{(m)}, \xi^{(m)}) + Premium(r). \quad (\text{A4.31})$$

Using the same gross samples for each programme creates a paired comparison and reduces simulation noise. A constrained decision problem is:

$$r^* = \arg \min_{r \in \mathcal{R}} \{ \mathbb{E}[S_r^{net}] + \lambda_C EC_\alpha(r) + \lambda_L Q_k(r) \}, \quad (\text{A4.32})$$

subject to counterparty, budget, coverage and operational constraints. Varying the weights produces a frontier across cost, capital and liquidity; governance selects a point consistent with risk appetite.

Dimensional check. Damage ratios and probabilities are dimensionless. V , L , deductibles, limits, pure premium and tail quantiles share a declared currency and price basis. Event rates have inverse-exposure units; multiplying by exposure produces an expected count. OEP and AEP are dimensionless probabilities over a stated occurrence or annual horizon.

Appendix G: Mathematical supplement to Part 5 - insurance and resilience finance

G.1 Parametric payout and basis risk

Let index J_e activate a payout schedule $P(J_e)$. A simple layer with threshold k , scale $s > 0$ and cap ℓ is:

$$P(J_e) = \min\{s(J_e - k)_+, \ell\}, \quad (x)_+ = \max(x, 0). \quad (\text{A5.1})$$

Equation (A5.1) is the increasing-index form. A low-index trigger, such as a vegetation or rainfall deficit convention, first transforms the index direction or uses $(k - J_e)_+$.

Its expected payout is obtained by integrating over the index distribution:

$$\mathbb{E}[P(J)] = \int \min\{s(j - k)_+, \ell\} dF_J(j). \quad (\text{A5.2})$$

Equation (5.1) defines monetary basis error $BR = P(J) - L$. The expected basis error is:

$$\mathbb{E}[BR] = \mathbb{E}[P(J)] - \mathbb{E}[L]. \quad (\text{A5.3})$$

Mean-squared basis risk decomposes as:

$$\mathbb{E}[BR^2] = \text{Var}(P(J) - L) + (\mathbb{E}[P(J)] - \mathbb{E}[L])^2. \quad (\text{A5.4})$$

Equation (A5.4) follows from $\mathbb{E}[X^2] = \text{Var}(X) + \mathbb{E}[X]^2$. Expanding the variance shows the importance of payout-loss association:

$$\text{Var}(P - L) = \text{Var}(P) + \text{Var}(L) - 2 \text{Cov}(P, L). \quad (\text{A5.5})$$

A well-designed trigger therefore considers both mean payout alignment and covariance with experienced loss. Geographic and livelihood heterogeneity can make one aggregate trigger perform differently across groups.

G.2 Discounted cash-flow derivation

Let net benefit in year t be $NB_t(a) = \mathbb{E}[\Delta L_t(a)] + B_t(a) - O_t(a)$. One unit of currency invested today grows to $(1 + r)^t$ after t periods, so a future amount NB_t has present value $NB_t/(1 + r)^t$. Adding the initial outflow gives Equation (5.2), equivalently:

$$NPV(a) = -C_0 + \sum_{t=1}^T \frac{NB_t(a)}{(1 + r)^t}. \quad (\text{A5.6})$$

The intervention creates positive discounted economic value under the stated assumptions when:

$$\sum_{t=1}^T \frac{NB_t(a)}{(1 + r)^t} > C_0. \quad (\text{A5.7})$$

This is an algebraic rearrangement of $NPV(a) > 0$. It does not establish who receives the benefit or who can repay finance.

G.3 Benefit-cost ratio

Separate discounted gross benefits from discounted costs:

$$PV_B = \sum_{t=1}^T \frac{\mathbb{E}[\Delta L_t(a)] + B_t(a)}{(1+r)^t}, \quad PV_C = C_0 + \sum_{t=1}^T \frac{O_t(a)}{(1+r)^t}. \quad (\text{A5.8})$$

The benefit-cost ratio is:

$$BCR(a) = \frac{PV_B}{PV_C}. \quad (\text{A5.9})$$

For positive PV_C , $BCR > 1$ is equivalent to $PV_B - PV_C > 0$, which is the same sign test as NPV. NPV measures absolute value; BCR measures value per discounted cost and can rank projects differently when scale differs.

G.4 Financial risk layers

For loss L , attachment a and exhaustion $b > a$, the payout of a layer is:

$$X_{a,b}(L) = \min\{(L - a)_+, b - a\}. \quad (\text{A5.10})$$

For any non-negative loss, the stop-loss identity gives expected layer loss:

$$\mathbb{E}[X_{a,b}(L)] = \int_a^b \Pr(L > x) dx. \quad (\text{A5.11})$$

To derive Equation (A5.11), write $X_{a,b}(L) = \int_a^b \mathbf{1}\{L > x\} dx$ and interchange expectation and integration. The result connects financing layers directly to the exceedance curve. Pricing then adds expenses, risk margin, capital cost, counterparty terms and market conditions.

G.5 Liquidity requirement

If funding sources $F_1, \dots, F_J$ become available with delays τ_j , emergency liquidity over horizon k is:

$$Q_k = \left[L_k^{cash} - \sum_{j=1}^J F_j \mathbf{1}\{\tau_j \leq k\} \right]_+. \quad (\text{A5.12})$$

Equation (A5.12) distinguishes ultimate economic loss from cash required before slower financing arrives, which is central to contingency funding and claims-payment readiness.

G.6 Four-ledger reconciliation

Let R_0 be baseline risk, R_a residual risk after intervention a , C_a intervention cost, F_a financing cash flows and M_a observed performance. The avoided-loss distribution is:

$$\Delta L(a) = L(R_0) - L(R_a). \quad (\text{A5.13})$$

The finance ledger requires its own cash-flow identity:

$$Sources_t + OperatingCash_t + Support_t = Uses_t + DebtService_t + ReserveChange_t. \quad (A5.14)$$

Equation (A5.14), not Equation (A5.13), establishes the ability to fund or repay. The MRV ledger updates beliefs about performance:

$$p(R_a \mid M_{1:t}) \propto p(M_{1:t} \mid R_a) p(R_a \mid Design, Delivery, Maintenance). \quad (A5.15)$$

Shared intervention and version identifiers connect the three equations without merging economic benefit, financial cash flow and evidence.

G.7 Cash-flow stress and debt-service coverage

For scenario m , Equation (5.4) gives $DSCR_t^{(m)} = CFADS_t^{(m)} / DS_t$. The probability of covenant breach over project horizon T is estimated by:

$$\hat{p}_{breach} = \frac{1}{M} \sum_{m=1}^M \mathbf{1} \left\{ \min_{1 \leq t \leq T} DSCR_t^{(m)} < k \right\}, \quad (A5.16)$$

where k is the contractual threshold. A hazard changes revenue, operating cost, restoration cost and timing through different channels; scenario attribution shows which channel creates the breach.

Where independently calibrated credit components are available, expected credit loss follows the conditional identity:

$$\mathbb{E}[CreditLoss] = \mathbb{E}[PD(H) LGD(H, C, T) EAD]. \quad (A5.17)$$

Multiplying unconditional point estimates is valid only under restrictive independence and horizon assumptions. Sample-level calculation retains dependence between catastrophe, default, collateral impairment and exposure drawdown.

G.8 Internal rate of return and staged adaptation

IRR in Equation (5.6) is any root r^* of the project NPV polynomial. Multiple sign changes in cash flow can create multiple or absent roots, so NPV at a declared discount rate remains the primary value measure.

For a two-stage adaptation, invest C_1 now and choose expansion a_2 at time τ after observing information $Y_{1:\tau}$. Its value is:

$$V_{stage} = -C_1 + \mathbb{E} \left[\max \left\{ 0, \max_{a_2 \in \mathcal{A}_2} NPV_\tau(a_2 \mid Y_{1:\tau}) \right\} \right]. \quad (A5.18)$$

The maximum with zero represents the option to wait or stop. The value of flexibility is $V_{stage} - NPV(a_{fixed})$ under consistent scope and discounting. Learning has value when future design can respond to it; irreversible urgent protection requires a different timing decision.

G.9 Performance-linked payment and verification

Let verified KPI K_t have target k_t and financing margin m_t . A symmetric illustrative adjustment is:

$$m_t = m_0 + \lambda [k_t - K_t], \quad (\text{A5.19})$$

with contractual floors, caps, direction and verification rules. Measurement error is carried explicitly. If $K_t^{obs} = K_t + \varepsilon_t$, performance probability is:

$$\Pr(K_t \geq k_t \mid K_t^{obs}) = \int_{k_t}^{\infty} p(K_t \mid K_t^{obs}) dK_t. \quad (\text{A5.20})$$

The contract determines whether payment uses a point estimate, verified threshold or dispute process. SCRI preserves the observation, calculation and correction record for the appointed verifier.

Dimensional check. Parametric payout, experienced loss, NPV, benefits, costs and liquidity are expressed in the same currency and price basis before subtraction. The discount rate is a dimensionless rate per stated period. BCR is dimensionless. Attachment and exhaustion share the loss unit.

Appendix H: Mathematical supplement to Part 6 - validation and governance

H.1 Expected operational decision loss

For scenario s , let an error generate cost C_s . Expected loss is the probability-weighted average across mutually exclusive outcomes. Restricting the displayed baseline to false negatives and false positives gives Equation (6.1). A fuller confusion-matrix form is:

$$R(m) = \sum_{s \in \mathcal{S}} \pi_s \sum_{o \in \{TP, TN, FP, FN\}} c_{s,o} \Pr(o \mid s, m). \quad (\text{A6.1})$$

Equation (6.1) follows from Equation (A6.1) when correct-decision costs are set to zero or removed as a common constant. The weights π_s define the evaluation population; changing them changes the product objective and therefore requires ownership.

H.2 Champion-challenger evidence

Let $\Delta R = R(m_b) - R(m_c)$ be improvement over the baseline. Equation (6.2) can be rewritten:

$$\Pr(\Delta R > \delta \mid \mathcal{D}_{val}) \geq 1 - \alpha. \quad (\text{A6.2})$$

A paired bootstrap approximates this probability by resampling validation events, preserving within-event spatial dependence, recalculating both models and recording $\Delta R^{(b)}$. The empirical estimate is:

$$\widehat{\Pr}(\Delta R > \delta) = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{\Delta R^{(b)} > \delta\}. \quad (\text{A6.3})$$

Event-level rather than image-level resampling prevents thousands of correlated frames from creating artificial precision.

H.3 Brier score and calibration-resolution decomposition

For binary outcome $Y \in \{0, 1\}$ and forecast probability p , the Brier score is:

$$BS = \frac{1}{n} \sum_{i=1}^n (p_i - y_i)^2. \quad (\text{A6.4})$$

Group forecasts into bins k with count n_k , mean forecast $\bar{p}_k$, observed frequency $\bar{y}_k$, and overall event rate $\bar{y}$. The empirical decomposition is:

$$BS = \underbrace{\sum_k \frac{n_k}{n} (\bar{p}_k - \bar{y}_k)^2}_{\text{reliability}} - \underbrace{\sum_k \frac{n_k}{n} (\bar{y}_k - \bar{y})^2}_{\text{resolution}} + \underbrace{\bar{y}(1 - \bar{y})}_{\text{uncertainty}} + \epsilon_{bin}. \quad (\text{A6.5})$$

The identity follows by adding and subtracting $\bar{y}_k$ and $\bar{y}$ inside the squared error and collecting terms; ϵ_{bin} records finite-bin approximation when individual probabilities vary within bins. Lower reliability error and higher resolution improve the score.

H.4 Subgroup calibration

For group a , calibration error over bins is:

$$ECE_a = \sum_{k=1}^K \frac{n_{a,k}}{n_a} |\bar{p}_{a,k} - \bar{y}_{a,k}|. \quad (\text{A6.6})$$

An average metric can improve while one group worsens. The release scorecard therefore reports ECE_a , sample size and uncertainty for each operationally relevant group rather than relying only on a pooled ECE.

H.5 Distribution drift

For categorical or binned feature distribution p_k in the reference period and q_k in the monitoring period, the population stability index is:

$$PSI = \sum_{k=1}^K (q_k - p_k) \log \left(\frac{q_k}{p_k} \right). \quad (\text{A6.7})$$

Each term is non-negative in aggregate because Equation (A6.7) is the sum of two Kullback-Leibler divergences, $D_{KL}(q||p) + D_{KL}(p||q)$. It is zero only when the binned distributions match. Small smoothing constants are declared where empty bins occur. Drift prompts diagnosis; it does not by itself prove performance loss.

H.6 Safe degradation and uncertainty expansion

Let σ_{full}^2 be posterior variance with all expected source classes and let source class j fail. Under an information-form Gaussian approximation, precisions add:

$$\sigma_{full}^{-2} = \sigma_{prior}^{-2} + \sum_{j=1}^J \mathcal{J}_j, \quad (\text{A6.8})$$

where $\mathcal{J}_j \geq 0$ is source information. Removing failed source r gives:

$$\sigma_{-r}^{-2} = \sigma_{full}^{-2} - \mathcal{J}_r \leq \sigma_{full}^{-2}, \quad \sigma_{-r}^2 \geq \sigma_{full}^2. \quad (\text{A6.9})$$

Thus a missing informative source widens uncertainty in the approximation. This supplies the mathematical rationale for degraded-mode intervals and for suppressing uses whose evidence threshold is no longer met.

H.7 Sequential monitoring boundary

Let E_t be a non-negative test martingale under the no-drift null with $\mathbb{E}[E_t] \leq 1$. Ville's inequality gives:

$$\Pr\left(\sup_{t \geq 1} E_t \geq \frac{1}{\alpha}\right) \leq \alpha. \quad (\text{A6.10})$$

This provides an optional always-valid alert boundary for repeated monitoring without treating every daily check as an independent fixed-sample test. The chosen monitoring method, assumptions and response playbook remain part of the model card.

Dimensional check. Error probabilities, Brier score, calibration error, PSI and posterior promotion probability are dimensionless. In Equations (6.1) and (A6.1), multiplying an error probability by its monetary or utility cost gives that cost unit; scenario weights are dimensionless and sum to one under the declared evaluation design. The materiality threshold δ therefore has the same unit as operational decision loss.

Appendix I: Canonical loss interface

event_id
hazard_type
valuation_time
observation_cut_off_time
forecast_horizon
scenario_or_posterior_state_version
exposure_snapshot_version
currency_and_price_basis
ground_up_loss_distribution
gross_insured_loss_distribution
net_retained_loss_distribution
reported_and_ultimate_claims_views
mean_and_quantiles
occurrence_or_aggregate_exceedance_probabilities
uncertainty_components
policy_terms_version
reinsurance_terms_version
model_version
data_quality_and_degradation_status
accountable_owner
permitted_use

Every interface consumer must preserve the valuation time, gross/net basis, currency, horizon, uncertainty and model version. A dashboard may simplify the presentation; it may not discard the meaning.